

**ELABORATING A 'NEW KIND OF FAMILY':
FORDIST SOCIAL POLICIES AND RUBBER WORKERS IN THE GLOBAL SOUTH**

BY

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[Committee Page, Acknowledgments, and Dedication to be completed]

Abstract

Gramsci claimed that central to the rise of Fordism was an attempt to 'elaborate a new kind of man' to work in Ford's factories. This 'new kind of man' was fundamentally a 'family man,' and Fordist social policies such as the the Five Dollar Day explicitly encouraged the development of single-wage nuclear families. The Fordist pattern of production was accompanied by a Fordist pattern of social reproduction. In the traditional historiography of Fordism, the productive relations of Fordism, after being perfected in America, were exported - first to Europe during the World Wars, and then to the Global South in the 1970s and 1980s. According to Alain Lipietz, however, by the time Fordist production landed in the Global South, it had been shorn of its social-familial aspect and took the form of a 'Bloody Taylorism'. My dissertation challenges this stagist notion of Fordism and demonstrates that the rise of Fordism in fact had an immediate effect on social relations of families in the Global South. By examining the rubber producing regions of the Amazon and British Malaya, I will show that the increased demand for rubber effected a reconfiguration of family and community relations; a 'new type of man' and corresponding new family forms were elaborated in these regions, albeit on very different terms than in North America. My dissertation will draw on the work of world-systems analysts and of Marxist feminists to examine the social reproductive dynamics that emerged in the growing rubber industry. I will answer a series of related questions: How were existing family relations transformed and exploited by rubber producers in service of increased production? What were the different forms of labor employed in the rubber fields, and how did they correspond to different regimes of social reproduction? and How did workers draw on their families to resist exploitation?

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Introduction

This dissertation began its life in 2019 as what I thought was a simple question: why are all of the houses in Michigan so terrible? I was working with a local tenants union and was confronted with the abject condition of the housing stock in a state that was once home to America's second and third largest companies: General Motors and the Ford Motor Company.¹ The homes I saw around me were once state of the art construction built for workers of the world's most advanced corporations. Why did such wealth devolve into such poverty? In the course of my research, I have not exactly succeeded in answering this question, and indeed the next two hundred and some-odd pages spend rather little time on matters of housing or in the auto factories of Michigan. Instead, each of the elements of this question, 'housing' and 'the Michigan auto industry', has evolved into a larger phenomenon. In this dissertation, in the place of housing I look at the household and the domestic lives of workers, and my conception of the auto industry has expanded to include one of its most important inputs: rubber production in the Global South, without which the automotive industry would have been impossible.

I examine the social reproductive dynamics of working households as they appear in Michigan, the Amazon, and Colonial Malaya. I envision the early automotive industry as a globe-spanning operation connected by a chain of commodities that combines raw materials from around the world with radical new production methods. In the first decades of automotive production saw massive changes not only in the organization of factory work, but in the

¹Gary Hoover, "Largest Companies 1917 and 1929," American Business History Center, September 27, 2018, <https://americanbusinesshistory.org/largest-companies-1917-and-1929/>.

organization of working class households. However, this influence is not limited to the well-paid workers of Michigan. The changes initiated by the automotive industry were global, and in the Global South they led to the immiseration of millions of workers. Through diverse labor forms from slavery to debt-bondage to semi-proletarian plantation work, rubber workers in the tropics were worked to death by planters and merchants eager to get rich by supplying rubber to the industrial North.

My thesis in three parts: 1) The dynamics that have been analyzed under the heading of Fordism were not local to the Midwest, nor limited to the productive and reproductive characteristics in the US and Europe. Rather they were, from the very start, global dynamics. This is especially clear in the connection between the automotive industry and the rubber industry in the Global South. 2) At the heart of their operation is the creation and maintenance of a stratified labor force, manipulated by differentiated but interlocking relations of reproduction. This most important site is the household, but firms and the state play a crucial role here. 3) The dominant effect of Fordist development was not the creation of a large class of well paid proletarians, but a great mass of what may be termed 'proletarian detritus' - footloose migrants, maimed and injured former-workers, and the dead, all used up by and cast out of production.

Background

My initial investigation seven years ago into the housing stock of Michigan quickly led, as all investigations in Michigan do, to the automotive industry. Many of the houses I was looking at - old homes, rotting and sinking in their foundations, connected to city water mains via lead pipes - were originally built to house the employees of automotive companies. Oldsmobile and REO Motor Company had once produced cars in Lansing, where I lived at the

time; General Motors employed one in four residents of Genesee county in its Flint plant in the 1970s; the Ford Motor Company was headquartered in Dearborn, next to Detroit. Each of these cities grew rapidly in the first half of the twentieth century as migrants came from all over the world to work in the auto factories, and the automotive companies developed housing schemes to provide shelter for their new employees. After the 1970s, automation and offshoring decimated the local labor market, and each of these cities has since experienced a housing crises as the old housing stock decays. The most famous images of deindustrializaiton and urban decay come from Michigan: urban blight in Detroit and the 2014 water crisis in Flint. Indeed, the abject state of housing in Michigan today can be traced to the very companies that once brought the state great wealth.

Early 20th century employee housing initiatives at automotive companies were part of a larger system of labor management. One of the key innovations associated with Fordism is capital's growing interest in the domestic conditions of its laborers - the conditions of both the reproduction of labor and the extended realization of value by means of working class consumption. Programs such as Ford's Five Dollar Day and General Motors' Modern Housing Corporation were not expressions of altruism, but were capitalists' (rather successful) attempts to extend their influence over more of their workers' lives. Workers were included or excluded in these programs based not only on their performance at work, but also on their race, the structure of their households, and their adherence to arbitrary moral codes set by their employers.

In other words, these programs were concerned with the social reproduction of the conditions of capitalist society. Social reproduction in capitalist society refers to "the reproduction of capitalist society as a whole, a process shaped by the imperative of endless

accumulation."² This includes the reproduction of labor, but also the reproduction of capital, capitalists, and the conditions of continual accumulation. The social programs of automotive companies were especially interested in ensuring a continual supply of labor. Labor is not a one-time input, but once expended must renew itself and re-enter into production on both a daily and a generational basis. In order to produce the wealth of commodities that circulate in society, workers must be not only present but healthy and educated according to their particular task in production. Following Tithi Bhattacharya we must ask:

If workers' labor produces all the wealth in society, who then produces the worker? Put another way: What kinds of processes enable the worker to arrive at the doors of her place of work every day so that she can produce the wealth of society? What role did breakfast play in her work-readiness? What about a good night's sleep? We get into even murkier waters if we extend the questions to include processes lying outside this worker's household. Does the education she received at school also not "produce" her, in that it makes her employable? What about the public transportation system that helped bring her to work, or the public parks and libraries that provide recreation so that she can be regenerated, again, to be able to come to work?³

The actual work of renewing labor power takes place largely (but not entirely) outside of the sphere of production and in the home, and just like the tasks that workers perform, the structure of households differ according to their position in society: only the most productive employees are paid well enough for their family to survive on a single wage, or to hire a domestic worker to help around the house. The families of workers in lower-paid jobs are less likely to be able to afford to own homes in nice neighborhoods, and the rest of the family is more likely to work to supplement the household income. My project takes up an analysis of the differentiated domestic sphere and how its structure relates to capital's need for differentiated labor inputs.

²Kirstin Munro, "Chapter 12: Social Reproduction," in *Marx: Key Concepts* (Edward Elgar Publishing, 2024), 214.

³Tithi Bhattacharya, "Introduction: Mapping Social Reproduction Theory," in *Social Reproduction Theory: Remapping Class, Recentering Oppression* (Pluto Press, 2017), 1-2.

There is a second role that the household plays in capitalist society: it is the endpoint of commodity circulation, the final stage in the valorization of value. The purpose of capitalist production is accumulation, which requires the purchase and consumption of commodities. Commodity production is generally aimed at the ultimate goal of consumption, either directly through the production of domestic goods like food, clothes, and gasoline, or indirectly through the production of machines that other companies will use to make these goods. The necessity of household consumption also factors into the structure of households, as workers must be paid well enough to purchase some of the means of subsistence from the market. The tension between lowering the cost of labor while still maintaining working class consumption is a principle tension in social reproduction.

This tension was a primary concern for firms in the early automotive industry. When they were first invented, cars were luxury items, available only to the very wealthy. The major automotive companies in Michigan, led by the Ford Motor Company, transformed production through a campaign of rationalization and vertical integration. By increasing the productivity of labor through automation, they lowered the price of cars while still maintaining a well paid class of workers who could afford to buy what they produced. Another crucial element of what made this strategy so successful was establishing control over the materials inputs required for automotive production; Ford owned copper mines and forests in Michigan and coal mines in West Virginia and Kentucky.⁴ The one input that could not be produced domestically was rubber, which could only be produced in tropical regions. In order for automotive production to be viable, rubber had to be sourced from these regions at a low enough price to enable both the high wages of factory workers and the low price of finished cars. The automotive industry relied on

⁴*The Ford Industries: Facts About the Ford Motor Company and Its Subsidiaries* (Ford Motor Company, 1924), 6.

the severe exploitation of Global South Workers, primarily in the Amazon and Colonial Malaya, to produce this cheap rubber.

Amazonian and Malayan production, and indeed all capitalist production, connect to what are called commodity chains. Commodity chains are the linked production processes that result in commodities. Automotive commodity chains, for example, include the assembly of the finished car, but also the production of raw materials, their transport, mediate processing of the materials into automotive components, and also the reproduction of labor power at each of these points, including the food eaten by workers and other commodities consumed in their private lives. Automotive companies were forced to source rubber from the Global South, and the long chains of transit added considerable cost to the final produced commodity. This is offset - indeed the whole system is made possible - by the considerably lower price of labor in the Global South. The lower price of labor is, in turn, made possible by household structures that supplement lower wages with other types of informal income or subsistence labor.

The development of global infrastructure in the capitalist world-system is crucial to the extension of commodity chains. Advances in infrastructural technology such as steamboats, railways, and even guns, facilitated the "penetration of Asia and Africa by Europeans and the conquest of colonial powers."⁵ These technologies are the means by which the capitalist system maintains the global division of labor and conditions the corresponding value differential in labor and goods. The incorporation of peripheral zones into the capitalist world-economy required the development of an extensive system of shipping routes, transit lines, and roads. This system connected markets in the core with cheap labor supplies in the periphery, but they also

⁵Daniel R. Headrick, *The Tentacles of Progress: Technology Transfer in the Age of Imperialism, 1850-1940* (Oxford University Press, 1988), 5.

maintained the cheapness of these labor supplies. The study of labor regimes along commodity chains must take into account the introduction of infrastructure, and the exploitation of existing transit ways such as navigable rivers.

A Critical Method

My methodological approach draws on several traditions, but is basically animated by Moishe Postone's approach to critical social theory.⁶ In his reinterpretation of Marx's critical theory, he attempts to historicize the central category of capitalism: labor. Rather than treating labor as a transhistorical category that is acted upon and mistreated by capitalists, he critiques labor as constitutive of capitalism itself. In this project, I aim to undertake a parallel analysis of the household: I do not treat the household as some domestic sphere separate from capitalism that is acted upon by capitalists or the state, but as itself a constitutive category of the global capitalist system. With the spread of capitalism, families were of course destroyed and traditional bonds dissolved, but they were then recomposed according to capitalist logic. What are the specifically capitalist forms of households? What are the roles of the state and the firm in developing and maintaining them? I hope to investigate why capital has such an intense interest in cultivating diverse types of households, and what purpose they serve in propping up global structures of exploitation and accumulation. I believe that the early automotive industry provides an ideal case to examine these dynamics.

In this project, I primarily draw on the literature of historical sociologists and Marxist feminists to historicize the category of the household. I am especially influenced by the work of scholars who study the household as the unit of the reproduction of labor, and the various forms

⁶Moishe Postone, *Time, Labor, and Social Domination: A Reinterpretation of Marx's Critical Theory* (Cambridge University Press, 1993).

it takes in different parts of the world. The world-systems definition of the household as an income-pooling unit which combines various forms of labor and income to survive is useful.⁷ I am inspired too by the attempt of Social Reproduction Theorists to "make visible labor and work that are analytically hidden by classical economists and politically denied by policy makers,"⁸ and by Kirstin Munro's work demonstrating the role of household production in the reproduction of capitalism itself.⁹ Each of these approaches to analyzing the household is in some way critical - rather than accepting the received category of family as a transhistorical, coherent, and self-contained phenomenon they interrogate the category itself, shedding light on the obscured but manufactured role that the family-as-household plays in capitalist society.

The cases of rubber producers in the Amazon and Colonial Malaya also demonstrate the futility of isolating local phenomena from their global context. The rubber industries in these regions - indeed even the automotive industry in Michigan - cannot be analyzed as national phenomena. Wallerstein called for "the substitution of a unit of analysis called the 'world-system' for the standard unit of analysis, which was the national state."¹⁰ In other words, national phenomena such as economies and industries, can not be divorced from the global economy. By analyzing the various systems of rubber production that emerged in the late 19th and early 20th centuries in the context of the automotive industry, we can better understand how and why they took the historical form they did. This is true also of the domestic lives of their workers: my analysis shows that the household dynamics of labor reproduction are in fact global dynamics. I

⁷Immanuel Wallerstein, *World-Systems Analysis: An Introduction* (Duke University Press, 2004), 32.

⁸Bhattacharya, "Introduction: Mapping Social Reproduction Theory," 2.

⁹Kirstin Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," *Science & Society* 83, no. 4 (2019): 451-68.

¹⁰Wallerstein, *World-Systems Analysis: An Introduction*, 16.

want to show, not merely that the nuclear family which reproduces formally free proletarian labor is but one among many possible household organizations, but that it is in fact dependent on the existence of other forms of family organization which reproduce unfree forms of labor.

One of Marx's important contributions was to denaturalize so-called 'free' wage-labor - to demonstrate that its generalization is not a natural state of humanity, but a historical artifact of the modern period. Vulgar Marxists have taken this too far, insisting that it is the only artifact of modernity, and any other form of labor is pre-modern or pre-capitalist. I take issue with this strict identification, as the development of capitalism as a world system has been accompanied by a great increase in unfreedom, but my interest is not in litigating which of waged or unwaged labor is more 'capitalist'. My central concern is not whether the workers I study sold some x amount of time for some y amount of currency. My concern is the generalization of exchange, and the extent to which a system of generalized exchange which holds all things to be somehow commensurable shapes the ways in which they lived. In other words, as the global market expanded along the automotive commodity chain, how was labor subordinated to commodity production - production for exchange.

The Structure of the Dissertation

Chapter 1 serves to develop part of the theoretical framework I will employ in this dissertation. I will begin by surveying the existing literature around Fordism and its implementation in the Global South starting in the 1960s. Using the concept of commodity chains, I hope to complicate this narrative and demonstrate that the development of Fordist practices in the early automotive industry required an immediate restructuring of productive relations in the Global South, especially in the rubber producing regions. From here I will

introduce my main research question: if Fordism developed a new norm of working class nuclear households in Michigan, what effect did it have on households in other parts of the world? Or more pointedly, in what ways did the reconfiguration of peripheral households make the American nuclear family possible?

Chapter 2 introduces the second part of my theoretical framework, drawing on Marxist feminist literature on households dynamics and social reproduction. This chapter investigates the domestic dynamics of social reproduction in the Michigan automotive industry from roughly 1914 through the mid 1930s. I will first focus on the Five Dollar Day program implemented at the Ford Motor Company, with attention to housing policy. I hope to establish the firm's interest in propagating single income households among their white workers, and to question why this was never generalized to the entire population. I will then look at the Modern Housing Corporation, a division of General motors which built and sold homes to its employees. This program differed from earlier company town models in that it fostered home ownership among workers rather than reserving all property rights with the firm. As with the Five Dollar Day, this program was administered on a segregated and racialized basis, and contributed to the development of a deeply stratified workforce in which the value of Black labor was depressed.

Chapter 3 looks at Amazonian rubber production, which dominated the global market through the 19th and into the 20th centuries. This chapter will examine the patterns of wild-rubber collection in the Amazon and the household formations that supported them. First I will look at transhumant rubber workers in Brazil who moved seasonally between the port cities and the rubber fields deeper inland. What are the social-reproductive dynamics of transhumant labor? As working men were separated from their families for a season, how did they arrange the daily

renewal of their labor power, and how did their families deal with the long gaps in income? This chapter will also look at Henry Ford's attempt to start a rubber plantation in the Amazon in the 1920s. His plantation, which was a colossal failure, was based on the notion that the social-reproductive regime in Michigan could be imported to other parts of the world. Fordlandia was a mirror image of working class neighborhoods in the US, complete with prefabricated houses, schools, parks, and libraries. Administered life in Fordlandia met with resistance from local workers in Brazil and the town was abandoned in the mid 1930s without producing any commercially significant amount of rubber. This chapter will attempt to answer the questions: why was the establishment of a town based on Fordist productive and reproductive norms unsuccessful in Brazil? and In what ways did Ford misunderstand his own innovation, and why were his production and social practices unfit for rubber production in Brazil?

Chapter 4 focuses on the Peruvian Amazon Company, a rubber firm based in Iquitos that notoriously kept workers in near-slavery, and that carried out the Putumayo genocide. During the course of several decades, the indigenous population of the Putumayo was depleted by 90 percent. This chapter looks at the social, political, and geographical differences between the Putumayo and the rest of the Amazon that led to different labor regimes. In the course of the chapter I ask several questions: How does genocide fit into the global world-system? How was a continual supply of labor secured in the midst of mass slaughter? How did the rest of the world react once the genocide was made public?

Finally Chapter 5 traces the rise of Asian rubber. In the 1910s, wild harvested Amazonian rubber was superseded in the global market by rubber produced on plantations in Colonial Malaya. This chapter will chart the rise of Malayan rubber, as well as the development of its

specific labor regime. The workers on Malayan rubber plantations were primarily ethnic Tamils from Madras who migrated to Malaya for multi-year labor stints. I outline their working and living conditions on the plantations, and contextualize this within the broader context of their household groups. As with transhumant workers in Brazil, I ask how daily life was reproduced among the largely male workers at the work sites, and what effect this migratory pattern had on their families who were left behind.

Chapter 1: Fordism, Social Reproduction and the Global Division of Labor

Fordism has been a slippery concept within historical writing over the last 100 years. Although labor historians and economists agree that it is a central moment in both American and global economic history, there is little agreement on what Fordism actually *is*. In 1990, in a talk humorously titled “What in the F—’s name is Fordism”, British sociologist Simon Clarke noted the imprecision in discussions of Fordism and whatever non-Fordist capitalist epoch is supposed to be its successor: “Much energy has been expended in debating the diffuse characterizations of ‘Not-Fordism’. However, much less attention has been paid to the characterization of Fordism.”¹ Even the popular legacy of Henry Ford himself is deeply ambivalent. He was a benefactor who improved the lives of his workers and a tyrant whose Sociological Department spied on their personal lives; he created the modern middle class but also initiated the culture of consumption that is currently eating the planet; he was a humanitarian who founded schools for the his mostly immigrant employees and he was a vociferous anti-Semite. Only one thing is uniformly agreed upon: Fordism represented a genuine break with existing patterns of production in capitalist society; it was a revolution.

The boundaries of this revolution are blurred and ill-defined. Attempts to define it find themselves faced with irreducible contradictions. Daniel Watson identifies two main approaches to Fordism: a narrow view of Fordism as a system of mass production centered around Henry

¹Simon Clarke, “What in the F—’s Name Is Fordism,” in *Fordism and Flexibility: Divisions and Change*, ed. Nigel Gilbert et al. (Palgrave Macmillan UK, 1992), 13.

Ford's innovative assembly line, and a broader view of Fordism as a "phase of capitalist development taking place several decades after the heyday of the Ford Motor Company" that is characterized by a "careful balance in the American economy between mass production and mass consumption and an increase in the wage rate and social provisions for workers".² In this broader view, the Fordist era was initiated by Henry Ford's productive innovations, but was not fully realized until decades later when his policies were generalized to the rest of the industrialized world.

Each of these definitions is unsatisfactory. In the first place, "[t]here was nothing original in either the detail or the general principles which Our Ford applied to the production of motor vehicles,"³ and so it makes little sense to isolate them from their social component. But if Fordism is to be defined by the unique articulation between these general production principles and his social policy, then why have analysts found it "difficult to discover any examples of 'Fordism' in its pure form, even in the United States"?⁴ Inflation quickly caught up to the Five Dollar Day, and the Sociological Department was dissolved in 1920-21, effectively ending Fordist paternalism and morality bonuses.⁵ In the US, the generalization of social provision to workers was not effected by private capital but by the state, with Franklin D. Roosevelt's New Deal. This flew in the face of both Henry Ford's ideology and his policies, as he was a fierce

²Daniel Watson, "Fordism: A Review Essay," *Labor History* 60, no. 2 (2019): 2.

³Simon Clarke, "What in the F—'s Name Is Fordism," in *Fordism and Flexibility: Divisions and Change*, ed. Nigel Gilbert et al. (Palgrave Macmillan UK, 1992), 16.

⁴Simon Clarke, "Overaccumulation, Class Struggle and the Regulation Approach," *Capital & Class* 12, no. 3 (1988): 62.

⁵Stephen Meyer III, *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921* (State University of New York Press, 1981), 197.

opponent of the welfare state,⁶ and had placed restrictions on his worker's access to his social benefits.

There is a sense in which the primary legacy of Fordism is one of stratification. Simple narratives about the Fordist worker and the family wage do not hold up to scrutiny. In Michigan's auto factories in the first decades of the twentieth century, local workers were increasingly divided into two camps. On the one hand, manufacturing tasks were decomposed into minimal parts, rationalized to the extreme and voided of intellectual stimulation. On the other hand, a great class of managers and engineers was cultivated to design the machines and processes that would combine the decomposed labor of others into finished products - following a managerial revolution that Alfred D. Chandler called the 'visible hand' of management.⁷ Outside the factories, the living conditions of workers were similarly divided. Skilled and semi-skilled workers, mostly white, were transformed into large-scale commodity consumers and inducted into a class of homeowners, while racialized and menial laborers often lived in crowded slums and still relied significantly on informal labor for survival.

Of course, the labor that contributed to the manufacturing of automobiles was not confined to factory work. The operations of the Ford Motor Company were embedded in a global network of production. Raw materials had to be imported - especially rubber which could only be sourced from the Global South. The stratified labor market in industrial Michigan was mirrored by a global division of labor, in which wild rubber harvesters in the Amazon and the Congo, and to a lesser extent the later rubber plantation workers in southeast Asia, occupied an intensely degraded position. Wage labor was not the dominant labor relation in the rubber

⁶Sidney Fine, "The Ford Motor Company and the N.R.A.," *The Business History Review* 32, no. 4 (1958): 353-85.

⁷Alfred Chandler, *The Visible Hand: The Managerial Revolution in American Business* (Belknap Press: An Imprint of Harvard University Press, 1977).

regions of the world, and when Northern capital intervened in the rubber industry, it created quite different modes of exploitation. These modes of exploitation were accompanied by different patterns of consumption as well. The rubber workers were made to be dependent on the market for the means of survival, and especially on the importation of canned goods and preserved food from the Global North. But they were not consumers of domestic commodities, sheltered by a commodity house and supported by de-commodified reproductive labor. Their precarity was exploited by capital to effect an extension of the market into the private lives of producers on very different terms than in the Global North.

This chapter takes up early labor stratification in the automotive industry and the methods used to create and discipline it. Keeping the complications listed above in mind, I hope to situate early Fordist production within the birth of what Giovanni Arrighi called the “US systemic cycle of accumulation” - the period of historical capitalism covering much of the twentieth century when the United States dominated global capitalist accumulation. According to Arrighi, this cycle of accumulation was distinguished by “an internalization of transaction costs,”⁸ in which firms vertically integrated in order to retain the surplus value otherwise lost through the purchase of inputs. The scope of this integration was explicitly global, as accumulation in the core has historically been dependent on the exploitation of peripheral regions, and as the working conditions in the periphery responded to demands of accumulation in the core. However, the methods of production and labor control closely related to Fordism, such as assembly lines, the decomposition of tasks, and a family wage, were *not* generalized globally. This raises the question of how vertical integration was effected across national lines, or perhaps more

⁸Giovanni Arrighi, *The Long Twentieth Century: Money, Power, and the Origins of Our Times*, 2nd ed. (Verso, 2010), 247.

accurately, what were the effects of Fordist production practices in the core on labor conditions in the periphery?

The second purpose of this chapter is to introduce the dynamics of social reproduction within this global context. Linking these two concepts - the global scope of early Fordism and its regime of social reproduction - is the commodity chain. Wallerstein and Hopkins used the term commodity chain to conceptualize the interconnectedness of global capitalism: “take an ultimate consumable item and trace back the set of inputs that culminated in this item - the prior transformations, the raw materials, the transportation mechanisms, the labor input into each of the material processes, the food inputs into the labor.”⁹ The commodity chain is the axis along which Fordist firms tried to integrate, but this chain includes more than just raw materials and the labor that works them: it is imbricated with “the reproduction of the labor forces involved in these productive activities.”¹⁰ Along the commodity chain, different production processes can be seen as part of the axial division of labor which “divides production into core-like products and peripheral products.”¹¹ The difference in core-like and peripheral products relates to the profitability and the market leverage of producers: automotive assembly in Michigan was much more profitable than rubber extraction which was defined by fierce, cutthroat competition. The axial division of labor has as a corollary an axial division of social reproduction, by which lower-waged laborers and high-waged laborers are supported by different household structures.

The specific ways in which social reproduction was effected at various points along the Fordist commodity chain will be examined more closely in chapters 2-5, and this chapter is

⁹Terence K. Hopkins and Immanuel Wallerstein, “Patterns of Development of the Modern World-System,” *Review (Fernand Braudel Center)* 1, no. 2 (1977): 128.

¹⁰Hopkins and Wallerstein, “Patterns of Development,” 128.

¹¹Immanuel Wallerstein, *World-Systems Analysis: An Introduction* (Duke University Press, 2004), 28.

mainly concerned with making the case that the reproductive dynamics in the rubber producing regions of the Global South are not externally articulated with Fordist reproduction, but internal to it. Automotive production required several different types of material and labor inputs, and these correspond to different regimes of social reproduction. In particular, they take advantage of a different articulation between the rationalization of labor, and the expansion of commodity consumption among the workers. The scope of this chapter will focus primarily on the origins of Fordism: the production of automobiles in the first decades of the twentieth century.

The Contradictions of Fordism

In the popular imagination, Fordism comprises two moments of personal genius: Henry Ford's implementation of the automated assembly line and his institution of a family wage with the Five Dollar Day. Faced with the mutually exacerbating problems of rising demand and increasing employee turnover, Ford took the innovative approach of intensifying and de-skilling labor while at the same time increasing its pay.¹² These principles were combined in Ford's notorious River Rouge factory, a single complex that took raw materials as inputs turned them into cars. In the meantime, the laborers who worked in this factory were middle class workers who could afford to purchase the commodities they were producing. At one and the same time the efficient production methods transformed automobiles from a luxury good to a near-necessity, and transformed the workers from struggling laborers into participants in the bounty of production. The company's win-win solution to the antagonism of labor and capital transformed American life, and facilitated the dominance of US industry on the world stage. The combination of innovative shop floor management with a reconfiguration of family life ushered in the modern

¹²Carl H. A. Dassbach, "The Origins of Fordism: The Introduction of Mass Production and the Five-Dollar Wage," *Critical Sociology* 18, no. 1 (1991): 77–90.

middle class and an era of mass consumerism. In this view, Fordism solved once and for all the antagonism between labor and capital.

This popular conception is problematic on several counts. Ford's production practices were enormously successful, but the core idea of a powered assembly line was borrowed from Chicago slaughterhouses and had first been employed in the automotive industry by a competitor of Ford. In 1911 management at EMF company "installed a mechanized assembly line to boost chassis output to 100 per day."¹³ Ford would not install his first line (also for chassis production) for another two years, in 1913. Even after Ford's unprecedented success with rationalizing production, his methods would often be taken and improved upon by competitors - most notably General Motors. Ford's real innovation was the extreme to which he took the rationalization of production, and its combination with an aggressive social policy. His famous Five Dollar Day was not a universal wage increase, but a profit share given only to employees whose private lives were approved by agents of the Ford Sociological Association. In addition to investigations into the stability of the workers' families, "the company wanted to know whether or not the workman owned or was purchasing property or a home, whether or not he had a savings account in a bank, and whether or not he was in debt."¹⁴ In other words, in order to qualify for a family wage, Ford's workers had to demonstrate that they were sufficiently rooted in the community surrounding the factory. When combined with wages that exceeded those offered by competitors, this meant that Ford's prized workers were less able to seek out employment elsewhere. The question was not

¹³Wayne Lewchuck, "Fordism and the Moving Assembly Line: The British and American Experience, 1895-1930," in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 20.

¹⁴Stephen Meyer III, *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921* (State University of New York Press, 1981), 131.

one of boosting the freedom of workers by increasing their pay, but of limiting their freedom and compelling them to stay in their jobs.

As we can see, Fordism is a deeply ambiguous concept, full of contradictions. This is part of why it has been studied and written about so extensively - almost any conceivable position on the relationship between labor and capital can find support in the empirical cases of Fordism. But the explanatory power of Fordism as a category in the history of capitalism is exaggerated. Scholarly attempts to resolve the contradictions of Fordism have usually involved a sort of analytical surgery, either removing the social-reproductive aspect and focusing on Henry Ford's specific management practices, or excising Ford almost entirely and treating 'Fordism' as synecdoche for larger structural changes in the global accumulation of capital in the mid twentieth century. Each of these approaches is untenable. The first reduces Fordism to a loose collection of stolen ideas that were implemented with a ruthless intensity, but which explain very little about the history of capital. The second attempts to establish an ideal type of capitalist society that bears little resemblance to any actually existing world. It inevitably devolves into a list of precursors, exceptions, and modifications that overwhelm the type itself: Taylorism, Sloanism, Toyotism, Keynesianism, Post-Fordism, Neo-Fordism, etc. Sociologist Simon Clarke criticizes this tendency, saying "to explain these variants as failed versions of one ideal-type implies that the latter is in some way more real or more fundamental than the former."¹⁵ Indeed, it is hard to see why Fordism would stand out in this list as the determining element.

The irresolvability of these contradictions are made clear in the work of the regulation school, a group of French scholars who devoted considerable time to defining and delimiting the

¹⁵Clarke, "What in the F—'s Name Is Fordism." This phrase does not appear in the version of this paper published in *Fordism and Flexibility: Divisions and Change*, but in an earlier (unpaginated) draft presented at the 1990 British Sociological Association Conference available at this web location: <https://scispace.com/pdf/what-in-the-f-s-name-is-fordism-2bwbsf1e0m.pdf>

concept of Fordism. Their scholarship was rigorous, but flawed. Within this group, Michel Aglietta provides the classic structuralist account of Fordism as a generalized set of production principles, historicized within the long cycles of historical capitalism. For Aglietta, although Fordist practices may have originated as specific policies within the auto industry, Fordism represents a societal-level shift in the relations of production and the reproduction of capital. In *A Theory of Capitalist Regulation*,¹⁶ he criticizes the dominant economic theories of capitalist equilibrium for being, among other things, ahistorical and abstract. Against this, Aglietta denies “that what exists does so automatically”, necessitating a theory of social reproduction, and he claims that “when actual social systems are studied, historical experience confirms that *transformation means rupture, qualitative change*.”¹⁷ In other words, although Henry Ford's specific policies may have been solutions to the immediate problems in his own factories, their eventual widespread adoption does not indicate that they tapped into some transhistorical and abstract need for workers to receive higher wages. Rather, it indicates that these policies addressed a *historical* condition that may not have existed before, and may change again in time. Aglietta is inspired by “Marx's refusal to postulate an immutable essence underlying the variability of phenomena”¹⁸ and so he attempts to situate Fordism as a specific moment within the turbulent history of capitalist production.

Taking the US since 1870 as his spatial-temporal object, Aglietta elaborates a theory of capitalist regulation as an attempt to “formulate in general laws the way in which the determinant structure of a society is reproduced.”¹⁹ These are not transhistorical laws for society in general,

¹⁶Michel Aglietta, *A Theory of Capitalist Regulation: The US Experience* (NLB, 1979).

¹⁷Aglietta, *Capitalist Regulation*, 12.

¹⁸Aglietta, *Capitalist Regulation*, 15.

¹⁹Aglietta, *Capitalist Regulation*, 13.

but laws structuring a specific society. In the decades after the American Civil War, they were oriented towards the development of heavy industry. Claiming that “the fundamental economic condition for industrial capitalism to develop is the formation of a growing agricultural surplus product and its realization in commodity form,”²⁰ Aglietta highlights 19th century America’s privileged situation. The incorporation of vast frontiers precipitated the development of rail infrastructure, irrigation, farming and mining. This ‘regime of extensive accumulation’ was funded and aided by the development of western urban centers and their growing financial powers, but it did not initiate a fundamental restructuring of the life of laborers. Labor was becoming more productive through intensification and mechanization, but the laborers' wages were generally low, and they were often reliant on a significant amount of subsistence labor rather than commodity consumption at home.

In the first decades of the twentieth century, this regime fell into crisis. Aglietta positions the economic collapse of the 1930s as a result of overaccumulation and underconsumption. Specifically, the average worker was paid so little that they had to support themselves through a significant amount of informal labor, and therefore could not contribute to the valorization cycle by purchasing and consuming commodities. As capital entered a crisis in the 1930s, producers sought out new ways to maintain profitability. This would mean restructuring society to expand the domestic market and increase the class of consumers. As producers cast about for a model, Fordism, at this point a decade and a half old, stood out as a generalizable answer to changing economic conditions.

Fordism figures into Aglietta’s account as an ‘intensive regime’ which “creates a new mode of life for the wage-earning class by establishing a logic that operates on the totality of

²⁰Aglietta, *Capitalist Regulation*, 73.

time and space occupied or traversed by its individuals in daily life.”²¹ In other words, Fordism is characterized by an unprecedented expansion of capitalist interest and control over the lives of its waged workers beyond the shop floor. In order to stave off another crisis of overproduction and underconsumption, the daily lives of workers would have to be rebuilt on the foundation of commodity consumption. Aglietta terms this reorganization of social life a ‘mode of consumption’ which is oriented towards the renewal of the energy expended in labor processes and the maintenance of a continual supply of labor power.

As an articulation between production and consumption, Fordism comprised several specific innovations linked to more general social reconfigurations. On the production side, Aglietta identifies the automated assembly line as Fordism’s “characteristic labour process.”²² The assembly line separated mental and manual labor to a previously unseen degree. Instead of managing the relationship between several related tasks, workers performed simplified rote tasks over and over: “the labour process [was] converted from a dense network of relationships between jobs, with intermediate products passing back and forth ... into a straightforward linear flow of the material under transformation.”²³ In Ford’s factories, the production of a car was decomposed into 7,882 separate tasks, each of which was carefully documented and classified. The workers were similarly rationalized - not humans, but various assemblages of limbs: “The lightest jobs were again classified to discover how many of them required the use of full faculties, and we found that 670 could be filled by legless men, 2,637 by one-legged men, 2 by armless men, 715 by one-armed men, and 10 by blind men.”²⁴

²¹Aglietta, *Capitalist Regulation*, 71.

²²Aglietta, *Capitalist Regulation*, 117.

²³Aglietta, *Capitalist Regulation*, 118.

²⁴Henry Ford and Samuel Crowther, *My Life and Work* (Doubleday, Page & Company, 1922), 108.

The mental component of labor was excised from the ambit of the general laborer and given to a growing class of managers and process engineers. Workers became fixed to clearly and rigidly defined roles that were “determined by the configuration of the machine system”²⁵ which was “planned in engineering departments which were themselves subject to the division of labor.”²⁶ In this sense, the Fordist revolution went hand in hand with the managerial revolution described by Alfred Chandler.²⁷ As the scale and scope of production increased, and as jobs were broken up into smaller and smaller elements of an intricately designed whole, the “division between the labour of head and hand”²⁸ was deepened. Thought was no longer a part of the act of labor itself, but something that had already been completed in the past and invested into the creation of a proscribed list of infinitely repeatable movements. The Fordist workplace began to look more and more like that described by Marx: living labor ruled by the dead labor that was embodied in the planning and manufacturing of the machine system.

This process of rationalization extended beyond the shop floor. Along with the revolution in production, Fordism reconfigured the social and private lives of the workforce. Aglietta notes that “for the first time in history, Fordism created a norm of working-class consumption in which the general ownership of commodities governed the concrete practices of consumption.” Two commodities stand out as governing this norm: the automobile as a means of private transportation, and standardized housing. The adoption of these two major commodities transformed other patterns of household production. Households began to substitute the purchase

²⁵Aglietta, *Capitalist Regulation*, 118.

²⁶Aglietta, *Capitalist Regulation*, 119.

²⁷Chandler, *The Visible Hand*.

²⁸Alfred Sohn-Rethel, *Intellectual and Manual Labour: A Critique of Epistemology*, trans. Martin Sohn-Rethel, Historical Materialism Book Series (1978; Brill Academic Publishers, 2020), 3.

of ready-made products like canned food and clothing for space-intensive home production, such as butchering and tailoring. Furthermore, the nineteenth century pattern of roaming peddlers bringing such necessities as pots, pans or medicines, directly to the tenement buildings where workers lived, was replaced by workers taking their now-indispensable cars on trips to the grocer and hardware store.²⁹ All of these commodities transformed the urban spaces in which the working classes were renewed, and their relatively high cost “presupposed a vast socialization of finance, and correlatively a very strict control over workers’ resources and expenditures.”³⁰ However, Aglietta notes that the rise of certain family norms within a consuming class of workers was never quite general: Fordism also deepened the stratification of the working classes, which “forms part of the concrete conditions of reproducing the wage relation.”³¹

Aglietta’s account of Fordism as an historical articulation of means of production and consumption responding to the global needs of capital accumulation is a compelling one, but it fails on two counts, both related to his idealist conception of Fordism. First, Aglietta considers the intensive regime of accumulation to be an ideal type of capitalist society embodied by Fordist America, against which other societies can be measured, rather than the outcome of a struggle between the concrete interests of specific classes and actors. Simon Clarke criticizes this position, deeming Aglietta’s perspective structural-functionalist. Clarke claims that for the Regulationists, “in the relation between structure and struggle, it is the class struggle which takes the subordinate role, its significance constrained by the structural imperatives of the regime of

²⁹Ruth Schwarz Cowen, *More Work for Mother: The Ironies of Household Technology from the Open Hearth to the Microwave* (Basic Books, 1983), 71-85.

³⁰Aglietta, *Capitalist Regulation*, 159.

³¹Aglietta, *Capitalist Regulation*, 137.

accumulation.”³² On the question of working-class struggle this charge is perhaps more applicable to Aglietta’s followers than to his own work, which deals at length with working class organization. But the working classes are not merely structural features of a capitalist society. Workers are agents with the power to impose their demands on capital, and their organizations are not subordinate to the structure of capital accumulation, but constitutive of it.

This calls into question the regulationist framework of extensive vs intensive regimes of accumulation. Clarke criticizes the concept of a regime of accumulation, which first assumes a structural equilibrium and then attempts to explain it with recourse to a relatively stable configuration of institutional forms. For Clarke, the dynamic between production and consumption associated with Fordism are “best seen not as aspects of the regulation of a regime of accumulation, but as related aspects of the institutionalization of a particular balance of class forces in the immediate post-war period.”³³ In other words, the dynamic pioneered by Ford in the 1910s and 1920s was established primarily through concrete social and political struggles. The economic policy of Keynesianism was dictated by “the balance of class forces, institutionalized in the systems of industrial relations, electoral politics and the welfare state.”³⁴ In short, the regulation approach “overestimates the stability and duration of ‘Fordist’ modes of regulation” and “the extent to which Fordism can be considered a systematic ‘regime of accumulation’.”³⁵

The second shortcoming of Aglietta's work is his (non)treatment of imperialism, and his abstraction of US capitalism from the system of global capitalism. In the introduction to *A*

³²Clarke, Simon. 1988. “Overaccumulation, Class Struggle and the Regulation Approach.” *Capital & Class*. Vol 12:3.

³³Clarke, “Overaccumulation,” 24.

³⁴Clarke, “Overaccumulation” 18.

³⁵Clarke, “Overaccumulation” 16.

Theory of Capitalist Regulation, Aglietta defines imperialism as a system “through which one state manages to influence a series of other states to adopt a set of rules that are favorable to the stability of a vast space of multilateral commodity relations guaranteeing the circulation of capital.”³⁶ After posing this anemic definition, he pushes it aside, claiming that “there could be little question of analyzing in the present work the system of contemporary states and the transformation that the crises of the 20th century have produced in inter-state relations.”³⁷ But he is unable to escape the weakness of his conception of capitalist hegemony - it undermines his entire project. In his attempt to analytically isolate America from the Global South, he misapprehends the extent to which American industry both exacerbated and was dependent on an unequal global division of labor. The intensification of exploitation across state lines undergirds capitalist accumulation in Global North states. His insufficient definition of imperialism as management inflicted by one state on several with the goal of broadening the circulation of capital prevents Aglietta from appreciating the global scope of capitalist regulation and reproduction.

In “Towards Global Fordism?” Alain Lipietz, another scholar from the regulation school, attempts to more adequately theorize the connection between the Fordist core and the periphery, but his analysis shows the same weaknesses as Aglietta's. Writing in 1982, he notes that

in three ‘glorious’ post-war decades, a crucial change was introduced by the system of capitalist accumulation that various French economists, following Gramsci, have baptized ‘Fordism.’ A ‘centre’, displaying a cohesive internal logic at the level of blocs (North America, the EEC) if not individual countries, came to stand against a ‘periphery’ which, whatever the internal dynamic of its component parts, was ever more disconnected from this ‘autocentred’ logic.³⁸

³⁶Aglietta, *Capitalist Regulation*, 32.

³⁷Aglietta, *Capitalist Regulation*, 32.

³⁸Alain Lipietz, “Towards Global Fordism?,” *New Left Review*, no. 132 (1982): 33–47.

However Lipietz immediately questions whether we can even “speak of a world-wide extension of Fordism,” noting that during the late 1960s and early 1970s, the economic periphery experienced uninterrupted growth while the economic core went into a production crisis. Indeed, the industrialization of the periphery was not a slow march following in the footsteps of the core, but was its own specific development, conditioned by and conditioning the dynamics of accumulation in the United States and Europe. By the time 'Fordism' reached the Third World, it had devolved into another one of its non-ideal, mutated forms. Lipietz describes a Global South Fordism stripped of its impetus to consumption as "sub-Fordism" and "'Bloody' Taylorization."³⁹

Although Lipietz's work is rigorous and empirically rich, it is undergirded by a flawed methodology. He first maintains Aglietta's idealist assumption that the Fordism of the US can be analytically extracted from its own contemporary global context. He then looks at the structures of accumulation in the Global South decades later and compares them to this ideal type. The implicit question is not 'how did the structure of accumulation in the South develop?' but 'why did Fordism fail to be exported as-is?' It completely sidesteps the question of how production in the South is contributed to the development of Fordism in the core.

A more appropriate approach to the global scope of Fordism is to trace the specific ways in which the automotive industry created new relations of production and consumption in peripheral countries, and how these relations contributed to the accumulation of capital in the core countries. A useful framework for such an analysis is the commodity chain - the global network of extraction and production processes that transform raw materials into finished goods. As an raw essential material unavailable in the US and Europe, rubber and its conditions of production expose the fundamentally global nature of the production of automobiles. Cars could

³⁹Lipietz, “Towards Global Fordism?” 41.

not be produced if rubber was not available - and their production would not be profitable if it was too expensive. In the drive to procure sufficient quantities of cheap rubber, European and American capital pulled the rubber producing regions in the Amazon, Africa and Southeast Asia into their sphere of influence, but on deeply unequal terms.

Several factors contributed to the underdevelopment of the rubber regions: the demands of Northern industry for cheap rubber, Colonial governments with no interest in the development of stable and autonomous economies in the Global South, and unscrupulous local actors happy to exploit their countrymen for a marginal share in the profits of capital. Each of these factors interacted with the same categories that were so important to Henry Ford: the possibilities of rationalizing processes of production and the substitution of commodity consumption for subsistence labor. These two elements of re/production were manipulated in different ways than in the US Midwest, but they were unified as part of a "logic that operates on the totality of time and space occupied or traversed by its individuals in daily life."⁴⁰ To sideline the exploitation of rubber workers, and the exclusion of Global South industry from the profits of the automotive industry is to misrecognize Fordism as a category of the capitalist world-system.

If we have questioned the definition of Fordism as a systematic regime of accumulation, and its validity as a mode of regulation divorced from both state action and the global contours of capital, we might reasonably ask what use the concept of Fordism provides, and what value we can gain from the regulation school's treatment of it. Two of Aglietta's core interventions - Fordism as a specific qualitative change in the articulation of production and consumption, and the increased stratification of the wage-earning class - are crucial, but they must be reformulated as questions. We are interested in asking what concrete struggles led to the particular relationship

⁴⁰Aglietta, *Capitalist Regulation*, 71.

between production and consumption in the early automotive industry, as well as interrogating the limits of this relationship. Furthermore, we cannot define Fordism so narrowly as to exclude the conditions of production in the Global South. The production of rubber in the Amazon, Africa and Southeast Asia was not external to the automotive industry, but internal to it. With this in mind, we will investigate the articulation between production and consumption in the rubber industry, and the stratification of the rubber workers. We will pay special attention to the development of commodity consumption and attempts at the rationalization of rubber gathering. In the following sections, we will look more closely at these phenomena both in the early Michigan automotive industry, and in the rubber industry abroad.

A Provincial Study: Fordism in Michigan

As the birthplace of Fordism, however defined, industrial Michigan bears the imprint of Fordism's early development. Since the Second World War, American labor historians have been arguing over how to analyse the productive dynamics it brought to light. Marxist historians in the US emphasized the labor process itself as an essential component of labor history. Harry Braverman's *Labor and Monopoly Capital: The Degradation of Work in the Twentieth Century* centered the deskilling of the workforce as a method of labor control. Nelson Lichtenstein writes in the introduction to *On the Line: Essays in the History of Auto Work* that "Braverman broke new ground by alerting historians and sociologists to the social context within which the technology of production was embedded."⁴¹

⁴¹Nelson Lichtenstein and Stephen Meyer III, eds., *On the Line: Essays in the History of Auto Work* (University of Illinois Press, 1988), 5.

Braverman's view was oversimplified, however, and labor historians later debated "the extent to which craft skills were actually eroded in mass production industry."⁴² Rather, the labor market was bifurcated between low-paid and low-skilled workers and highly paid engineers and managers. The partial deskilling of the workforce was a point of tension between labor and capital. At times capitalists strategically degraded the skill of workers in order to disempower them, while the unions resisted this degradation.

It is often repeated that unions in late 19th and early 20th century Michigan were weak and listless - supposedly this contributed to the rise of Fordism as a new method of labor exploitation. However, this claim is challenged by Thomas Klug, in an essay in *On the Line*. As the lumber industry declined at the close of the century, economic activity in the state shifted towards the growing cities. In the 1890s, Detroit was the stove producing capital of the world, with a significant presence of other types of metalwork that would be crucial to the development of the auto industry. Most of the organizations representing workers were craft unions, and they admitted only skilled labor to their ranks. However, the assertion that these unions were powerless is misguided. Klug notes that "although only a few employers of iron molders formally agreed to the exclusive employment of union men, many foundries were de facto closed shops."⁴³ The reason for this is the Iron Molders Union's strategy of organizing the "best molders" so that any employer wanting good labor had to go through the union.

The craft unions in fact posed enough of a threat to employers that in 1902, "metal trade employers organized the Employers' Association of Detroit (EAD)."⁴⁴ They felt that "the brief

⁴²Lichtenstein, "Introduction," 5.

⁴³Thomas Klug, "Employers' Strategies in the Detroit Labor Market, 1900-1929," in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 44.

⁴⁴Klug, "Employers' Strategies," 48.

accommodation with labor unions in the late 1890s and early 1900s had gained them few benefits Rather, strikes became more frequent, and unions gained additional recruits.”⁴⁵ The unions had developed significant leverage in eastern Michigan and prevented employers from undermining labor rights to increase productivity and profitability. As a unified bloc, the EAD was able to enforce what they considered to be their right to pay workers based on the quality of their individual work, and to hire from outside the local labor pool.

Detroit employers were also troubled by what they saw as a decrease in control over the labor process itself. As the unions had a monopoly on skilled labor, which was crucial to the operation of heavy manufacturing, they were able to exercise significant influence on the operations of firms. John J. Whirl, a member of the EAD, said that “in some of our industries the proprietors were merely the financial agents to purchase the supplies, pay the wages, and sell the manufactured products ... As to the operations of the factory, they had no more real control than if they were in no way connected with it.”⁴⁶ The attempt by employers to band together and undermine the craft unions was joined with an understanding that to truly gain control of their businesses, they would need to reduce their reliance on a small pool of skilled labor: an understanding that would animate the coming Fordist revolution.

And it was a revolution. The automobile was not invented in Michigan – in fact the first two viable gasoline powered cars were built in Massachusetts and Indiana in 1892 and 1894.⁴⁷ But Michigan arrived early on the automotive scene, propelled by easy access to machining, raw materials, and capital: in 1896 Ransom E. Olds of Lansing and Detroit’s Charles B. King and

⁴⁵Klug, “Employers’ Strategies,” 48.

⁴⁶Klug, “Employers’ Strategies,” 49.

⁴⁷Willis F. Dunbar, *Michigan: A History of the Wolverine State* (Eerdman, 1980), 552.

Henry Ford demonstrated viable motor powered carriages. By the early twentieth century, there were dozens of auto companies in Michigan, but they faced significant issues.

Initially, auto production was a craft industry: each car was handmade by skilled craftsmen. Due to the competitive market in Michigan, it was difficult for employers to keep workers from leaving for greener pastures at another company. The turnover rate was over 200% in Detroit plants, reaching a high of 382% at Ford's Highland Park plant, and the costs of training new workers was estimated to be between \$35 and \$100.⁴⁸ Meanwhile, the cars were also expensive, and so there was a limited market for the product being produced. Henry Ford's management approach targeted both of these problems, and ultimately transformed the industry.

In 1908, Ford introduced the Model T, a mass produced car with standardized parts that required less skill to make and cost far less than competitor's cars, contributing to a rise in sales. In 1913 he introduced his version of the automated assembly line, further standardizing production and reducing costs. And in 1914, he initiated the Five Dollar Day program, a wage increase that solved the problems of employee turnover and bloated training costs. In order to compete, rival firms had to adopt and even advance Fordist innovations. In the 1920s, General Motors introduced the yearly model change to stimulate consumer demand even further.

The challenges that Fordism introduced to the labor market and to the shop floor have been of particular interest to labor historians working in the Marxist tradition. From the early suppression of union activity by auto companies to the 1936 sit-down strike in Flint and the rise of the most militant unions in American history, the Fordist factory is a towering edifice in American labor historiography. But a full accounting of the social change wrought by Fordism requires a labor history that is critical of the notion of labor itself.

⁴⁸Klug, "Employers' Strategies," 54.

Liberal and Marxist historians in the twentieth century disagreed about the political nature of their work and the goal of the labor movement, but there is a fundamental similarity in their analysis. The key category for both traditions is labor, but while labor historians interrogated its relations with other social forms such as capital or law, they were less critical of the category of labor itself. Labor and by extension laborers are taken to be self-evidently coherent classifications: generally wage-laborers, generally men, generally working for someone else.

In 1971, Thomas Krueger, in “American Labor Historiography,” narrates a partial break with the shared paradigm of the liberal and Marxist historiography, but at the time of his writing that break was still incomplete. He is concerned with a shift away from the traditional conception of labor history, citing Hyman Berman’s opinion that labor history “is more than the rise or fall or the failures and successes of organizations,” and David Montgomery’s call to research “the cultural and intellectual life of the working classes.”⁴⁹ This new direction follows the work of the British Marxists in expanding the category of the working classes to include a greater sample of the relations of working class life, including community support of labor against capital during strikes, the influence of religion, and the permeability of class categories.

Krueger notes that in 1971, this development was still immature, and had to “solve some of its intellectual problems.”⁵⁰ His framing of the task ahead, however, demonstrates just how partial the break with old labor history still was. He lists ten “imposing questions” that need to be answered by new labor historians, and while they demonstrate an expansive view of labor history, they hew to a decidedly traditional view of the category of labor itself. One question is

⁴⁹Berman and Montgomery quoted in Thomas A. Krueger, “American Labor Historiography, Old and New,” *Journal of Social History* 4, no. 3 (1971): 281-2.

⁵⁰Krueger, “American Labor,” 283.

representative: “What happened to blacklisted workers, to men over 40 who got fired, to families whose head was mangled or killed on the job.”⁵¹ Implicit in this question is the orientation of a nuclear family around labor and the elevation of specifically male waged labor.

In the 1970s, a significant challenge was posed to this traditional notion of labor, and to both Marxist and classical labor historiography, by feminist organizations and historians. Prominent among these was the Wages for Housework campaign, and the challenge they posed changed the way that labor history in the US could be written. Writers and activists in this tradition questioned the notion that only waged labor contributed to the economy and took traditional Marxism to task for accepting the capitalist devaluation of feminized work.

Sylvia Federici, in the 1975 essay “Counterplanning from the Kitchen” lays out the feminist challenge to a wage-centric notion of labor posed by Wages for Housework:

Beginning with ourselves as women we realize that working for capital does not necessarily produce a paycheck and does not begin and end at the factory gates.... Housework, in fact, is much more than housecleaning. It is servicing the waged workers physically, emotionally, and sexually and getting them ready to work day after day for the wage.... This means that behind every factory, every school, every office, and every mine there is the hidden work of millions of women, who consume their life reproducing those who work in those factories, schools, offices, and mines.⁵²

The family life of workers, in other words, is not a realm separate and apart from their labor but is constitutive of it. The unpaid work of women (and children) forms the base on which the institution of waged labor stands. Implicit in Federici’s quote (and explicit in her other work) is the accusation that labor history and labor activism, by ignoring the work of women, reproduces capital’s own devaluation of gendered and non-waged work.

⁵¹Krueger, “American Labor,” 284.

⁵²Sylvia Federici, *Patriarchy of the Wage: Notes on Marx, Gender, and Feminism* (PM Press, 2021), 11-12.

The point of this intervention is not to merely add a new type of labor to an old conception: the new category is not labor plus gender. A labor history that merely traces struggles and changes in the structure of housework would do little more than transplant the old framework of analysis onto a new subject. Likewise, a treatment of gender as a variable that affects only women would be counterproductive. Instead, the task for feminist labor history is to interrogate the very process of categorization. Elizabeth Faue, introducing a volume of essays on gender and labor history, claims that “labor historians’ use of collectivities as units of analysis – whether classes, communities, trades, or families – masked both intraclass division and an understanding of conflicting interests that was at times one-dimensional.”⁵³ To remedy this, we must examine not how the category of gender *acts on* the category of labor, but how it permeates and constitutes it.

The work of Marxists in theoretically elaborating the role of the household in capitalist reproduction is useful here. Marxist feminists have critically investigated the categories of the family and the household, and how they have been manipulated to serve the interests of capital. Within the household, labor is reproduced through a combination of inputs, chief among them gendered and devalued domestic labor. The structure of labor reproduction is not constant, but changes in response to different circumstances/locations. In the work of Social Reproduction Theorists, devalued domestic labor takes center stage as the condition of the reproduction of wage laborers, and so they make a general distinction between productive labor (generally that which is done for a wage) and unproductive/reproductive labor (that which is concerned with providing for the base needs of life, on either a waged or unwaged basis). As Nancy Fraser puts

⁵³Elizabeth Faue, “Gender and the Reconstruction of Labor History: An Introduction,” *Labor History* 34, nos. 2–3 (1993): 171.

it, the unwaged work of (usually) women supports the renewal of the life of the waged worker and forms “one of [the] background conditions of possibility” of capitalist accumulation.⁵⁴

The household, however, serves more purposes than just the reproduction of wage-laborers; they are also the ultimate sites of commodity consumption. Kirstin Munro writes that “[w]orking class households are compelled to engage in household production not only to survive, but so their members can appear continually on the market as renewed labour-power in order to continually buy more commodities.”⁵⁵ Because households in capitalist societies are severed from traditional means of subsistence, they are forced to resort to commodity consumption in the maintenance of their lives. This means that “the household and its production processes are implicated in the reproduction of capitalist society”⁵⁶ - households are not merely the objects of a devastating transformation effected by capitalist society. Through commodity consumption, they continually reproduce the necessary conditions for the existence of capitalism. This is not an assignment of blame, but a diagnosis of the mutually conditioning nature of the relationship between households and the capitalist society in which they are embedded.

Marxist Feminist perspectives on the household can grant us insight into the management of labor in Detroit around the turn of the century. Prior to 1914, when Henry Ford instituted the famous Five Dollar Day in an attempt to reduce employee turnover, it was uncommon for Michigan families to live on the wage of a single earner. Working-class families cobbled together livings from various sources, fighting both to make higher wages and to carve out more time to

⁵⁴Nancy Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” in *Social Reproduction Theory: Remapping Class, Recentering Oppression*, ed. Tithi Bhattacharya (Pluto Press, 2017), 23.

⁵⁵Kirstin Munro, “THE CAPITALIST FORM OF HOUSEHOLD PRODUCTION: SUBSIDY, ANACHRONISM, OR SOMETHING ELSE,” *Etica & Politica / Ethics & Politics* 27, no. 1 (2025). 266.

⁵⁶Kirstin Munro, “‘Social Reproduction Theory,’ Social Reproduction, and Household Production,” *Science & Society* 83, no. 4 (2019). 454.

do necessary subsistence labor. In “The Detroit Story: the crucible of Fordism,” Kathleen Stanley and Joan Smith examine the extent to which working-class families were transformed by the family-wage in Detroit from roughly 1890 through the middle of the twentieth century. They build on the category of the household as an income-pooling unit from world-systems analysis and take the family as a site of wage-contestation. Working-class households piece together subsistence from various sources: in addition the consumption of commodities purchased with a wage, they rely on the unpaid subsistence labor of women and children, charity or community assistance, income from boarders, informal commodity production, and many other sources. The need to depend on these extra-wage sources of income would be alleviated with a family-wage.

Smith and Stanley point out that in the 1890s, wages in the Detroit area were much lower than the national average. This was partly due to the depressed wages of immigrants who made up close to 70% of the city, and due to the relatively large percentage of the workforce that were women. While noting that there was much variation among family incomes, they cite that “The [average] laborer whose budget was reported by the Michigan Bureau of Labor, for example, faced an annual shortfall of \$86, or 27 percent of his total income.”⁵⁷ Over a third of the city’s workforce was made up of “semiskilled, unskilled, and service workers” who earned considerably less than this average.

In addition to the type of labor performed, a major factor in the ability of Detroit workers in 1890 to survive on a wage was family composition, which correlated strongly with race. Polish immigrants generally had the largest households of any ethnic group, much greater than American-born whites. While most of the households across all ethnic groups were organized as

⁵⁷Kathleen Stanley and Joan Smith, “The Detroit Story: The Crucible of Fordism,” in *Creating and Transforming Households: The Constraints of the World-Economy*, ed. Joan Smith and Immanuel Wallerstein, Studies in Modern Capitalism (Cambridge University Press, 1992), 36.

nuclear families, Black families were far more likely to either not have children, or to have only one parent. Stanley and Smith note that while smaller families are generally considered to be more sustainable in times of depression (such as the 1890s), the significantly larger-than-average Polish households were able to leverage their family composition into more effective “patterns of subsistence.”⁵⁸

It appears that “Polish families brought with them skills associated with their recent past as peasants, which could be put to use outside the wage nexus.”⁵⁹ While it was uncommon for married Polish women to work outside the home for a wage, they were more likely than other groups to maintain livestock and vegetable gardens. When cash flow was an issue, Polish children were sent to work at higher rates than children of other families. The various forms of subsistence labor and multiple sources of wage income combined to insulate Polish families from both economic downturns and a reliance on a single wage-earner.

Other ethnic groups in the 1890s had their own strategies for survival that combined wage and non-wage incomes. Black families in Detroit generally had fewer dependents, which lowered the cost of living, but also meant that they could not send children to work. As a result, “married Black women were twice as likely to work for wages as other women”⁶⁰ and over a third of all Black women had waged jobs. Additionally, by 1900, Black families had the highest proportion of boarders in the city – non-family members who earned their keep either through rent payments or through contributing subsistence labor to the family.

⁵⁸Stanley and Smith, “The Detroit Story,” 37.

⁵⁹Stanley and Smith, “The Detroit Story,” 38.

⁶⁰Stanley and Smith, “The Detroit Story,” 39.

American born whites also depended on boarders, although to a lesser extent than other ethnic groups. They generally had children but pulled them out of school to work at a lower rate than other ethnic groups. Stanley and Smith note that “this was possible in part because of the higher wages received by native White men, 72 percent of whom were either professional or skilled workers.”⁶¹ These families rarely sent married women into the workforce, although occasionally unmarried white women did work outside of the home.

After the development of the Five-Dollar Day in 1914, household patterns in Detroit changed, but not necessarily in the direction of single-income nuclear families. Stanley and Smith note that in the decades following Ford’s family-wage campaign, “even in Detroit, this presumed bastion of Fordism, the family wage was never a reality for the majority of wage-earners.” Except for a small elite strata of the working classes, “the male wage was always supplemented by the wages of children and sometimes of wives, as well as by considerable amounts of subsistence labor.”⁶² This picture looks much the same as pre-Fordist Detroit, so what changed? Was the social transformation of Fordism as radical as its labor transformation?

In short, yes, but to understand exactly how we need to look more closely at patterns of race. Between 1900 and 1940, the population of Michigan more than doubled, from 2.4 million to 5.3 million. During this same period, the Black population of the state increased by a factor of more than *thirteen*, from just over 15,000 to more than 208, 000.⁶³ Most of this increase was driven by Black migrants from the southern states, looking for better working conditions and wages in the industrial Midwest.

⁶¹Stanley and Smith, “The Detroit Story,” 38.

⁶²Stanley and Smith, “The Detroit Story,” 34.

⁶³US Census Bureau. *Michigan*. 1940.

To an extent they were able to find them, but they did not enter the Michigan workforce on equal footing with whites. According to the 1930 census, Black men were significantly overrepresented in low paid manufacturing jobs, while their white counterparts were more often employed either as shop foremen or skilled machinists.⁶⁴ The cities themselves mirrored this, with white workers more often owning free standing homes in newly developed neighborhoods, and Black workers and poor whites confined to the slums. In other words, the social significance of Fordism is defined as much by who it excluded from prosperity as by who it included.

Disparities in income are compensated for in many ways, and as Stanley and Smith demonstrate, this is an essential element in labor history. It is not enough to note that Black laborers generally earned less than whites – we must look at the household forms that propped up this disparity and that allowed low-income groups to survive despite their condition of degraded labor. Stanley and Smith noted that Black families were much more likely to rely on boarders to make ends meet, and Black women were much more likely to work outside the home. Even within the “white” ethnic groups, there are various forms of family composition that contributed to the types of labor performed. Polish women laboring informally in vegetable gardens or caring for chickens are as much a part of labor history as American born unmarried white women working outside the home for a wage. And both are inextricable from the histories of the generally male, generally white worker that take up so much space in liberal and Marxist labor histories.

⁶⁴US Census Bureau. *Michigan*. 1930.

Global Complications: the Rubber Industry

Fordism did not develop in isolation, nor exclusively from local dynamics. Just as the existence of Fordist production methods was dependent on racial and gendered divisions in the local labor market, it was also built on a global division of labor, whereby the more brutal exploitation of workers in peripheral states propped up the productive relations in the core. Wallerstein refers to this global dynamic as the “axial division of labor,” a dynamic which ensures a “constant flow of surplus-value from the producers of peripheral products to the producers of core-like products.”⁶⁵ The success of industry in the core cannot be isolated from this global network of production that supports it.

The nature of this global dynamic between the capitalist core and periphery was characterized by Andre Gunder Frank as the development of underdevelopment.⁶⁶ In his account, the supposed economic backwardness of the Global South is not a failure to keep pace with the capitalist core, but rather “is the result of its centuries-long participation in the process of world capitalist development.”⁶⁷ In other words, the underdevelopment of the Global South is a condition positively created by the dynamics of global capitalism. Samir Amin’s *Accumulation on a World Scale* deals with this concept as well, attempting to counter the prevailing stagist narrative of underdevelopment which isolates the poverty of the periphery from world history and assumes that “laissez-faire was certain to develop the colonies just as it had developed the industrialized metropolitan countries.”⁶⁸ Amin insists that it is the capitalist core’s economic growth itself that creates the impoverished condition of the periphery. He claims that “as

⁶⁵Wallerstein, *World-Systems Analysis: An Introduction*, 60.

⁶⁶Andre Gunder Frank, “The Development of Underdevelopment,” *Monthly Review* 18, no. 4 (1966): 17–31.

⁶⁷Frank, “The Development of Underdevelopment,” 21.

⁶⁸Samir Amin, *Accumulation on a World Scale* (Monthly Review Press, 1971), 13.

economic growth proceeds, none of these features by which the structure of the periphery is distinguished lessens; on the contrary, each increases.... In the periphery growth is *not* development, for its effect is to disarticulate.”⁶⁹

Economic disarticulation in the periphery is central to Amin’s analysis. In the core, national economies are largely integrated: different sectors “carry out substantial exchanges between themselves, which may be called “interindustrial” or “intersectoral” exchanges.”⁷⁰ The outputs of extractive industries within developed economies directly support domestic manufacturing and agriculture. For so-called “underdeveloped” economies, internal trade between industrial sectors is stifled: “The different kinds of mineral wealth exploited by these great concerns - metals, oil, etc. - are not destined to supply domestic industries on the spot, but are exported in order to supply complex industrial groups in the advanced countries.”⁷¹ Whereas in the core, rationalization and growth in one industry is quickly generalized to others, in the periphery industrial growth has minimal effect on the domestic economy as a whole, transferring the benefits to its industrial trading partners.

This dynamic can be traced along what are called commodity chains: the global networks of labor and raw-material inputs that feed into the production of particular industrial and consumer goods. The term commodity chain was coined in a 1977 paper by Wallerstein and Hopkins, who propose it as an analytical tool to combat a “developmentalist” perspective in which “the world is said to consist of a number of related but basically autonomous ‘societies’ ...

⁶⁹Amin, *Accumulation*, 18.

⁷⁰Amin, *Accumulation*, 16.

⁷¹Amin, *Accumulation*, 16.

each moving upward along an essentially similar path of development.”⁷² Criticizing an analytical nationalism which attempts to abstract autonomous societies from each other, Wallerstein and Hopkins insist on analyzing the world-system *in toto*: “a definite ‘world’, a spatio-temporal whole, whose spatial scope is coextensive with the elementary division of labor among its constituent regions or parts and whose temporal scope extends for as long as the elementary division of labor continually reproduces the ‘world’ as a social whole.”⁷³ Within this global social body, commodity chains form a sort of vascular system, transporting raw materials and ossified labor inputs from their sites of production, combining them, and bringing them to market where they can be valorized and turned into profit.

Commodity chains bring local and global production processes into contact. As commodity production expanded over the course of capitalism’s global history, the need for vast amounts of raw materials to be transported across large distances increased. Bunker and Ciccantell claim that the “solutions to the raw materials problem require the coordination of multiple physical and social processes across geopolitical and physical space with domestic relations between firms, sectors, the state, labor, and new technologies.”⁷⁴ As a sort of vascular system for raw and processed materials within the capitalist world-system, commodity chains are a primary axis of contestation for actors across ‘geopolitical and physical space.’ The state acts on commodity chains through trade agreements, or through nationalizing or privatizing reserves of raw materials. Labor asserts its power to strike or withhold itself at various points along the chain - labor inputs being the stuff that connects various links. And finally the firms act to either

⁷²Hopkins, Terence K., Immanuel Wallerstein. 1977. “Patterns of Development of the Modern World-System.” *Review (Fernand Braudel Center)*, 111.

⁷³Hopkins and Wallerstein, “Patterns of Development,” 112.

⁷⁴Stephen G. Bunker and Paul S. Ciccantell, *Globalization and the Race for Resources* (Johns Hopkins University Press, 2005), 19.

subsume more and more of the production processes that feed into the final consumable as in Fordist vertical integration, or they outsource moments in the production process to locales with a cheaper labor supply as we see in the trends of post-Fordist globalization.

Commodity chains are a useful concept for countering perspectives which treat capitalist accumulation as a local phenomenon. Analyses, such as Aglietta's, that treat capitalist production in core regions as a self-contained whole fail to account for how productive relations in the core are propped up by quite different productive relations in the periphery. Smith, Ciccantell, and Sowers note that this problematic has been taken up at length by world-systems analysts who examine the racialized nature of labor in "resource frontiers and extractive peripheries."⁷⁵ The effect of these "extractive peripheries on existing population, and especially indigenous populations, are almost universally negative, engendering often strong and violent opposition from indigenous groups."⁷⁶ The remainder of this section will examine production and reproduction in this context along the Fordist commodity chain.

For early automobile producers in Michigan, a significant portion of the commodity chain was entirely internal to the US. Many of the raw materials needed for the big three automakers to produce automobiles were readily available within the United States. Michigan's vast lumber and mineral resources were brought to Detroit by a robust rail system, and Ford even owned "mines in Kentucky and West Virginia, which gave him control over every natural resource needed to make a car - save rubber."⁷⁷ To acquire the rubber necessary for the

⁷⁵David A. Smith et al., "Reconstructing Commodity Chain Analysis as World-Systems Analysis," in *World-Systems Analysis at a Critical Juncture*, ed. Corey R. Payne et al. (Routledge, 2023), 62.

⁷⁶Smith et al, "Reconstructing Commodity Chain Analysis," 62.

⁷⁷Greg Grandin, "Empire's Ruins: Detroit to the Amazon," in *Imperial Debris: On Ruins and Ruination*, ed. Ann Laura Stoler (Duke University Press, 2013), 118.

production of automobiles, Ford and his competitors had to look outside the US - at first to the Amazon and the Congo, and then later to Malaya and Sumatra in Southeast Asia.

Rubber has been known and used by Mesoamerican peoples for thousands of years - it was used to make bottles, balls or waterproof shoes. Despite this, in the 19th century the rubber industry in Europe and the US mainly provided for products with limited industrial practicality until the development of the vulcanization process in the late 1830s. Vulcanization allowed for the creation of stable and hardened rubber industrial products, and by 1845 carriage wheels were equipped with solid rubber tires.⁷⁸ The demand generated by tire production, along with rubber insulation for electrical wires, drove the rubber industry to new heights. In the US, the rubber industry was centered in Ohio - by 1900 Akron was the home to Goodrich, Goodyear, and Firestone. Business was good and all three were multi-million dollar companies by 1910. Firestone was especially successful, thanks to exclusive contracts with Ford to provide tires to the Model N in 1906 and to the famous Model T in 1908.⁷⁹ Fortunes were made in the manufacture and sale of tires, but down the commodity chain the story was quite different.

Large scale extraction of rubber began in Brazil and central Africa in the 1800s, where latex producing plants grew natively. Bunker and Ciccantell claim that since “the Amazon provided the technically most useful and abundant kinds of rubber in the world, states and firms in the rapidly industrializing, trade-dominant nations of Europe and America were particularly interested and active in the Amazon.”⁸⁰ Initial production consisted of wild-collection by contract laborers who made long treks through the jungle to collect rubber from sparsely distributed trees.

⁷⁸Austin Coates, *The Commerce in Rubber: The First 250 Years* (Oxford University Press, 1988), 49.

⁷⁹Coates, *Commerce in Rubber*, 134.

⁸⁰Bunker and Ciccantell, *Globalization*, 39.

Rubber plants were quickly exported to tropical regions around the globe such as southeast Asia, although there was little commercial production in Asia until the second decade of the twentieth century.⁸¹ Although plantations would dominate the global rubber production by the second decade of the twentieth century, for the first several decades of the auto industry there were few rationalized rubber plantations, and rubber was sourced from wild collectors in the Amazon and the Congo. By the end of the nineteenth century, the Amazon was supplying the world market with 15,000 tons of rubber annually.

The working conditions in rubber fields of the Amazon were a world away from those in the US. In the late 1870s a severe drought struck Northeastern Brazil, destroying the means of subsistence for a large portion of the population. Many left their homes and found work in the growing Brazilian rubber industry, but under the harshest of conditions. Forced to pay for their own provisions on the way to the work locations, rubber tappers started working in severe debt. Further, the workers were cut off from their previous communities, and so to survive they were forced to rely on local rubber agents. Their debt was perpetuated by the sale of “the necessary working tools and basic goods such as foodstuff, which were forcibly acquired at the rubber patron’s storehouse or station.”⁸² The rubber companies held this debt on account against future rubber yields, but “every time [the worker’s] rubber (his credit) was collected, it was found that his debt had mounted higher than his earnings.”⁸³ The rubber agents and traders ensured that the prices of necessary provisions were higher than the piece-wages for rubber, creating a vicious

⁸¹Coates, *Commerce in Rubber*, 115.

⁸²Louise Cardoso de Mello and Sven Melkebeke, *From the Amazon to the Congo Valley: A Comparative Study on the Violent Commodification of Labour During the Rubber Boom (1870s–1910s)*, Palgrave Studies in Economic History (Palgrave Macmillan, 2019), 148.

⁸³Coates, *Commerce in Rubber*, 94.

cycle of debt peonage. It was this debt relation, rather than a wage, which ensured the continual supply of labor for the rubber industry.

The labor supply in the Congo was ensured through similar means, but this time with a tax burden rather than a debt burden. In a comparative study of Amazonian and Congolese rubber extraction, de Mello and Van Melkebeke note that from the 1890s on, an “in kind rubber tax ... proved to be the solution to increase revenue.” They go on to say that “this huge increase in exports received by in-kind taxation would result in severe and violent labour mobilisations and would cause devastation throughout the Congolese countryside.”⁸⁴ Whereas in the Amazon, workers migrated to the rubber regions due to harsh economic and environmental conditions back home, in the Congo the state and private firms sent expeditions into the countryside to capture and enslave Indigenous men. “Expeditions often ended up in plunder, destruction of villages and capturing of people: initially, to collect ivory (by the Congo Free State) and subsequently, to collect rubber (by the Congo Free State and private sector).”⁸⁵ These workers were then forced to collect and deliver “six to eight kilograms per person every two weeks” and “when they did receive some sort of remuneration, this was usually in-kind (salt, knives, beads, textiles, etc.).”⁸⁶

Unlike in the US Midwest, few provisions were made for generational reproduction. It is true that the automotive industry in Michigan employed large amounts of immigrant labor, but if they conformed to Ford's social expectations then allowances were made to support their families. The Five Dollar Day was explicitly an encouragement to generational reproduction. In

⁸⁴de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 143.

⁸⁵de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 147.

⁸⁶de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 152-153.

the Amazon and the Congo, however, existing communities and their social relations were violently disrupted by rubber barons and state actors who saw no need to recreate conditions for the renewal of laborers. They preferred to exploit indigenous and migrant populations as an external pool from which new labor could be drawn. The pay given to rubber workers (if it can be called pay) was insufficient to provide for the renewal of the conditions of labor on an open labor market, and so the gap was filled by coercion, and by the workers' informal reproductive labor. Furthermore, although workers along the commodity chain were increasingly commodity consumers, and not merely producers, in Michigan this took the form of household appliances purchased with an earned wage. In the rubber fields these commodities were obtained from closed company shops, and in both the Amazon and the Congo, roving rubber tappers lived not in their own houses, but in temporary shelters in the rubber fields.⁸⁷ When debt and tax burdens were insufficient to compel labor in these conditions, violence was used, and when violence was insufficient workers were killed, and new workers brought in.

In the wild rubber fields, gendered labor dynamics differed significantly from those in industrial Michigan. Many Michigan women performed unpaid reproductive labor to support and supplement their husband's 'family wages' and in the poorer communities they took on piecework or service work in addition to this unpaid care work. In the core, this gendered dynamic was structured in part by the housing policies of the industries, but in the periphery the gendered division of labor manifested quite differently. Maria Mies' analysis of women and colonialism helps us contextualize the dynamics of gendered labor in the periphery. Writing about the 18th and 19th centuries, she notes that where unfree labor was secured through slaving raids and trade, "slave women were actively discouraged from bearing children or forming

⁸⁷de Mello and Melkebeke, "From the Amazon to the Congo Valley," 153.

families.”⁸⁸ As long as the slave trade was thriving and the problem of reproducing the labor force was resolved through external means, enslaved women were expected to perform hard labor alongside men. When the Atlantic slave trade was abolished, this dynamic shifted, and enslaved women were forced to bear and raise children - future workers. Mies notes that colonial administrators enacted “incentives in the slave codes of the late eighteenth and nineteenth centuries to encourage local breeding of slaves by slave women on the plantations.”⁸⁹

How did these dynamics play out in the twentieth century in the rubber fields? De Mello and Melkebeke seem to indicate that the 19th century practices of slaving raids in the Congo, in which “people were captured and enslaved, usually men, but also women, who were incorporated into the raiders’ harems,” were continued by rubber producers in the 20th century.⁹⁰ However, women also occasionally worked in the rubber fields. As a result of this system, the Congo witnessed a significant demographic decline, with “contemporary observers and present-day scholars [agreeing] on rural depopulation and decimation of the Congolese,” along with “numerous cases of depopulated villages”⁹¹ - homes standing empty save for ghosts. The situation in the Amazon was analogous: “In general terms, the demographic post-boom scenario is marked by the emptying of villages or whole regions, being caused either by workforce recruitment, the resulting unsustainable livelihood, death by diseases, ethnocide, demographic replacement and miscegenation or encroachment.”⁹²

⁸⁸Maria Mies, *Patriarchy and Accumulation on a World Scale: Women in the International Division of Labour* (Zed Books, 1986), 91.

⁸⁹Mies, *Patriarchy and Accumulation*, 91.

⁹⁰de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 145-146.

⁹¹de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 160.

⁹²de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 161.

The wild rubber industry was quite volatile, in large part because of the decentralized nature of the production system. Rubber in Brazil was not produced on plantations, but was gathered from wild rubber trees by contract workers. Although the exploitation of tappers by traders and barons was significant and the labor productivity of wild tapping was low, there were several factors that impeded the rationalization and industrialization of rubber production in Brazil. Scholars note two explanations for the relative failure of proletarianization among Brazilian rubber tappers. The first emphasizes the dispersed nature of the workforce alongside tapper resistance: the desire among tappers to maintain the relative autonomy of piece work and their aversion to wage-labor. Along with the traders, tappers actively resisted the rationalization of their production.⁹³ The second explanation was the notorious South American leaf blight which had limited spread in the wild, but thrived on plantations with rows of close planted trees.⁹⁴ This disease, native to South America, was not present in Asia, and so had no deleterious effect on the development of plantations in Colonial Malaya.

As wild harvested rubber declined on the world stage, hampered by the South American blight in the Amazon, King Leopold's personal finances in the Congo, and the low productivity of wild rubber collection, it was replaced by plantation rubber in southeast Asia. Rubber plants were first brought to Malaya in the 1870s, but were not cultivated on a large scale until 1897-8 when plantation owners were looking for a replacement for failing coffee crops.⁹⁵ With the Malay Reservations Enactment in 1913, foreigners were allowed to purchase land in certain

⁹³Barbara Weinstein, "Capital Penetration and Problems of Labor Control in the Amazon Rubber Trade," *Radical History Review* 27 (1983): 121-40.

⁹⁴Warren Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History* (Cambridge University Press, 1987).

⁹⁵J. H. Drabble, "The Plantation Rubber Industry in Malaya up to 1922," *Journal of the Malaysian Branch of the Royal Asiatic Society* 40, no. 1 (1967): 54.

areas and large rubber plantations developed, requiring a large immigrant workforce. To supply this demand, the government of the Federated Malay States proposed a policy centered around three points: the development of a free workforce; the recruitment of immigrant workers from India, and regulation of employment conditions (e.g. hospitals, water, housing etc.).⁹⁶

In practice, this policy led to significantly better working conditions than those in the Amazon and the Congo, but worse conditions than those of auto workers in Michigan. Between 1840 and 1940 in one of the largest migratory systems on earth some 28 million people crossed the Bay of Bengal, many of them low-caste Tamils looking to work on the rubber plantations in Malaya.⁹⁷ Their legal status fluctuated between indentured, debt-bonded, and free, but these labels often had little influence over their day-to-day working conditions. J. Norman Parker, in *Colonial Labor Policy*, notes that in Malaya, “to be a free laborer often had little real meaning since the working conditions of a free laborer and an indentured laborer differed little or not at all.”⁹⁸ Most of the workers, indentured or free, worked for a wage that was differentiated on type of job and the sex and race of the worker. Women generally received a fraction of their male co-workers wages, and Chinese laborers received more than Indian or Javanese workers. For all workers, the workday was limited, although planters struggled to extend it.

In Malaya, a steady supply of workers was ensured by the enticement of good pay, and “in the possibility of his being able, during his two or three years average stay in this country, to remit small amounts of money to relations.”⁹⁹ An interesting feature of the Malayan rubber

⁹⁶Drabble, “Plantation Rubber Industry,” 59.

⁹⁷Sunil S. Amrith, *Crossing the Bay of Bengal: The Furies of Nature and the Fortunes of Migrants* (Harvard University Press, 2013), 2.

⁹⁸J. Norman Parmer, *Colonial Labor Policy and Administration: A History of Labor in the Rubber Plantation Industry in Malaya*, *Monographs of the Association for Asian Studies*, IX (J. J. Augustin Incorporated Publisher, 1960), 16.

⁹⁹Parmer, *Colonial Labor Policy*, 177.

industry is the recruiter/overseer called a *kangany*.¹⁰⁰ These individuals would travel back to India to recruit family members as rubber workers, and would later serve as their managers on the plantations, employing a combination of coercions based on debt and familial relations. As in the Congo and the Amazon, the labor force was not reproduced locally and generationally, but through tapping an external source of potential workers. In this case, the government of India worked with Malaya to facilitate the migration of workers from southern India to the Malayan rubber fields. Unlike in the wild cultivation fields across the world, workers in Malaya were paid in cash, and employers were directed to provide for workers' housing, sanitation and medical care in addition to wages.¹⁰¹ In the 1920s, the level of acceptable wage and quality of social provisions was a fixed point of contention between the government of India, who facilitated the supply of immigrant workers, and the government of Malaya and the planters, who wanted to reduce costs.

The question of social reproduction is complicated by the introduction of migrant and indentured labor. As with seasonal/transhumant labor in the Amazon, the social reproductive dynamics of migrant workers in colonial Malaya were stretched across vast distances. The existing communities of workers in India and Malaya were disrupted by migration, but do not seem to have been destroyed as in the Congo and among indigenous Amazonians. Rather, the social reproduction of the workforce became bifurcated, operating locally within the amenities of the rubber plantations and regionally through cash transfers across borders. This introduces significant complications to the question of labor reproduction, but provides an exciting

¹⁰⁰ Tirthankar Roy, "Sardars, Jobbers, Kanganies: The Labour Contractor and Indian Economic History," *Modern Asian Studies* 42, no. 5 (2008): 971–98.

¹⁰¹ Parmer, *Colonial Labor Policy*, 177-8.

opportunity to examine the fundamental relationship between labor and capital as it is attenuated by geographical and political space.

A steady supply of laborers is a central problem for industry everywhere. The social-reproductive conditions of laborers varied widely along the global automotive commodity chain. Even at the point of the chain where rubber entered the production process, the living conditions of workers in different regions were structured differently by global and local pressures. The conditions of the reproduction of labor responded to factors such as the availability of cheap labor from an outside source, the resistance of workers to changes in their social and communal patterns, and the type of labor required by capital.

Conclusion

The concept of Fordism is useful in that it highlights several fundamental dilemmas facing capital at the turn of the twentieth century. The expansion of the production of commodified domestic goods was combined with an expansion in the market for their consumption. However, although the networks of consumption and production were global, the techniques used to effect their expansion were not uniform across the globe. As we see in the early automotive industry, capital employed several distinct methods of exploitation that corresponded to different cultural, political, and ecological conditions. Recent theoretical work on the household allows us to understand the role that diverse structures of reproduction play in this global network of production.

The primary question posed by this chapter is that of the articulation between capital's need for specific kinds of labor, and the specific ways in which it reproduces the corresponding different kinds of laborer. How did global configurations of social reproduction respond to the

Fordist revolution in production? Following from the insights of Marxist Feminists, I posit that capital made use of different household patterns to structure the reproduction of labor, but this is not a simple dynamic to investigate. It is clear that neither conditions of production nor reproduction were exported globally wholesale by Fordist producers, and yet the entire world of production and reproduction was certainly transformed.

In the Amazon and the Congo, the reproduction of the debt relation and of the tax relation is inextricable from the reproduction of labor. Although commodity consumption was introduced through the closed provision shops, household production was destabilized, rather than supported. The razing of homes by slaving expeditions in the Congo, and the “kidnapping of indigenous infants ... to be raised within the families of enforcers or patrons to work the rubber fields”¹⁰² in the Amazon are the logical parallels of the planned communities in Michigan. In Michigan a labor supply was ensured and renewed through the new family patterns of the middle class; in the Global South this reproduction of labor was ensured and renewed through the destruction of indigenous families.

The supply of labor on the Malayan plantations was also secured from external sources, but through economic enticement rather than brute force. Plantation workers were paid a wage, but as their housing was provided for by the planters, and the state enforced labor laws regarding the length of the workday and workweek. The tenure of migrant workers was short, and for many workers a portion of his/her wage was used to support families back home in India.

These patterns of reproduction are connected across the world-system by commodity chains. A finished car needed inputs of iron, wood, coal, and rubber, and each of these had to be gathered and prepared by laborers far from auto factories. The labor processes, and the

¹⁰²de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 151.

conditions of its renewal, were conditioned by various local factors - the position in the global economy, the extant patterns of life and labor, and the family formation in which laborers lived. While factory workers in America could be said to be wage-earning proletarians, slave laborers and informal workers in the rubber fields could not. And while production for the world market dominated the lives of workers and their families, it was not the only type of work they engaged in: they were supported by unique combinations of wage labor, unpaid reproductive work, and subsistence labor. If Fordism can be seen as a reconfiguration of the articulation between patterns of consumption and production, then the changes that we have traced across the Global South form a part of the history of Fordism and warrant further study within this framework.

Chapter 2: Social Reproduction in Fordist Michigan

As demonstrated in the previous chapter, the Fordist revolution was a watershed moment in the history of capitalist production. It set, in the 1910s, a standard pattern of production that would be taken up across industries in the core regions of the capitalist world-economy. However, very few of the technical innovations attributed to Henry Ford actually originated in the Ford Motor Company. Sociologist Simon Clarke notes that “the decomposition of tasks, the specialization of tools, the assembly of tools into the machine, and even of machines into the machine system”¹ were already common in US industry prior to Ford’s project. Even the automated assembly line, Ford’s most notorious invention, was actually developed in 1911 by EMF Company, a rival automobile firm.² For Clarke, Ford’s real innovation was that he “applied these principles to a new branch of production ... with such single-minded ruthlessness that he transformed the conditions of production of motor vehicles almost overnight.”³

The ruthlessness with which Ford reconfigured the labor process was mirrored in his approach to controlling labor inputs. After the establishment of the moving assembly line and the consequent reconfiguration of the labor process, the primary remaining barrier to efficient production at Ford was employee turnover- 400% in 1913. In addition to training new workers

¹Simon Clarke, “What in the F—’s Name Is Fordism,” in *Fordism and Flexibility: Divisions and Change*, ed. Nigel Gilbert et al. (Palgrave Macmillan UK, 1992), 16.

²Wayne Lewchuck, “Fordism and the Moving Assembly Line: The British and American Experience, 1895-1930,” in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 20.

³Clarke, “What in the F—’s Name Is Fordism,” 17.

hired to support growing production, Ford also was constantly training replacements for workers who left. With training costs between \$35 and \$100 for each new worker, Ford was hemorrhaging money that could not be recouped by suppressing wages.⁴ Ford's solution? Pay better. But not just better, a lot better. In January 1914, to the outrage of other regional employers, Ford doubled the salary of auto workers from \$2.50 to \$5.00 a day. The name of this program, the "Five Dollar Day," belies the scope of Ford's concern with labor input: workers were only eligible for higher wages if they met certain moral standards in their daily lives, to be confirmed by a team of sociologists hired by Ford to investigate his employees. Ford's management practices, as well as those of other automobile firms such as GM, intentionally blurred the line between work and private life: private life and the home came to be seen as an extension of the shop floor, crucial zones for the management of the relations of production and reproduction of capital.

There has recently been a wealth of scholarship on the social reproduction of the conditions of capital accumulation - both on a high theoretical level and in specific historical situations. The work of Marxist feminists as well as world-systems theorists have gone a long way in demonstrating the centrality of processes outside the wage relation to capital accumulation. The household is crucial as a site of both the reproduction and socialization of laborers, and as the most important site of commodity consumption. In this paper I will examine the specific conditions of social reproduction in Fordist Michigan. The configuration of working-class life underwent a profound change at the beginning of the twentieth century, and industrial Michigan was, if not necessarily the epicenter of this change, at least one of its clearest

⁴Thomas Klug, "Employers' Strategies in the Detroit Labor Market, 1900-1929," in *On The Line: Essays in the History of Auto Work*, ed. Nelson Lichtenstein and Stephen Meyer (University of Illinois Press, 1989), 54.

examples. As the tasks of production were divided up between decomposed, rote tasks and the planning and supervision of engineers and managers, the working classes were similarly divided. High wages for some workers were established alongside a growing matrix of informal labor and unpaid domestic work. I am interested in the articulation of the relations of reproduction in this specific historical situation, and especially the stratified and segregated ways in which specific types of labor were produced and reproduced in industrial Michigan. In other words, I ask not merely how labor power was reproduced, but how the matrix of stratified concrete labors was reproduced.

In choosing to focus on industrial Michigan, I am influenced by both the Marxist geography of Michael Storper and Richard Walker, and by the insights of world-systems scholars, especially Giovanni Arrighi. In his analysis of the rise of the US as global capitalist hegemony, Arrighi credits two aspects of American capitalism embodied by Ford: the “internalization within a single organizational domain of activities and transactions previously carried out by separate businesses”⁵ – that is, vertical integration – and a new relationship with space. Crucial to the rise of US hegemony was a massive continent full of rich reserves of labor and raw materials. Accessing these materials required, alongside a genocidal campaign of dispossession of indigenous communities, a material transformation of the relationship between capital and space: “as soon as the Revolution had freed the settlers’ hands, they set out to conquer as much of the North American continent as was profitable and to reorganize its space in a thoroughly capitalistic manner.”⁶

⁵Giovanni Arrighi, *The Long Twentieth Century: Money, Power, and the Origins of Our Times*, 2nd ed. (Verso, 2010), 247.

⁶Arrighi, *The Long Twentieth Century*, 62.

Industrial Michigan was central to both of these developments in US capital accumulation. It sat at the entrance to the American west, and after the opening of the Erie Canal Detroit was transformed into a crucial shipping port. Prior to the rise of the automobile industry, Michigan was at the center of US production in ship engines, rail-cars, and lumber – all industries crucial for the development of land west of the Mississippi river. The rail industry in the late 19th century, and then the auto industry in the early 20th were established on the principle of vertical integration as a way to minimize transaction costs and maximize production. Most notorious is Ford’s River Rouge plant which turned raw materials into cars within a single massive complex. For Storper and Walker, this was the “long wave of early Fordism” in which “the Midwest became the country’s industrial core led by the automobile ensemble centered on Detroit and the machine-tool complex focused on Chicago.”⁷

The aim of this chapter is to show that this historical shift in the relations of production was accompanied by very deliberate management of the conditions of the reproduction of the conditions of capital accumulation. This brings us to the central questions of this paper: what role does private life outside the firm play in capitalist production, and how has capital attempted to exercise control over it? To approach this question, the chapter will outline at the conditions of reproduction in industrial Michigan in the early decades of the Fordist revolution, paying special attention to the policies of both the Ford Motor Company in Detroit and its rival General Motors in nearby Flint. The Five Dollar Day, implemented by Henry Ford, and GM’s Modern Housing Corporation were both concerted efforts to reconfigure the relationship between public and private life, and to bring this relationship under the control of the capitalist firm.

⁷Richard Walker and Michael Storper, *The Capitalist Imperative: Territory, Technology, and Industrial Growth* (Blackwell, 1989), 207.

It is important to note that the automotive industry in the US Midwest was not isolated from other industries or regions. Before the dramatic ascent of Ford and General Motors, many regions of the US had thriving automotive industries. Many of these industries implemented social reproductive policies similar to those we find in Michigan - although usually on a smaller scale. Likewise, industrial firms were not the only agents in the reconfiguration of reproductive relations at the beginning of the twentieth century. A clear line can be drawn from the social-reproductive policies of Ford and GM and the welfare policies implemented by the federal government under Roosevelt. The reason that Fordist Michigan is centered is that the policies developed there prefigured later developments in the history of American capitalism.

This examination begins with the elaboration of a theoretical apparatus which will contextualize the facts and figures. Two distinct traditions serve this purpose: the work of world-systems analysts and of Marxist feminists' research into the domestic sphere, especially recent work in Social Reproduction Theory. Combining the insights of these two very different traditions allows us to identify the relationship between global dynamics of production and the local reproduction that takes place within working households. Bringing recent scholarship in SRT to bear on our empirical study of labor reproduction in Fordist Michigan will have the secondary effect of allowing us to identify both the strengths and limitations of current research on social reproduction and the family. The next section will look at the of social reproduction as it first appears in Marx, as well as its uptake by Marxist feminists

The Reproduction of Capitalist Society

The question of reproduction is central to the historical study of capitalism. In order to persist as a social formation, capitalism requires that certain conditions be met, and continually

renewed. Some of these are basic to every society - as humans, we require the natural materials that form the stuff of subsistence, and we require human labor to work them into a usable state. It is through this metabolism with nature that we can survive, but this relationship can be mediated by quite distinct social formations. This variety of social formations have been analyzed under the category 'modes of production', defined by Louis Althusser as "a way or manner (a mode) of producing. Of producing what? The material goods indispensable to the material existence of the men, women, and children living in a given social formation."⁸ Althusser breaks down this category further into two constituent elements, the forces of production and the relations of production. The forces of production comprise: the *objects of labor*, such as raw materials or sources of energy; the *instruments of production*, such as tools or machines; and the *agents of production*, or the laborers who set upon the objects of labor with their instruments.⁹ Access to each of these three factors in the forces of production must be continually renewed; we must cultivate raw materials or find new sources where they can be extracted, we must maintain our machines and tools, and we must ensure a steady flow of laborers to work them. The structure of these factors - their internal relationships and the conditions of their reproduction - is effected by the relations of production.

Under the capitalist mode of production, there is a strict separation between the workers (agents of production) and the instruments and objects they employ in the production process. In other words, the laborers do not own the objects with which they labor - the tools and raw materials are owned by another party who stands outside the forces of production: the capitalist.

This means, firstly, that production is controlled not by those who will make use of the

⁸Louis Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, trans. G. M. Goshgarian (Verso, 2014), 22.

⁹Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 25.

commodities produced, but by those who are concerned only with profit. Capitalist production is commodity production, production for the market. It subordinates the creation of useful things to the creation of profitable things; it is a "mode of production whose overriding objective is to produce, not objects of social utility, but surplus-value and capital itself."¹⁰ Secondly, under the capitalist mode of production, workers cannot directly labor to meet their own needs, but must labor for a capitalist in the production of commodities. In exchange for this labor, the capitalist who then pays them enough to survive, but less than the value of what they produce. This relation, the exchange relation, is the center point or determining element of what Alfred Sohn Rethel calls the 'social synthesis': "the network of relations by which society forms a coherent whole."¹¹

What this means is that the reproduction of capitalist society is effected through exchange. On the one hand, laborers are dependent on the act of exchanging for their own survival. They cannot grow food because they have no land, and they cannot eat the commodities they produce because these commodities belong to the capitalist. They must resort to exchanging their labor for wages, and their wages for the objects of the labor of others. On the other hand, the other elements of capitalist society are also reproduced through exchange: the capitalist must purchase the raw materials and tools used in the production process, and the profit that is his *raison d'être* requires that the commodities he produces be purchased by some end consumer.

This requires some clarification, as these conditions are not universal within the capitalist world-system. On the one hand, Marx's 'so-called primitive accumulation,' the violent plunder of the Global South and the dispossession of producers from their land, seems to be not merely the

¹⁰Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 33.

¹¹Alfred Sohn-Rethel, *Intellectual and Manual Labour: A Critique of Epistemology*, trans. Martin Sohn-Rethel, Historical Materialism Book Series (1978; Brill Academic Publishers, 2020), 4.

starting point of capital, but an ongoing structural feature.¹² The imposition of the exchange relation as social synthesis is based on a regime of continuous violence.¹³ Furthermore, even within the core countries, forms of non-waged production persist. The enslaved laborers in American prisons are not wage-workers, neither are small farmers and agricultural producers, nor petty commodity producers and gig workers that work inside their homes. In what sense do these non-waged social configurations of production belong to the capitalist mode of production? This question has bugged scholars of capitalism who have come up with endless contradictory explanations. Althusser refers to them as "archaic forms" that have been dominated by capital "by virtue of the capitalist market's nearly total domination of the 'small independent producers'".¹⁴ In other words, although the empirical characteristics of non-waged labor seem to be holdovers from a "pre-capitalist" era, they have nonetheless been reoriented towards production for the market.

Jairus Banaji takes a similar approach, emphasizing the difference between modes of *production*, and modes of *exploitation*. In a discussion of what he calls vulgar Marxism, focusing on Maurice Dobb, Banaji critiques the view that "modes of production were deducible, by a relation of 'virtual identity', from the given forms of exploitation of labour."¹⁵ His critique of Dobb and the vulgar Marxists who posit a relation of identity between wage-labor and capitalism itself borrows from Lenin's analysis of the development of capitalism in Russia:

¹²William Clare Roberts, "What Was Primitive Accumulation? Reconstructing the Origin of a Critical Concept," *European Journal of Political Theory* 19, no. 4 (2020): 532–52.

¹³Werner Bonefeld, "The Permanence of Primitive Accumulation: Commodity Fetishism and Social Constitution," *The Commoner* 2 (2001): 1–15.

¹⁴Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 29.

¹⁵Jairus Banaji, *Theory as History: Essays on Modes of Production and Exploitation* (Brill Academic Publishers, 2010), 53.

Lenin argued, in other words, that as simple-commodity producers are subordinated to the power of capital, in town or village, and specifically bourgeois relations of production develop, far from transforming bondage and serf-forms of exploitation into specifically capitalist forms of exploitation, i.e., those forms which correspond to the 'classical, adequate mode of production of capital', these new relations of production, founded on capital, intensify the existing backward forms of exploitation: these forms remain 'feudal' or 'semi-feudal' in character, while the relations of production acquire a bourgeois character.¹⁶

The argument that the development of capitalism intensifies the so-called 'pre-capitalist' forms of exploitation is itself not new to Lenin, but is also found in Marx. For Marx wage-labor was not an isolated phenomenon, divorced from any other social configuration. Rather, "the veiled slavery of the wage-labourers in Europe needed the unqualified slavery of the New World as its pedestal."¹⁷ Crucially, the relationship between wage-labor in the core and 'unqualified slavery' elsewhere is not unidirectional, with unfree labor providing cheap inputs for production based on the wage. The development of capitalist relations transforms other types of exploitation: "as soon as peoples whose production still moves within the lower forms of slave-labour, the corvee, etc. are drawn into a world market dominated by the capitalist mode of production, whereby the sale of their products for export develops into their principal interest, the civilized horrors of over-work are grafted onto the barbaric horrors of slavery, serfdom etc."¹⁸ What emerges are new and more extreme configurations of slavery, serfdom, etc that did not exist before.

My issue with the above analyses is the characterization of non-waged labor formations as "archaic," "pre-capitalist," "backwards," or "lower forms." While it is true that the history of capitalism has led to an unprecedented expansion of wage labor, it also has created novel modes of exploitation based on extra-economic forms of coercion. To consign these to some sort of

¹⁶Banaji, *Theory as History*, 56.

¹⁷Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Fowkes (Pelican Books, 1976), 925.

¹⁸Marx, *Capital*, vol. 1, 345.

capitalist pre-history is to deny their specificity to the capitalist world, as well as to overstate the freedom of wage workers. Philip Corrigan writes that "[t]hese forms of coercion should not be conceptualized as residuals from former modes of production and essentially irrational; but as indications of capitalism's expansion."¹⁹ In this context the thinkers we have cited are useful. The question is not why non waged forms of coercion have persisted, or if they will be swept aside by the progress of history, but how they are transformed and subordinated to the logic of capital: the nearly total domination of the capitalist market.

How is this discussion of modes of production relevant to the reproduction of capitalism? An analysis of social reproduction must look at the conditions for the creation and perpetuation of wage-labor, but it cannot be limited to this. It must include an analysis of the forms of unwaged labor that are internal to the capitalist world system, how these support and are supported by the system of waged labor. Furthermore, it must look at the conditions of accumulation and the domination of the market, and how the conditions of this domination are effected and sustained. Keeping these complications in mind, this section will look at three aspects of social reproduction: the daily and generational reproduction of the abstract category of labor-power divorced from the means of subsistence; the reproduction of specific types of concrete labor; and the broader reproduction of market domination.

The Reproduction of Labor Power

The primary site of the reproduction of labor power is the household/family unit. The family, in its modern form, is not a transhistorical human organization, but a constituent

¹⁹Philip Corrigan, "Feudal Relics or Capitalist Monuments? Notes on the Sociology of Unfree Labour," *Sociology* 11, no. 3 (1977): 441.

institution of capitalist production. Immanuel Wallerstein, in his critique of traditional Marxist understandings of proletarianization, implicitly questions the highly idealized view of the family that would later be endorsed by some social reproduction theorists. Noting that capitalism requires and expands the condition of the alienated wage-labor through the historical process of proletarianization, Wallerstein insists that throughout the globe the object of this process has not been the individual wage worker, but the household or family unit. Throughout history, “individuals have tended to live their lives within the framework of relatively stable structures which share a common fund of current income and accumulated capital.”²⁰ These structures are generally (but not always) kin-based, and their boundaries are always shifting, in part due to the demands of the market: “It is in the context of such a household structure that a social distinction between productive and unproductive work began to be imposed on the working classes.”²¹ In this formulation, the modern family is not a “bulwark against capitalism” but is an artifact of capital, formed by the pressures of accumulation.

For Wallerstein, the capitalist family/household primarily serves as an income aggregator. As an income pooling unit, the household structures various forms of contribution to the means of subsistence for its members: income refers to waged income, but also cash transfers, petty commodity production, rent, and subsistence activity. These five categories are combined in various proportions throughout the world, but generally wage-income is counterposed to the other four: Wallerstein considers households that depend primarily on wage-work to be proletarian households and considers those which depend primarily on other forms of income to be “semiproletarian households” (by far the larger group). Semiproletarian households are

²⁰Immanuel Wallerstein, *Historical Capitalism* (Verso, 1983), 23.

²¹Wallerstein, *Historical Capitalism*, 24.

attractive to capitalists because “whenever wage-labor constitutes a substantial component of household income, there is necessarily a floor for how much the wage-earner can be paid.”²²

The semiproletarian household organization is promoted by capital in most of the world, as the matrix of non-waged income acts as a subsidy to the reproduction of labor: the developed reproductive sphere depresses wages. However, there are two forces resisting semiproletarianization. “One is the pressure of the wagedworkers themselves who seek to be ‘proletarianized,’ because that in effect means being better paid.”²³ Working families throughout the history of capitalism have preferred family structures that require less unremunerated reproductive labor and have struggled for this in unions and social movements. The second pressure comes from employers themselves, who have a “collective [long-term] need to have a large enough effective demand in the world-economy to sustain the market for their products.”²⁴ In Wallerstein’s conception, the family is a site of many conflicting tensions, but also as a structure that has been organized to aid capitalist accumulation.

Marxist feminists have long emphasized the ways in which the category of waged labor is reproduced and maintained by unpaid domestic labor within the household, often performed by women. In fact, it is difficult to analyze the category of labor power in modern society without also discussing the unique exploitation of women in its reproduction. Lise Vogel ascribes the phenomenon of the oppression of women in capitalist society to the role they play in the generational reproduction of laborers. In her 1983 book *Marxism and the Oppression of Women*, Vogel argues “for the power of Marxism to analyse the issues that face women today in their

²²Wallerstein, *Historical Capitalism*, 35.

²³Wallerstein, *Historical Capitalism*, 35.

²⁴Wallerstein, *Historical Capitalism*, 35.

struggle for liberation” but she “rejects, however, the assumption made by many socialists that the classical-Marxist tradition bequeaths a more or less complete analysis of the problem of women’s oppression.”²⁵ Rather, she contends that the writings of Marx and Engels, as well as certain high-points in the twentieth century Marxist tradition, though they “never adequately addressed the question of women,” contain the kernel of a theory that can be developed into a “theoretical framework in which to situate the problems of women’s oppression and women’s liberation.”²⁶

Vogel identifies two dominant perspectives on the problem of women’s oppression, one which divorces women's oppression from the context of capitalism (the dual-systems perspective), and one which identifies women's oppression with their structural position in capitalist society (the social reproduction perspective). These approaches are basically incompatible. In the first approach, “women’s oppression derives from their situation within an autonomous system of sex-divisions of labor and male supremacy.”²⁷ In other words, the oppression that women in modern society face as women differs from the oppression that they face on the basis of class. There are obviously overlaps and intersections between the two, but logically, “the dual-systems perspective implies that women’s oppression follows a course that is essentially independent from that of class oppression.”²⁸ This has been the perspective of much radical feminism in the west, which has been at best agnostic towards anti-capitalism, and often hostile to it.

²⁵Lise Vogel, *Marxism and the Oppression of Women: Toward a Unitary Theory*, Revised Edition, Historical Materialism Book Series 45 (BRILL, 2013), 2.

²⁶Vogel, *Marxism and the Oppression of Women*, 2.

²⁷Vogel, *Marxism and the Oppression of Women*, 134.

²⁸Vogel, *Marxism and the Oppression of Women*, 134.

Vogel is more sympathetic to the social reproduction approach, although she considers it undertheorized. This approach grounds the oppression of women “in their differential position with respect to generational replacement-processes.”²⁹ This situates the analysis of the position of women in society within the “central dynamic of social development in societies characterized by exploitation.”³⁰ As capitalism is a total social system, transforming every aspect of modern life, a historically grounded critique of the oppression of women must take as a central concern the drive for endless accumulation. As the exploitation of labor-power drives accumulation under capitalism, the social-reproduction approach to the so-called woman-question examines the unique role of women in the production and maintenance of labor-power.

However, Vogel points to a much more fruitful analysis of social reproduction. Her book deals extensively with the history of Marxist approaches to the liberation of women, but in the last chapters, she interrogates the question itself: “the first requirement for further motion is to abandon the idea that the so-called women-question represents an adequate category of analysis.”³¹ She elaborates the useful Marxist framework not as a theory of the position of women in society, but of the “reproduction of labour-power in the context of overall social reproduction.”³² Within this dynamic, women occupy contradictory positions for capital – on the one hand, the potential capacity to produce future workers is necessary for continued accumulation in the long term, but on the other hand bearing children reduces a person’s capacity to perform productive labor for capital: “childbearing potentially entails a costly decline in the

²⁹Vogel, *Marxism and the Oppression of Women*, 135.

³⁰Vogel, *Marxism and the Oppression of Women*, 135.

³¹Vogel, *Marxism and the Oppression of Women*, 142.

³²Vogel, *Marxism and the Oppression of Women*, 142.

mother's capacity to work, while at the same time requiring that she be maintained during the period of diminished contribution.”³³

Although her conception of ‘woman’ as simply those who have the capacity to bear children is perhaps dubious,³⁴ Vogel is making a broader point about gender and capital. For Vogel, women are oppressed in capitalist society due to their specific role in generational reproduction, i.e. the specific way in which they are dominated in service of the needs of capital. This consideration brings to the fore the articulation between productive labor and reproductive labor beyond just the question of childbearing. Reproduction entails not only the creation of new and future workers, but also the day-to-day maintenance of present workers and their dependents.

Vogel identifies this day-to-day reproductive labor as domestic labor. Although domestic labor is most often associated with feminized labor in private households, it “also takes place in labour-camps, barracks, orphanages, hospitals, prisons, and other such institutions.”³⁵ In other words, the oppression of women through the imposition of unwaged domestic labor is a specific manifestation of the larger phenomenon of reproductive labor within capitalism. Key to Vogel’s insight is that different forms of reproductive labor are utilized to reproduce different forms of labor. It is this larger phenomenon of social reproduction that has been elaborated in a recent trend in Marxist Feminism called Social Reproduction Theory.

³³Vogel, *Marxism and the Oppression of Women*, 151.

³⁴ For a theoretical critique of Vogel’s transhistorical association of the category of woman with the process of childbearing, see chapter 7, “Capitalism and Difference,” in Søren Mau, *Mute Compulsion: A Marxist Theory of the Economic Power of Capital* (Verso, 2023). For a historical explication of the contingent nature of this association, see Maria Mies, *Patriarchy and Accumulation on a World Scale: Women in the International Division of Labour* (Zed Books, 1986), especially chapter 3, “Colonization and Housewifization.”

³⁵Vogel, *Marxism and the Oppression of Women*, 159.

Building on the work of Lise Vogel, Social Reproduction Theorists have staked out a formidable project: to reclaim fully half of the capitalist system as an object of study for Marxists. This project is elaborated in the recent celebrated volume *Social Reproduction Theory: Remapping Class, Recentering Oppression*, which defines Social Reproduction Theory (SRT) as "an approach that is not content to accept what seems like a visible, finished entity – in this case, our worker at the gates of her workplace – but interrogates the complex network of social processes and human relations that produces the conditions of existence for that entity."³⁶ For the scholars working within SRT, this complex network of the relations of reproduction is crucial for the functioning of capital, and they are fundamentally relations of human labor.

This formulation is in direct contradiction to the self-conception of capital, which “acknowledges productive labor for the market as the sole form of legitimate ‘work,’ while the tremendous amount of familial as well as communitarian work that goes on to sustain and reproduce the worker, or more specifically her labor power, is naturalized into nonexistence.”³⁷ Indeed, Marx notes that value in capitalist societies is based not merely on labor-time, but on ‘socially necessary labor time’ an abstraction that elevates the importance of commodity producing labor and devalues all other types. The determination of what instances of labor are considered socially necessary is of course a contingent social process that responds to historical conditions. We will discuss this later in the paper, but for now it is enough to emphasize SRT’s insight that to the extent that reproductive labor happens outside the realm of commodification, it is “analytically hidden by classical economists and politically denied by policy makers.”³⁸

³⁶Tithi Bhattacharya, ed., *Social Reproduction Theory: Remapping Class, Recentering Oppression* (Pluto Press, 2017), 2.

³⁷Bhattacharya, *Social Reproduction Theory*, 2.

³⁸Bhattacharya, *Social Reproduction Theory*, 2.

Social Reproduction Theory, then, is a theory of this invisible reproductive labor, and an attempt to understand it as constitutive of capitalism. As its state of ‘invisibility’ would indicate, this is far from a simple task. Bhattacharya notes the distinction between public/private to delineate reproduction from production. The life of the worker is divided into two sections: the working day during which the worker is engaged in the production of value, and the space of reproduction, in which the laborers reproduce themselves, or at least their labor-power. The distinction is not so cut and dry, however. These spheres, though “they may be separate in a strictly spatial sense are actually united in both the theoretical and operational sense. They are particular historical forms of appearance in which capitalism as a process posits itself.”³⁹ In fact, production and reproduction are quite messily intertwined.

Quoting an earlier article by Johanna Brenner and Barbara Laslett, Bhattacharya defines social reproduction as inclusive of “how food, clothing, and shelter are made available for immediate consumption, how the maintenance and socialization of children is accomplished, how care of the elderly and infirm is provided, and how sexuality is socially constructed.”⁴⁰ By this definition, reproductive labor includes the unpaid activity of mothers as well as waged nannies and daycare workers. It includes ‘unproductive’ teachers in public education (‘unproductive’ in the technical sense that the object of their labor is not a commodity) as well as ‘productive’ teachers in private schools (whose teaching *is* sold as a commodity). The network of reproductive relations is complex, pulsing just under the surface – unified by little other than its relation to labor-power and its devaluation by capital. It cuts across traditional Marxist understandings of labor: it includes both waged and unwaged work, it is sometimes commodified

³⁹Bhattacharya, *Social Reproduction Theory*, 9.

⁴⁰Bhattacharya, *Social Reproduction Theory*, 6-7.

and sometimes not, it is performed by both men and women on both formal and informal bases, and it permeates all institutions of society from the home to the state itself.

The work of SRT scholars is at its strongest when it takes an expanded view not just of the types of labor required for capitalist accumulation, but of reproduction itself. This expanded view takes seriously Vogel's consideration of the reproduction of labor-power that takes place in "labour-camps, barracks, orphanages, hospitals, prisons, and other such institutions."⁴¹

Reproduction is far more than mere biological reproduction, though that is a crucial aspect. Labor-power is reproduced at several different levels of social organization – in addition to the family, Bhattacharya notes that "[s]lavery and immigration are two of the most common ways capital has replaced labor."⁴² Societies can and have shifted the requirement to reproduce labor power from local families to the marginalized zones of the world, increasing the influx of workers to a region through invitation or theft. Additionally, the day-to-day reproduction of labor power in prisons and work-camps is performed not by family members, but by laborers themselves on an unwaged basis or by employees dedicated to reproduction, such as cooks or cleaners.

Capital has not, indeed cannot, commodify all aspects of reproduction. It may commodify aspects of care and subsume portions of the sphere of reproduction into production, but it requires as a foundation a matrix of unpaid and devalued life. Just what types of labor are commodified is historically contingent. Nancy Fraser's essay in *Social Reproduction Theory*, "Crisis of Care" articulates this complex relation: "Historically, this work of 'social reproduction' has been cast as women's work, although men have always done some of it too. Comprising both

⁴¹Vogel, *Marxism and the Oppression of Women*, 159.

⁴²Bhattacharya, *Social Reproduction Theory*, 7-8.

affective and material labor and often performed without pay, it is indispensable to society.”⁴³

Although the existence of this unpaid realm is crucial to the existence of capitalist accumulation, its boundaries are constantly shifting. In fact, its consolidation and denigration as “women’s work” has been a relatively recent (in historical terms) configuration: “Splitting off reproductive labor from the larger universe of human activities, in which women’s work previously held a recognized place, [industrial capitalist societies] relegated it to a newly institutionalized ‘domestic sphere,’ where its social importance was obscured.”⁴⁴

Fraser positions this domestic sphere not as a universal category, but a contingent one. It is identical neither to unpaid work, nor to reproductive work, but is the lever by which the relationship between unpaid work and reproduction is manipulated. Fraser treats the categories of SRT – domestic labor, reproduction, the family, etc. – not as fixed units of analysis, but as categories to be critiqued. How, specifically, have the configurations of production and reproduction been molded by the demands of value accumulation, and how have those configurations been constitutive of historical modes of capitalism?

Fraser tracks the shifting configuration of social reproduction through three eras of capitalism: the liberal capitalism of the nineteenth century, the state-managed capitalism of the early and mid 20th century, and the present moment of neoliberal financialization. In each of these periods, the relationship between production and reproduction was manipulated in order to relieve the tension of a systemic crisis. Liberal capitalism preferred a free labor market where the reproduction of labor power in the capitalist core was offloaded onto families, as production and

⁴³Nancy Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” in *Social Reproduction Theory: Remapping Class, Recentering Oppression*, ed. Tithi Bhattacharya (Pluto Press, 2017), 21.

⁴⁴Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” 24.

reproduction were forced into “separate spheres.” Drawing on Maria Mies notion of ‘housewifization,’ Fraser notes that this configuration of domestic labor in the core was dependent on the massive exploitation and devaluation of the labor of racialized women in the periphery.

After the ravages of the two great depressions in the 1870s-1890s and the 1930s, capitalist states reorganized the reproduction of labor-power on a model of state welfare. In this model, states assumed the burden of providing for reproduction, introducing in the US such measures as the minimum wage, social security, expanded public education and other programs of social protection and reproduction. State subsidization was limited, however. It was systematically withheld from workers in the periphery who were again exploited in order to subsidize social reproduction in the core. For the current moment of neoliberalization Fraser notes that there is “a new, dualized organization of social reproduction, commodified for those who can pay for it and privatized for those who cannot, as some in the second category provide care work in return for (low) wages for those in the first.”⁴⁵ Racialized immigrant workers are often employed in these low-waged domestic jobs – once again supporting the reproduction of other classes, but in this case often in a commodified manner.

At its best, SRT is a powerful critique of the foundations of the capitalist system: the too-often unpaid and devalued work of reproducing labor power. The consideration of reproductive labor as constitutive of capitalist accumulation is an important corrective to masculinist and wage-based analyses of capital. Although it is true that waged labor is crucial to the accumulation of value, the unpaid work of women and racialized men (and especially the work of abject laborers in the periphery) form the conditions of possibility of waged work. The

⁴⁵Fraser, “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism,” 32.

continual devaluation of certain types of labor, especially reproductive labor, is the historical foundation of capitalism.

Too often, however, instead of investigating the complex relationship between production and reproduction, social reproduction theorists have committed to an overly rigid distinction between the two spheres. Although noting that they are part of the same complex of social relations, Vogel insists that “[r]eproduction of labor-power is not, however, itself a form of production.”⁴⁶ Similarly, in Bhattacharya’s formulation: “SRT points out that labor power itself is the sole commodity ... that is produced outside of the circuit of commodity production.”⁴⁷ This is only partially true. Bhattacharya is correct that the reproduction of labor power requires activity outside of commodity production, but capital has always been interested in the commodification of reproductive care work. Activities of production and reproduction are not distinct but overlapping, and separating them so sharply reduces our ability to theorize, the privatization of reproductive work. Capital, though required to offload some portion of the costs of reproduction onto unwaged workers or distant communities, has always retained a keen interest in the commodification of all life. When it has been able to afford producing the requirements of reproduction as commodities, it has done so.

This overly rigid circumscription of the reproductive sphere leads to some significant oversights in Social Reproduction Theory. Its biggest shortcoming is the occasional credulous endorsement of the family as an intrinsically radical space due to its presumed logical separation from production. But the family is not merely an institution that has been harmed by capitalism, but it is a fundamentally capitalist institution. When Mohandesi and Teitelman claim that

⁴⁶Vogel, *Marxism and the Oppression of Women*, 144.

⁴⁷Bhattacharya, *Social Reproduction Theory*, 7.

“households have historically served as bulwarks against capitalism”⁴⁸ they accept at face value capitalism’s paradoxical glorification of the family as a sacred space, untouched by crass commercialism. In fact, the family was formed by crass commercialism, and it exists in (or rather, was disciplined into) its modern form to perpetuate the conditions of reproduction.

The Reproduction of Concrete Labor

In addition to its role in the generational and daily reproduction, the household has a secondary role as well. For Wallerstein, the household/family construct plays a vital role in capitalist society as the site of ideological reproduction. He argues that:

Households serve as the primary socializing agencies of the world-system. They seek to teach us, and particularly the young, knowledge of and respect for the social rules by which we are supposed to abide. They are of course seconded by state agencies such as schools and armies as well as by religious institutions and the media. But none of these comes close to the households in actual impact.⁴⁹

This socialization plays an important role in conditioning future (and present) workers to accept their role as either paid laborers or subsistence laborers, and “results in the internalization of the myths, the rhetoric, and the theorizing of the system.”⁵⁰ Wallerstein notes that occasionally this socialization can have an anti-systemic orientation, as households often “socialize members into rebellion, withdrawal, and defiance,” however, “antisystemic socialization can be useful to the system by offering an outlet for restless spirits, provided that the overall system is in relative

⁴⁸Salar Mohandesi and Emma Teitelman, “Without Reserves,” in *Social Reproduction Theory: Remapping Class, Recentring Oppression*, ed. Tithi Bhattacharya (Pluto Press, 2017), 40.

⁴⁹Immanuel Wallerstein, *World-Systems Analysis: An Introduction* (Duke University Press, 2004), 37.

⁵⁰Wallerstein, *World-Systems Analysis: An Introduction*, 37.

equilibrium.” Further, except in times of structural crisis, “negative socializations may have at most a limited impact on the functioning of the system.”⁵¹

Workers in capitalist society are not socialized uniformly. The division of labor in society is reproduced by differential socialization. Even within the working class, there are divisions and demarcations between e.g. managers, supervisors, engineers and unskilled laborers. The boundary is not strictly impermeable - the occasional worker who rises the ranks functions as "a museum piece exhibited to encourage belief in the 'possibility' of the impossible and the idea that there are no social classes...."⁵² This is itself part of the ideological reproduction of the system, for "the 'technical' division of labour is quite simply a mask hiding the fact that some people are permanently 'penned' in the situation of the working class while others can have either high-level posts that are immediately attributed to them, or fairly or (very) broadly open ended 'careers'".⁵³ Socialization is crucial for maintaining the social division upon which the technical division of labor is based.

The household may be the main site of ideological reproduction, but it is not the only site. In an explicit critique of Social Reproduction Theory, Kirstin Munro looks beyond the household/family structure and emphasizes the potentially repressive nature of reproductive labor in general. Munro’s critique of SRT is based in the claim that “Social Reproduction Theory overlooks the contradictory and antagonistic role of the state in the lives of people, as the reproduction of labor power in capitalism proceeds via antagonism and state repression.”⁵⁴ Not

⁵¹Wallerstein, *World-Systems Analysis: An Introduction*, 37.

⁵²Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 37.

⁵³Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 37.

⁵⁴Kirstin Munro, “Unproductive Workers and State Repression,” *Review of Radical Political Economics* 53, no. 4 (2021): 623.

only is the household itself a site of ideological reproduction, but the entire matrix of reproductive labor, whether performed for a wage or not, reproduces specific types of labor. Munro notes that the social role of reproductive and care workers such as nurses, teachers, and social workers “is the reproduction of not just any ‘life’ but that of a docile, exploitable worker.”⁵⁵ When this labor is done by workers employed by the state, it represents not the state’s interest in the well being of its subjects, but its interest in reproducing the specific types of labor historically necessary for the accumulation of capital.

Munro here focuses on non-household sources of reproductive labor: hospitals, schools, and social work organizations, among others. These institutions are generally a part of the state, although private versions of them exist in growing numbers. Though considered ‘unproductive’ by capital for the reason that it does not produce commodities, the labor performed by workers in hospitals, schools and social work agencies serves a valuable role for the capitalist system: it disciplines the labor of others. Though the aid this work provides is necessary for the survival of many working class families, Munro highlights what she perceives as the complicity of these laborers in state repression. In particular, she rejects the characterization of the labor done by these workers as virtuous “life-making,” preferring a more complicated formulation that situates the type of life being made within “a larger process that perpetuates the capitalist organization of society.”⁵⁶ In other words, although the labor performed within hospitals, schools, and other reproductive institutions is essential for the renewal of life, the context in which it happens reinforces a repressive social structure. Indeed, “if exploitation and domination are inherent to the organization of production and reproduction in capitalism, then the reproduction of capitalist

⁵⁵Munro, “Unproductive Workers and State Repression,” 624.

⁵⁶Munro, “Unproductive Workers and State Repression,” 625.

society proceeds by perpetuating these forms of organization.”⁵⁷ We see this play out explicitly in the continued perpetuation of structural racism in schools⁵⁸ and hospitals.⁵⁹

Munro’s critique of Social Reproduction Theory provides the impetus for the investigation central to this chapter. While welcoming the achievement of SRT in its attempt to “overturn Marxist orthodoxy and shed light on the importance of unproductive labor carried out by disproportionately women of all races and racialized men,”⁶⁰ she rejects an uncritical view of the reproductive sphere and care-work as virtuous or revolutionary that is common (though not universal) in the writing of Social Reproduction Theorists. In positioning the reproductive sphere as constitutive of capitalist accumulation and complicit in the repression of workers, she opens up the following question: How have the specific historical configurations of the reproduction of labor power – “the cultural norms that prescribe working class ‘needs,’ the specific proportions of inputs, the social assignment of tasks, and the arrangement of people in space”⁶¹ – been manipulated to fulfill the specific historical need of capital for certain types of workers?

As mentioned, the claim by certain SRT scholars that labor power is produced “outside the circuit of commodity production” is only partially true. Marx notes in volume 1 of *Capital* that “the ultimate or minimum limit of the value of labour-power is formed by the value of the commodities which have to be supplied every day to the bearer of labour-power, the man, so that he can renew his life-process.”⁶² In other words, the value of labor power is, as for all other

⁵⁷Munro, “Unproductive Workers and State Repression,” 625.

⁵⁸Rita Kohli et al., “The ‘New Racism’ of K–12 Schools: Centering Critical Research on Racism,” *Review of Research in Education* 41, no. 1 (2017): 182–202, <https://doi.org/10.3102/0091732X16686949>.

⁵⁹Ellesse-Roselee L. Akre et al., “Measuring Local-Area Racial Segregation for Medicare Hospital Admissions,” *JAMA Network Open* 7, no. 4 (2024): e247473–e247473, <https://doi.org/10.1001/jamanetworkopen.2024.7473>.

⁶⁰Munro, “Unproductive Workers and State Repression,” 623.

⁶¹Munro, “Unproductive Workers and State Repression,” 624.

⁶²Marx, *Capital*, vol. 1, 276.

commodities, defined precisely by the extent to which it can be produced within the circuit of commodity production. This overlap between production and the reproduction of labor power is crucial because it represents an axis on which the value of labor power can be manipulated. If the value of the commodities required for renewing life-processes can be cheapened by the offloading of reproductive labor to unpaid or underpaid workers, then the value of labor drops as well.

The process of devaluing labor by increasing the burden of uncommodified domestic work is not applied evenly to the working classes. Capital may value commodities based on an abstraction of undifferentiated labor power in various amounts, but it requires for the production process concrete acts of labor in various configurations and at various prices. Treating labor power as an undifferentiated mass may help us understand the logic of exploitation in a society based on generalized exchangeability - it is through the abstraction of 'labor-power' divorced from its immediate context that we can consider commensurate, for example, work done on a farm and that done in an office. However, in order to understand the historical configuration of production and reproduction in a given time and place, we must disaggregate this mass into its component parts. In other words, to build on Munro's critique, what is reproduced by domestic and unproductive laborers is not merely life, or labor power, but the specific kinds of concrete labor required by capital.

The Reproduction of the Market

Domestic and reproductive labor serves another crucial and much broader purpose for capital, in addition to its role in the reproduction of labor. The household is the primary

institution responsible for renewing the life-force of commodity producers, but it is also the final destination of most of the commodities produced. Althusser writes that the family "is a *unit of consumption* in our societies. It is not the only kind of unit of consumption there is, but, of those in existence, it is a kind that still plays an extremely important part and it is not about to disappear."⁶³ Although many industries are involved in the production of machines or tools that are not directly consumed in private households, these can be seen as intermediate links in the commodity chains which terminate in the home: e.g. tractors and looms are built and sold as commodities, but are subsequently employed in the production of grain and cloth for clothes. This grain and these clothes are then purchased and consumed in the household in what Munro calls the "household production process."⁶⁴ The household production process is not external to the circulation of capital, but is one of its main components.

Household production is the final bit of processing that commodities undergo to make them fit for consumption: cooking food, cleaning clothes, employing household appliances such as vacuum cleaners in domestic chores, etc. This process combines two necessary elements: commodities purchased with a wage and unpaid domestic labor. For Munro, the household production process is linked to both the state and firms, as it "is forced to rely on capitalist firms and the state to ensure the survival of its members, both day-to-day and intergenerationally; it in turn contributes to the perpetuation of capitalist society."⁶⁵ In other words, we cannot isolate the household from broader capitalist society and identify it exclusively with the reproduction of labor. It must be theorized as part of a larger process: "[t]here is no social reproduction without

⁶³Althusser, *On the Reproduction of Capitalism: Ideology and Ideological State Apparatuses*, 77.

⁶⁴Kirstin Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," *Science & Society* 83, no. 4 (2019): 451–68.

⁶⁵Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," 454–455.

'societal reproduction,' as all production and reproduction in capitalist society are indelibly shaped by accumulation."⁶⁶ This means that households are complicit in the reproduction of the very conditions of their unfreedom. They are separated from the direct means of subsistence and forced to rely on commodity consumption in order to survive; in so doing, they valorize capital and provide the conditions of profitability that make endless accumulation possible.

This framework has important implications for the study of class struggle. If capital can make use of both devalued labor that is heavily dependent on unpaid reproductive work, as well as highly-paid labor that is supported by more extensive commodity consumption, then a 'win' for the working class is not necessarily a 'loss' for capital. Postone writes that "manifestations of class struggle between the representatives of capital and the workers over working-time issues or the relationship of wages and profits, for example, are structurally intrinsic to capitalism, hence an important constitutive element of the dynamic of that system."⁶⁷ The implication is that such struggles over the distribution of surplus are internal to capitalism: if the goal of class struggle is merely higher wages for the workers, or the ownership of a larger portion of surplus, then the success of the working classes means a decrease in the amount of capital available to be re-invested, but at the same time an expanded base of consumption through which the capitalist relations of production are reproduced.

The remaining sections of this chapter will explore this tension in the context of industrial Michigan. As the Fordist revolution changed the way that capital is produced, it was accompanied by a revolution in the relations of reproduction. The relations of production were reproduced through transformed relationships between wage-labor, unpaid domestic labor, and

⁶⁶Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," 455.

⁶⁷Moishe Postone, *Time, Labor, and Social Domination: A Reinterpretation of Marx's Critical Theory* (Cambridge University Press, 1993), 36-37.

commodity consumption. We will examine the extension by capital of new types of control over the private lives of workers in an attempt to tease out how specific configurations of household reproduction supported the new productive regime.

Households in the Michigan Automotive Industry

Prior to the Great Depression, automotive manufacturing was widespread in the US, especially in the Midwest and northeast although there were also firms on the west coast. However, the production innovations that originated in Ford plants and that were picked up and improved upon by firms such as General Motors or Ransom Olds were less widespread and after the Great Depression 50% of motor vehicle manufacturers had closed.⁶⁸ The Big Three automakers headquartered in Michigan - Ford Motor Company, General Motors, and Chrysler - emerged as the dominant entities in the industry. In this section we will explore the conditions of social reproduction in Detroit and Flint that contributed to their success during the first decades of the Fordist revolution. I highlight what I see as two important institutions through which industrial employers took control of the realm of private life – Ford Motor Company’s Five Dollar Day, and GM’s housing subsidiary the Modern Housing Corporation. The first is well studied, but we will be looking to the extent that the Five Dollar Day was generalized among Detroit workers, and the types of household formations that it influenced. The second is less represented in the literature, and so we will rely on a combination of archival documents and secondary literature. We will look at the time period stretching roughly from 1910, just three years before Henry Ford’s assembly line, to 1930 and the collapse of the global economy.

⁶⁸Timothy Bresnahan and Daniel M. G. Raff, “Intra-Industry Heterogeneity and the Great Depression: The American Motor Vehicles Industry, 1929–1935,” *The Journal of Economic History* 51 (1991): 317–31.

At the end of the 19th century and the beginning of the 20th, American industry underwent significant changes. During and just after the first world war, as industry was booming in the Midwest, waves of workers moved both north and west, whether Black workers looking for relief from the racial regime of the American south, or white workers from the northeast looking for new opportunities in the west. In the 1910s and 1920s, an estimated 1.5-2 million African Americans, predominantly men, migrated from the American South to cities such as Detroit, Chicago, New York, and Los Angeles in what is known as the Great Migration.⁶⁹ Although the Midwest, and Michigan in particular, had been a destination for immigrants looking for work since the late 18th century, the population virtually exploded in the twentieth. From 1910 to 1930, the population of Michigan almost doubled from 2.8 million to 4.8 million residents. Michigan's cities received the lion's share of this boom as the Michigan economy shifted from extractive industries such as logging and copper mining to craft production and manufacturing. In these same two decades, Detroit more than tripled to over 1.5 million inhabitants, and Flint more than quadrupled to 150,000.⁷⁰

Much of this growth was driven by two demographic sources: European immigrants (listed as "foreign-born whites" in the censuses) with high birth rates, and Black workers looking for better working conditions than were available to them in the American south. While the mass migration of willing (and as-yet un-unionized) workers was welcomed by employers, it created several real problems. There were simply more people moving to and living in the cities than could be comfortably housed with existing housing stock, and these new workers were unfamiliar with industrial work and the lifestyles expected of them.

⁶⁹Jr. Trotter Joe William, "The Great Migration," *OAH Magazine of History* 17, no. 1 (2002): 31–33, <https://doi.org/10.1093/maghis/17.1.31>.

⁷⁰US Census Bureau, *Characteristics of the Population*, no. 2 (1940).

The change in the nature of work and in the demographic composition of workers led to the development of new social institutions, and to the rise of overcrowded slums in industrial cities. In the 1890s, Congress allocated funds for an investigation into the country's developing slums as a response to "civic leaders and voluntary associations [crying] out for action to improve the housing and surroundings of the 'new Americans' from Europe, and of the native poor."⁷¹ This investigation was one of the first steps toward the development of city planning, and was part of a concerted effort "aimed at abolishing sweatshops, strengthening the labor movement, providing more manual and trade instruction, and improving education."⁷²

The federal government would begin to play a larger part in providing for the welfare of its citizens during the Great Depression, with the bundle of social programs included in Roosevelt's new deal, but prior to that much of the work of dealing with the overcrowding and immiseration of urbanites was left to local and corporate actors. In Detroit in the 1920s, the "Detroit Bureau of Governmental Research, a private organization ... worked with a Mayor's Committee on Finances in estimating future increases in municipal revenues" in order to fund public services.⁷³ Even so, in 1927 the Detroit City Planning Commission and the Michigan Housing Association conducted "a series of surveys of Detroit and found 'what was to be expected'- that juvenile delinquency, deaths from tuberculosis and pneumonia, infant mortality, crime, and dependency were more prevalent in the blighted areas than they were in other parts of the city."⁷⁴ A major goal of the early automobile manufacturers was to insulate their workers from this blight, and so they developed private institutions of social control.

⁷¹Mel Scott, *American City Planning Since 1890* (University of California Press, 1969), 7.

⁷²Scott, *American City Planning Since 1890*, 11.

⁷³Scott, *American City Planning Since 1890*, 255.

⁷⁴Scott, *American City Planning Since 1890*, 256.

Immigrant workers coming from agricultural regions of the American South and Eastern Europe had to be actively conditioned to the industrial life of the Midwest. If Fordism was at its core the “transformation of production, according to strict criteria of technical rationality” then it requires for that transformation “new forms of social stratification, in strict accordance with the technical function, and corresponding new forms of morality and of personality, of socialization and education, to ‘elaborate a new type of man suited to the new type of work and productive process’.”⁷⁵ Gramsci refers to this social elaboration of a stratified society oriented towards the rationalization of work as ‘Americanism’ – imbricated with Fordism, but fundamentally distinct.

There are two aspects of Gramsci’s concept of Americanism that are crucial to our present project – its historical specificity and its function as the mechanism of social differentiation. Although he welcomed the application of Fordist principles to European production, Gramsci was skeptical of the possibility of establishing Americanism in Europe as it requires a fundamental preliminary condition. “This condition could be called ‘a rational demographic composition’ and consists in the fact that there do not exist numerous classes with no essential function in the world of production, in other words classes which are purely parasitic.”⁷⁶ European society, for Gramsci, is characterized precisely by the existence of ‘parasitic’ classes with no economic function beyond consumption. For Gramsci, it was precisely the lack of such a calcified social structure that allowed Fordism to succeed in America.

This brings us to the second point of interest in Gramsci’s schema – Americanism is not a system of reducing stratification by the elimination of ‘parasitic’ classes, but is the creation of a

⁷⁵Clarke, “What in the F—’s Name Is Fordism.” This phrase appears in the 1990 draft available at: <https://scispace.com/pdf/what-in-the-f-s-name-is-fordism-2bwbsf1e0m.pdf>

⁷⁶Antonio Gramsci, *Selections from the Prison Notebooks*, trans. Hoare Quintin and Geoffrey Nowell Smith (International Publishers Co, 1971), 281.

new configuration of stratified society in which every social class is defined by its relation to production. For industrial Michigan, the massive waves of immigration in the early 20th century, although posing the problems of overcrowding and resource strain, provided the perfect opportunity for an experiment in Americanism. Families from Europe and individual workers from the American South, in looking for life and work, were funneled into different types of both. In our glance at the social patterns emerging in industrial Michigan, we will see that the different types of life created and reproduced correspond to the different types of work required by local industry.

The Five Dollar Day – Patterns of Work

Enter the Five Dollar Day. As mentioned previously, the Five Dollar Day was a significant pay increase for skilled workers at Ford Motor Company designed to combat the region's high turnover rates. As most of the new laborers available to Ford came from agricultural backgrounds, training costs were so exorbitant that a 100% increase in wages was justified if it significantly reduced turnover. However, the Five Dollar Day was much more than just a raise. Half of the pay was categorized as a profit sharing scheme in which workers only qualified to participate if they met "specific standards of efficiency and home life."⁷⁷ As a part of this program, Ford employed a team of sociologists to ensure that employees spent their free time appropriately, and through his English school the Ford Motor Company "attempted to change an immigrant worker's life and culture to its preconceived ideal of an 'American standard of living,' which it felt was the basis for industrial efficiency."⁷⁸

⁷⁷Stephen Meyer III, *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908-1921* (State University of New York Press, 1981), 6.

⁷⁸Meyer III, *The Five Dollar Day*, 6.

Ultimately the Five Dollar Day was short-lived, interrupted by rising inflation and labor problems associated with the First World War. However, despite its early demise, it retains an outsized significance as an “early managerial strategy to match working-class culture to the requirements of the modern factory.”⁷⁹ But the ‘modern factory’ is not the whole of production, and even within the factory, labor and pay were not evenly distributed by this program. Although there was little pay variation based on race within the Ford factories, Black workers were segregated by job type. They may have received similar pay to whites, but in order to do so they had to work in “hot, dangerous foundry jobs where quit rates were generally high.”⁸⁰ In other words, Black workers were forced into the most precarious and dangerous jobs in order to preserve both the lives and careers of white workers. Even when Black workers received the same pay rate as their white peers, their position in the workplace was subordinate.

The Five Dollar Day was never evenly applied to an entire population of workers. Kathleen Stanley and Joan Smith, in their study of the household in Detroit at the turn of the century, claim that “the crucial element of Fordism was the presumption of the so-called family wage – a male earned wage sufficient for the support of a dependent wife and children. However, in fact, even in Detroit, this presumed bastion of Fordism, the family wage was never really a reality for the majority of wage earners.”⁸¹ Rather, the majority of working families relied on supplemental income from wives and children, as well as significant subsistence labor.

⁷⁹Meyer III, *The Five Dollar Day*, 7.

⁸⁰Christopher L. Foote et al., “Arbitraging a Discriminatory Labor Market: Black Workers at the Ford Motor Company, 1918–1947,” *Journal of Labor Economics* 21, no. 3 (2003): 493.

⁸¹Kathleen Stanley and Joan Smith, “The Detroit Story: The Crucible of Fordism,” in *Creating and Transforming Households: The Constraints of the World-Economy*, ed. Joan Smith and Immanuel Wallerstein, Studies in Modern Capitalism (Cambridge University Press, 1992), 34.

Moreover, after the economic collapse in 1929, working families depended increasingly on direct assistance from the state.

Stanley and Smith convincingly portray the position of households in Detroit as a textbook case of partial proletarianization. Although employers required wage laborers in increasing numbers for industrial production, they were loath to pay the full cost of that labor's reproduction. And so the burden of reproducing labor at the level of the household was meted out to various actors. If we follow Marx's analysis of the value of labor-power, then the Five Dollar Day, as it was granted only to portions of the working class, did not represent a general increase in the value of labor-power, but a new stratification of the value of labor-power. If the renewal of the life processes of some class of laborer can be supplied outside the commodity sphere by unpaid or devalued subsistence labor, then the value of labor-power for that class of laborer falls.

This stratification is reflected in the structure of Detroit households. Stanley and Smith look at three types of families: US-born white families, Polish families (as representative of white immigrant workers), and Black families. In 1920, wives and mothers in white immigrant families were the least likely to work outside the home (about 1 in 20), and their children most likely. US-born white women worked at rates that were slightly higher, but "still quite low," and their children were the most likely to stay in school. Stanley and Smith consider that the difference in family makeup between these two groups result from the higher wages of US-born white men, "72 percent of whom were either professional or skilled workers," and because "Polish families brought with them skills associated their recent past as peasants" such as the practice of women keeping livestock and gardens.⁸²

⁸²Stanley and Smith, "The Detroit Story," 38.

While the differences in household structure between immigrant families, and the families of American-born whites were far from negligible, they paled in comparison to their shared differences with Black families. Unlike US-born whites with existing family structures in Michigan and white European immigrants, who usually migrated as family units, Black workers more often migrated as individuals without family or community contacts; this mass arrival of single Black workers was “a strong encouragement to the phenomenon of boarding, which helped both the new migrants and old residents to hold down housing costs.”⁸³ While boarding, or the inclusion of non-family members in a household, was extant among all ethnic groups, it was almost required in Black households as most Black laborers were unskilled and received wages far below the subsistence level. Boarding was, for Black households, a strategy of survival. It was at once a result of the devaluation of Black labor in Detroit, and the conditions for the continuation of that devaluation.

Although boarding increased a household’s cash flow via an extra wage-earner, it also required an increase in the demand of domestic subsistence labor, usually performed by women. Indeed, “in larger households, the addition of at least one woman boarder became necessary at certain points simply to share in the additional housework.”⁸⁴ The increased domestic service load for Black women is illuminating when compared with white families within Detroit and Black families in other cities. Although married Black women worked outside the home at approximately half the rate within Detroit as Black women in other cities, they had outside work at five times the rate of immigrant whites: “In 1920, one in four married Black women worked

⁸³Stanley and Smith, “The Detroit Story,” 46.

⁸⁴Stanley and Smith, “The Detroit Story,” 47.

for wages compared to only one in twenty married immigrant women.”⁸⁵ Moreover, just as their burden of unpaid domestic labor was increased at home due to boarders, their waged labor outside the home was also domestic service. Black women were underrepresented in all occupations except service: whereas white women worked in not-insignificant numbers in manufacturing, trade, professional service and clerical work, 88% of Black women working outside the home in Detroit were employed as personal servants.⁸⁶ This is not to say that white women did not experience foreclosed employment opportunities relative to white men, but rather that the employment of Black women was overdetermined by both their race and their gender.

Home and communal life

The working conditions of Detroit workers were inseparable from their living conditions. One of the most significant objects of Ford’s sociological investigation, and prerequisite for increased pay, was workers’ communal and private life. Sociologists working for the company would drive into working class neighborhoods and find neighbors, friends, and family members to vouch for the employee’s character. Stephen Meyer notes that “the company held the worker responsible and accountable for the outside witnesses to his character. ‘In cases where it is impossible to find someone who knows the habits and home life of the employee,’ the home office instructed Ford branches, ‘we withhold the increase [in profits] otherwise to be granted him under the plan’.”⁸⁷ This would not necessarily be an issue for families with generations-long roots in the city, or for white European workers who migrated with their families, but for single

⁸⁵Stanley and Smith, “The Detroit Story,” 46.

⁸⁶US Census Bureau, *Occupations, by States*, no. 4 (1930), 805.

⁸⁷Meyer III, *The Five Dollar Day*, 130.

Black migrant workers, or workers whose wives and children were forced to work outside the home, this amounted to almost an automatic disqualification.

Workers were also judged on their living conditions, with properly maintained homes factoring heavily into the judged fitness of a worker to receive higher wages. Meyer recounts a story of a German Catholic worker who was investigated by Ford's sociologists, and rejected from the profit share:

He lived in a two-story frame house that was in "poor condition." The neighborhood was also unacceptable: "Very bad. This is an alley with one and two story houses in poor condition on one side only. Foreign." The investigator detailed his impressions of the Ford worker's standard of living: "This man lives in a dirty unsanitary hut and has a room full of boarders, who sleep 3 and 4 to a room. Some of the boarders go through the room where the man and wife sleep to reach their own room. Wife looks haggard from overwork. She and the children are as dirty as their surroundings."

Living under such conditions, the investigator did not approve the worker for a share in the profits. Instead, he "advised" the worker not to have boarders, to move to a better neighborhood, and generally to improve his living conditions.⁸⁸

Even having boarders was considered a sign of unfitness, an inability to manage one's home. The worker in this anecdote eventually received his part of the profit share, but only when he had evicted his boarders, moved to the suburbs, and (on his own meager salary) built a free-standing single family home. Labor that is supported by certain household and reproductive configurations was explicitly devalued, which then reinforced the need for those configurations. This anecdote highlights what I consider to be the most powerful factor in social stratification, and one of capital's greatest tools in disciplining the conditions of reproduction: housing, and its corollary, residential segregation.

Racial and residential segregation are extreme forms of social stratification. In industrial Michigan, residential segregation allowed for the housing shortage caused by increased

⁸⁸Meyer III, *The Five Dollar Day*, 133.

immigration to be addressed on a differential basis – more of the limited resources available to cities could be directed towards housing more ‘valuable’ populations. Of course, racial segregation was not invented in 20th century Michigan – it had been a crucial element of colonial policy for hundreds of years. Carl H. Nightingale notes that the words “white town” and “black town,” designating the sectors of a divided city, were first used by colonial city officials in Madras in South India.⁸⁹ This pattern was copied and reinforced by states and municipalities throughout the world for centuries, separating people of European ancestry from those they viewed as unequal. In the US, however, by the early 20th century, “black activists and their allies were able to convince the US Supreme Court to invalidate residential race segregation by government legislation.”⁹⁰ The state did, of course, still pass racial legislation such as race-based immigration restrictions and lending policies, but it was no longer able to formally enforce the residential color line with direct legislation.

In order to work around this federal restriction, segregationists “reformulated the crusade for urban color lines in the United States. They culled a set of ideologies and city-splitting tools from the two other global-scale sources of institutional power used by urban segregation movements elsewhere – urban reform politics and the real estate industry.”⁹¹ This allowed for the creation of overcrowded slums to house poor and non-white residents, separated from the well-served affluent neighborhoods of skilled white workers and managers. In industrial Michigan, the institutions of segregation were developed by private industry, providing the state with a blueprint to continue the maintenance of a stratified society through indirectly selective welfare.

⁸⁹Carl H. Nightingale, *Segregation: A Global History of Divided Cities* (University of Chicago Press, 2012), 47.

⁹⁰Nightingale, *Segregation*, 296.

⁹¹Nightingale, *Segregation*, 296.

From 1910 to 1930, as Detroit more than tripled in size, the Black population grew from 5,741 to over 120,000 – a 20x increase. Many of these workers had lost jobs with the mechanization of southern agriculture and were moving to the Midwest, where manufacturing was booming. However, what awaited them was a potent combination of second-rate jobs and derelict neighborhoods. Mark Jay and Philip Conklin note that Black workers were forced into the “worst paid, most degrading, and most dangerous work Blacks were largely confined to jobs as production assistants, janitors and foundry workers.” This was reflected in housing, as “African Americans were also forced to live in the segregated Black Bottom neighborhood, which contained the city’s worst housing stock, often owned by predatory absentee landlords.”⁹²

It is well known that Black workers were forced to live in a state of segregation and abjection, and that they were often forced into bad jobs or paid less than their white peers. However, the connection between these phenomena is more complicated than it initially appears to be. Analysis of Detroit, both its history and its present, often centers a two-city hypothesis – one Detroit is a slum, and the other a great economic boomtown. However, Jay and Conklin argue that the “‘Two Detroits’ consensus in fact represents two components of a dialectical unity: redevelopment and austerity are not distinct processes but *two elements of the same process of uneven development*.”⁹³ They are writing of recent gentrification, but the divided city has always been a complete, if complicated, whole.

Looking at the relationship between housing and demographic trends in employment, we begin to see how segregation served city employers, and particularly how residential segregation assisted in the process of Americanization. In Detroit in 1930, Black men were significantly

⁹²Mark Jay and Philip Conklin, *A People’s History of Detroit* (Duke University Press, 2020), 84.

⁹³Jay and Conklin, *A People’s History of Detroit*, 14.

underrepresented in trades and in professional service (architects, doctors, lawyers etc.) relative to their population, and significantly overrepresented in domestic and personal service. In manufacturing as a whole, Black workers were employed consistent with their relative population, but were largely absent from highly paid or skilled positions such as foremen, managers, or even machinists. Instead they were overrepresented in low-paid laborer positions: over a third of Black male industrial workers were employed as unskilled laborers for automobile factories, as opposed to less than 10% of US-born whites. When we speak of the socialization of new workers, it is clear that they were being socialized into different strata of the working class, separated largely by their race. This separation was reinforced and reproduced by housing and household patterns.

The Modern Housing Corporation

In nearby Flint, Michigan, General Motors took a different approach to Americanism than Ford did with the Five Dollar Day, one that even more explicitly emphasized housing as crucial to the elaboration of social stratification. This approach centered around the Modern Housing Corporation, a subsidiary of GM that purchased lots of land, developed them with houses and amenities and then sold them directly to employees. According to a retrospective report, written by a “Mr. Cowdrick” in 1930, the Modern Housing Corporation was created to conduct “all the more important housing activities” of GM.⁹⁴ These activities included the acquiring or building of more than 4,000 homes and 54 non-residential buildings, the vast majority in Flint, Michigan, but a small number also in Pontiac, Lansing, and Janesville, Wisconsin. The project was initiated in 1919 to combat limited housing due to “the expansion of automobile manufacture in the years

⁹⁴Cowdrick, *Operations of Modern Housing Corporation* (General Motors Library, 1930), 1.

immediately following the World War,” and ran until it was dissolved in 1936. It seems to have always been viewed as a short term project to resolve a discrete problem, rather than a long term undertaking, as Mr. Cowdrick notes in 1930 that it “is expected that when all the properties have been liquidated, probably about 1945, all expenses will have been paid and the company will have recovered all the money it put into the project, but without interest.”⁹⁵ That the corporation was liquidated 9 years before Mr. Cowdrick’s prediction hints at who was expected to continue the project: the global economic crisis of the 1930s saw the state take over the management of the social status quo to an unprecedented degree.

Indeed, direct return on investment seems to have been less of a concern for the Modern Housing Corporation than exercising control over their employees' living conditions. The small profit made on each house in the project’s mature years mostly funded the salaries of project administrators. After taxes and interest, if the company is to be believed (which is a big “if,” to be sure), it “about ‘[broke] even’ on its expenditures.”⁹⁶ For the first several years of the project homes were sold at a loss, and for the length of the project, “at the time a house [was] sold, the Modern Housing Corporation ... secure[d] a mortgage from an outside lender,”⁹⁷ reducing the burden of capital outlay, but also significantly reducing the potential financing profits.

If the only problem was a housing shortage, then there were far more simple and far more profitable solutions than the Modern Housing Corporation. But when we understand the goal to be parallel to Ford’s desire to “elaborate a new type of man,” – or, taking liberty with Gramsci, to elaborate a new type of stratified society – then the efficacy of the Modern Housing Corporation

⁹⁵Cowdrick, *Operations of Modern Housing Corporation*, 4.

⁹⁶Cowdrick, *Operations of Modern Housing Corporation*, 4.

⁹⁷Cowdrick, *Operations of Modern Housing Corporation*, 3.

is obvious. General Motors saw in the rapid growth of the auto industry and the rapid growth of Flint an opportunity to completely restructure the conditions of reproduction for life and capital. Flint's population was about 13,000 in 1900 and increased more than tenfold in thirty years. This forced the city to consider where and how to house the new workers, but more importantly, it had to decide what kind of city it would become. Tied up with this was the question of who would make the decisions about Flint's future, and who would provide the capital to fund the necessary development? The obvious answer seemed to be that Flint's biggest employer would fill both of these roles, and General Motors took to the task with enthusiasm. This was not just a chance to build houses for workers, but a chance to fundamentally shape every aspect of their lives.

General Motors did not create its housing division solely for the purpose of alleviating population issues, nor was the Modern Housing Corporation a venture into conventional real estate investment. The project was explicitly one of Americanization, in which the reproductive apparatus was brought under the company's control. Mr. Cowdrick's report hints at the scope of the project. The homes sold to employees were not placed outside of cities in small company owned communities, but were inside city limits. This provided the communities with "the benefits of customary municipal services, including police and fire protection and sanitary regulations,"⁹⁸ at little to no cost to General Motors, but it also gave GM enormous leverage over the geography of the city itself.

Framed as General Motors' cooperation "with the public authorities for the benefit of the homeowners,"⁹⁹ the Modern Housing Corporation's community administration plan was a collaboration with the municipality to allocate city resources to serve GM employees. The effect

⁹⁸Cowdrick, *Operations of Modern Housing Corporation*, 5.

⁹⁹Cowdrick, *Operations of Modern Housing Corporation*, 5.

was to establish a standard pattern of reproduction embodied by a neighborhood of homeowners and then to transfer the responsibility for the reproductive labor of street and neighborhood maintenance to other entities. GM would make initial investments into public infrastructure that would then be taken over either by the city or by homeowners themselves. After an early contractor paid for the initial pavement of “about ten miles of streets ... a considerable amount of additional paving has been laid at the expense of the property owners.” Additionally, the corporation planted trees along the streets, and noted that “home owners generally seem to be interested in improving their property and keeping it in good condition.”¹⁰⁰

General Motors’ administration of the Modern Housing Corporation showed a strong commitment to ensuring that employees had access to benefits far beyond mere housing stability. The corporation donated or purchased land for public schools, effectively choosing who would have ready access to education: “At Flint, there is one grade school and one junior high school inside the Modern Housing development Another grade school is contemplated, and will be located inside the Modern Housing development.”¹⁰¹ To add to this, schoolteachers were one of the few non-employee groups who were allowed to purchase homes in the Modern Housing developments. By investing more in education for the employees who lived in the development, GM exercised control over just what type of workers their families children were being raised as. In addition to considerations for education, the corporation “laid out and improved a golf course and presented it to the city. It also erected a community building for donation to the city.”¹⁰² The

¹⁰⁰Cowdrick, *Operations of Modern Housing Corporation*, 5.

¹⁰¹Cowdrick, *Operations of Modern Housing Corporation*, 5.

¹⁰²Cowdrick, *Operations of Modern Housing Corporation*, 5.

golf course and community center were both public and although they were built by the corporation, they were maintained at the expense of the city.

From the first five pages of Mr. Cowdrick's report, Flint seems like the apogee of the American dream: well-paid workers living with their families in owned homes on tree-lined streets; readily accessible public parks and community centers; schools and communities where teachers are given privileged status. One can imagine why Flint's population quadrupled from 1910 to 1930. But in Flint as elsewhere this dream was built on uneven ground. The last paragraph of the report starts: "The regulations provide that houses may be sold only to members of the Caucasian race."¹⁰³ Just as in Detroit, the project of Americanism was not the creation of a more affluent class of workers, but the reconfiguration of a stratified working class.

Whites, especially well-employed whites, were encouraged to live in the well-served company communities, but GM did not create designated neighborhoods for Black and other non-white workers. They were segregated by default, by exclusion. This racial differential in attention (and funding) provided to their workers neighborhoods is a clear example of the reproduction not just of labor, but of differentiated laboring classes. Although the Modern Housing Corporation was explicitly developed to address the city of Flint's inability to house its growing population, it did nothing to help the fastest growing demographic: Black workers. While the white population almost quadrupled from 1910 to 1930, the Black population increased more than fourteen-fold. By default, these Black migrants looking for a place to live were consigned, along with poor white families, to "informal working-class settlements, most of which lacked urban amenities such as parks, sewers, and paved roads."¹⁰⁴ Described by the city

¹⁰³Cowdrick, *Operations of Modern Housing Corporation*, 6.

¹⁰⁴Andrew R. Highsmith, *Demolition Means Progress: Flint, Michigan, and the Fate of the American Metropolis* (Chicago University Press, 2015), 28.

building inspector as “shanty towns,” these unserved and informal suburbs were “more gritty and undeveloped than those in many other metropolitan areas.”¹⁰⁵ These neighborhoods were simultaneously the poorest, most racially diverse, and least served by city resources.

General Motors’ project of Americanism was not a uni-directional imposition of norms: rather the makeup of the town and the makeup of the company were co-constituted. The Modern Housing project was a project of control over the conditions of capital's reproduction, both the reproduction of labor, and the realization of value in commodity consumption. GM sold houses to their best men, and then praised the men for their responsible homeownership, stating that “theoretically, foremen and superintendent do not favor customers of the Modern Housing Corporation in selecting men for retention, when reductions in force are necessary, but it is probable that the fact of home ownership exercises some influence.”¹⁰⁶ Even when a homeowner lost their job, they were given support by GM to keep their homes until they could resume making mortgage payments. The implication is clear: the valued worker is a homeowner, and a homeowner makes for a good worker. But this positive assertion of the value of a worker and their home life was predicated on the devaluation of a large mass of unskilled labor – and the consequent degradation of Black and poor neighborhoods. As GM funneled city funds and services to their Modern Housing developments, ultimately it was marginalized families that paid the price.

Conclusion

What emerges from this soup of demographic data is a complex hierarchy of labor that does not fit neatly into categories of waged/unwaged and productive/unproductive. A given type

¹⁰⁵Highsmith, *Demolition Means Progress*, 28.

¹⁰⁶Cowdrick, *Operations of Modern Housing Corporation*, 6.

of work could fit into any combination of the above categories depending on who was performing it. Housework is unwaged and unproductive labor if done by a housewife to feed her family, or by a worker in their free time, but it becomes both waged and productive if performed on a waged-basis by a domestic servant hired on the open market. The physical act of the housework is identical in either case; the only change is the social context in which it is performed. The same could be said for cooking meals, or growing food, or teaching: it is not the act itself, but the social context of the act that determines its relationship to capital accumulation.¹⁰⁷

In all of these scenarios, whether productive or not, the labor is reproductive. A worker needs clean clothes, a full belly, and skill in order to “arrive at the doors of her place of work every day so that she can produce the wealth of society.”¹⁰⁸ Sometimes these necessities are provided by the market, sometimes they are extracted from unpaid laborers, or from the devalued work of racialized communities. That the requirements of the reproduction of labor cannot be so easily disaggregated from the results of production does not mean that the sphere of “social reproduction” is incoherent. Social reproduction is a crucial category of analysis precisely because it is not simply the negative space of production: it is both a requirement of capitalist production, and a lever by which capital can reconfigure production inputs.

Historically the burden of unremunerated reproductive labor has fallen disproportionately on a gendered and racialized other. However this is not an absolute imposition, and the extent to

¹⁰⁷ For the question of whether service can be considered a commodity, and therefore whether service work can be considered productive labor, see Michael Heinrich, *An Introduction to the Three Volumes of Karl Marx's Capital*, trans. Alexander Locascio (Monthly Review Press, 2012), 44. A service provided for pay is simply a commodity whose production and consumption are not separated either spatially or temporally: they occur at the same time.

¹⁰⁸ Bhattacharya, *Social Reproduction Theory*, 1.

which racialized and gendered proletarians have had to labor for a diminished wage or no wage at all is a historical question – one that we have tried to answer for early 20th century industrial Michigan. We will briefly recapitulate the results of this study below.

Fordist production, though often considered to be a process of deskilling labor, was in fact a complex reorganization of the relationship between labor and skill. A large portion of the laboring force was assigned decomposed tasks, most of whose required skill resides in complex machinery as dead labor. In a small number of positions such as engineering, managing, or the manufacture of engines, a large amount of skill was required. However most jobs were relatively low-skilled positions such as janitorial work, jobbing or assembly. This stratification did not exist only within individual firms – the increased stratification of the working class is observed across society. As the cities of industrial Michigan grew, the types of available jobs grew as well. A very small portion of workers were considered worthy of a family wage, but most workers survived by combining multiple incomes, or by supplementing substandard incomes with large amounts of subsistence work.

This should not be seen, however, as a simple case of cause-and-effect. Different family and household configurations did not develop in response to set wages. Rather, the larger employers in industrial Michigan systematically created a stratified society by exercising control over the private lives of workers. In the case of the Five Dollar Day, the Ford Motor Company required changes in the configuration of households before increased wages would be given. In order for a worker's labor to be deemed valuable, the worker's family had to disengage from patterns of informal subsistence labor. If they failed to do so, i.e. if the reproduction of their

ability to labor continued to be subsidized by informal and non-waged income, then their labor was bought at a lower rate.

In Flint, General Motors took a more direct approach to structuring the private lives of their employees. Through the Modern Housing Corporation, they manipulated not only the family structures associated with higher-valued employees, but the very geography of the city. They used strategic investments to develop patterns of city life. By influencing the location of parks and schools, the planting of trees, and the paving of roads, they structured the reproductive environment of labor, answering Bhattacharya's questions "Does the education [the worker] received at school also not 'produce' her, in that it makes her employable? What about the public transportation system that helped bring her to work, or the public parks and libraries that provide recreation...?"¹⁰⁹ with an emphatic yes.

But not only the public schools, transit, parks and libraries – private institutions contribute to the reproduction of labor as well. If the domestic labor of a wife or child depressed the value of their husband's or father's labor power, that same work performed by any of the 88% of working Detroit Black women employed in domestic service had the opposite effect: now as a commodity contributing to the reproduction of the worker it contributes also to the value of his labor power. In Michigan, the value of domestic and ancillary workers' labor power was systematically depressed by the necessity of subsistence labor in underserved neighborhoods. At the same time, Black and female workers in reproductive jobs propped up the value of what industry considered to be the most important labor – that of white, male, and skilled workers.

¹⁰⁹Bhattacharya, *Social Reproduction Theory*, 1-2.

The creation of a particular configuration of stratified society is a totally different thing than its maintenance. In this instance, capital and its representatives created certain conditions of social reproduction, but were quick to offload the responsibility at the first opportunity. General Motors transferred the responsibility for paving streets and planting trees to homeowners, and the running of schools and parks to the city of Flint. The management of wage-rates slipped out of the hands of employers with the development of a federal minimum wage as a part of FDR's New Deal, and the rise of a unionized auto industry with the Flint sit-down strike of 1937. It is important to note that both of these institutions, both early 20th century unions and New Deal programs, were more than happy to maintain segregation and a stratified workforce.

The point of this paper is not to directly link a certain type of reproductive scheme with Fordist production. As Gramsci noted, Americanism and its particular configuration of social life would be neither desirable nor possible outside of America. Rather I have attempted to demonstrate ways in which social-reproductive institutions are integrated into schemes of production. The specifics of this integration will change according to the needs of capital, but Social Reproduction Theory provides an excellent, if limited, starting point for their excavation.

Earlier I noted that a full elaboration of industrial Michigan's role in the United States' hegemonic ascent was outside the scope of this chapter. The gap here is larger than I let on, primarily because 'industrial Michigan' is not a neatly bounded object – it is an incoherent object when divorced from the larger structure of the world-system and its globe-spanning commodity chains. Ford's River Rouge plant may have turned raw materials into cars in a single location, but the raw materials did not appear spontaneously at the factory gates. They had to be extracted from the earth by laborers outside of Detroit and shipped to the factory. A continuation of the line

of inquiry expressed in this study will have to look along these commodity chains to truly account for the interlinked relations of production and reproduction supporting early Fordist production. This commodity chain is global, and includes the tens of thousands of workers in the rubber fields of South America, Africa, and Colonial Malaya. The study of productive and reproductive relations in the core is incomplete without an accounting of their co-constitution with similar relations in the global periphery. It is in this direction that the following chapters turn.

Chapter 3: Commodity Society and the Amazon Rubber Industry

In the early twentieth century, the automotive industry, centered in Michigan, USA, massively stimulated the demand for rubber production. Rubber was the main raw material that could not be produced within the United States, and so Henry Ford and his competitors were required to look outside the US to tropical zones more hospitable to rubber trees. Until 1913, the Amazon was by far the largest producer of rubber for the world market. During the Amazon Rubber Boom the massive pressure of producing rubber for the emerging automotive industry transformed life in the Amazon. Investments in infrastructure connected the relatively isolated region to global markets, while tens of thousands of people migrated to the region looking for work. At the same time, the (under)development of Amazonian industry followed a very different trajectory than the rise of the automobile industry in the US. Despite the wealth generated by the exploitation of rubber, there was little long-term, stable investment into production, and the Amazon economy collapsed in the 1920s with the rise of cheaper rubber from plantations in Asia. Additionally, the workers in the Amazon experienced high mortality rates and treacherous working conditions, with little of the stability afforded workers in Ford's automotive factories.

The forms of labor exploitation in the Amazon differed significantly from those which prevailed in industrialized countries. Wage-labor was relatively uncommon in the Amazon, and rubber traders instead coerced tappers to work through usurious debt contracts, and sometimes outright force. Meanwhile, wealth disparity was enormous - while the urban centers of the

Amazon boasted spectacular wealth and imported fashion and commodities from Europe, some tappers lived close to starvation,. However, despite the difference in living and working conditions between the rubber workers of the Amazon and those in automotive factories in the US and Europe, they were two poles of a single system. The extreme exploitation of the rubber workers, and correspondingly the cheapness with which rubber was procured, was a crucial condition of profitability for the automotive industry, and thereby subsidized the higher wages granted to Ford's factory workers. At the same time, it was the success of the automotive industry that drove the growth of the rubber industry and demanded commodity production from a region whose characteristics made wage-labor near impossible. Each region, and the actors within them, formed part of a totality - and each is impossible to analyze without the other.

This chapter will analyze social reproduction in the rubber industry of the Amazon as a constitutive element of the capitalist world-system. Following the recent body of scholarship known as Social Reproduction Theory,¹ we will be looking at the way in which individuals were produced and reproduced as rubber workers within a commodity producing society - what Marx calls the "restoration, renewal and refreshment of the vital forces" of the worker.² In other words we are concerned with the conditions that bring laborers to the factory door every morning - or in this case to the rubber tree. This consideration is important for capital in both the short- and long-term. Reproducing a labor force means ensuring not only that workers are able to rest and renew themselves as working subjects from day-to-day, but also ensuring that a new stream of workers is ready and able to replace existing workers who retire, leave, die, or are incapacitated through injury or disease. This reproduction of labor is most often effected generationally, through the

¹ Tithi Bhattacharya, ed., *Social Reproduction Theory: Remapping Class, Recentering Oppression* (Pluto Press, 2017).

² Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Fowkes (Pelican Books, 1976). 376.

rearing of children in workers' homes, but also through institutions of migration, labor conscription and enslavement.³

Labor is not the only element of capitalist society reproduced socially, and so following Kirstin Munro's critique of Social Reproduction Theory we will also be looking at the way in which the consumption practices (or 'household production') of workers in the rubber industry contributed to the valorization of capital and thereby to the renewal of the conditions under which capital accumulates.⁴ The way in which commodities are consumed by workers and private persons both structures and is structured by the interplay of global and local markets. In many ways the household is both the starting point and end point of the cycle of capital accumulation - it is the well from which labor is drawn, and the place in which the exchange-value of commodities is finally realized as their use-values are consumed and demand is renewed.

The lives of the inhabitants of the Amazon - both established communities and newly arrived migrants and refugees - were transformed by the rubber industry. Settlers looking to exploit the natural resources of the region destroyed Indigenous communities and their ways of life. This was accelerated by the push to extract rubber for the global market. Not only did the capitalists and traders press Indigenous communities into serving as labor for the global automotive commodity chain, but they also interrupted indigenous patterns of reproduction - especially practices of food cultivation - and replaced them by commodity consumption. In this way the indigenous communities were incorporated into the world economy as both a labor

³ Lise Vogel, *Marxism and the Oppression of Women: Toward a Unitary Theory*, Revised Edition, Historical Materialism Book Series 45 (BRILL, 2013). 157-163.

⁴ Kirstin Munro, "'Social Reproduction Theory,' Social Reproduction, and Household Production," *Science & Society* 83, no. 4 (2019): 451-68.

source and as a market for finished commodities from the Global North, such as canned food, tools, medicine, and cloth.

This dynamic resembles the process of proletarianization as discussed in Marxist literature. Indigenous people's ability to reproduce themselves directly from the land was interrupted as they were dispossessed of their commons, and instead a commodity market was established. Only through the exchange of their labor-power could they access the social product necessary for their survival. In this sense, they were 'freed' from their connection to the land. In many cases they entered into contracts with their employers and were also 'free' in the eyes of the law. But for the most part this freedom was a legal fiction, and they were held in debt-bondage and servitude not only by the impersonal compulsion of the market but by force. At any rate, they were not free waged workers, and their conditions have been described by historians of the region as 'pre-capitalist' formations, residual and anachronistic social structures held over from the colonial period.⁵

This chapter re-evaluates the characterization of the Amazonian rubber industry as pre-capitalist. I agree that the rubber tappers were unfree, and that their conditions differed greatly those of wage-laborers in the core. Indeed their laboring circumstances could not have resembled Fordist factories less. But I disagree strongly that their conditions were in any way out of place in the capitalist world. Their conditions were the same as those experienced by the vast majority of the Global South after being incorporated into the world economy. They were what world systems analysts call part lifetime proletarians: "laborers who depend on non-market sources for

⁵ Barbara Weinstein, "The Persistence of Precapitalist Relations of Production in a Tropical Export Economy: The Amazon Rubber Trade, 1850-1920," in *Proletarians and Protest: Studies in Class Formation*, ed. Michael P. Hanagan and Charles Stephenson (Greenwood Press, 1986).

part of their livelihood rather than solely on wages."⁶ They combined income from wages, illicit sales of rubber hidden from their bosses, restricted subsistence labor, and off-season work in the urban areas of the Amazon in order to survive. Their living and laboring conditions benefited global capital in two ways - to produce cheap raw materials for growing industry, and to serve as a market for finished commodities. In both their labor inputs and in the patterns of their private lives, the conditions of rubber workers in the Amazon were not just dictated by the capitalist world-system, but were constitutive of it.

The Amazon Rubber Boom

Rubber has been used by humans for centuries. When Europeans arrived in the Americas in the fifteenth century, they witnessed Indigenous communities use rubber to make balls, bottles, syringes, and to waterproof cloth and shoes.⁷ However, despite its elastic and water-repellent properties, it was of little use to Western industrialists as it became sticky in the heat and rigid in the cold - its industrial application was largely limited to raincoats. In the 1830s Charles Goodyear developed and then perfected the process of vulcanization which stabilized rubber and made it more suitable for industrial and consumer applications.⁸ Vulcanized rubber proved quite useful in various industries, and would receive a massive boost in 1888 with John Dunlop's pneumatic bicycle tire and the creation of a rubber tire industry.⁹ Within 5 years, "the only tires [in use] on self-propelled vehicles [were] either solid rubber or pneumatic rubber tires;

⁶ Ravi Arvind Palat, "Dependency Theory and World-Systems Analysis," in *A Companion to Global Historical Thought* (2014). 377.

⁷ Austin Coates, *The Commerce in Rubber: The First 250 Years* (Oxford University Press, 1988).

⁸ Herbert Morawetz, "History of Rubber Research," *Rubber Chemistry and Technology* 73, no. 3 (2000): 405–26.

⁹ Burgess Darrow, "Pneumatic Tires - Old and New," *SAE Transactions* 27 (1932): 438–44.

no other type seems as the present time likely to engage the serious attention of engineers."¹⁰ It caught the attention of capitalists as well as engineers, and the number of firms producing pneumatic tires in the US grew from 1 in 1896 to 157 in 1917 and reaching a peak of 278 in 1922.¹¹ Driven by this growth in tire manufacturing, US rubber imports rose from 20,308 long tonnes in 1900 to 157,371 in 1917 and 301,499 in 1922. Imports would continue to grow, reaching half a million in the 1930s.

Prior to the development of rubber tires, and for the first few years of the new industry's growth, the majority of the world's rubber was sourced from the Amazon. However, by the third decade of the twentieth century the Amazonian rubber industry had collapsed and was replaced on the market by Asian rubber, especially from Malaya and Sumatra. In fact the automotive tire industry, though it initially boosted demand for Amazonian rubber, eventually spelled the end of Amazonian dominance in the industry. The exponential growth in demand for natural rubber could not be met by the inefficient harvesting methods used to tap the wild rubber trees that dominated the Amazon. Meanwhile both the rubber tappers and the ecology of the Amazon resisted the attempts at rationalization that would be necessary to produce rubber on a large scale. The sluggishness of Amazon rubber production gave way to a more efficient plantation system in Asia that lacked such labor and ecological issues.¹²

Various dates are given for the Amazon Rubber Boom, but we can generally consider it to have lasted from ~1870 to 1913.¹³ The first date is hard to establish with precision as rubber

¹⁰ P. W. Litchfield, "PNEUMATIC TIRES," *Transactions (Society of Automobile Engineers)* 8 (1913): 208–23.

¹¹ Guido Buenstorf and Steven Klepper, "Submarket Dynamics and Innovation: The Case of the US Tire Industry," *Industrial and Corporate Change* 19, no. 5 (2010): 1563–87.

¹² The rise of plantation labor in Asia will be the subject of Chapter 5.

¹³ Bradford L. Barham and Oliver T. Coomes, *Prosperity's Promise: The Amazon Rubber Boom and Distorted Economic Development*, *Dellplain Latin American Studies* 34 (Westview Press, 1996); John Tully, *The Devils Milk* (NYU Press, 2011), JSTOR, <http://www.jstor.org/stable/j.ctt9qfjqp>; Barbara Weinstein, *The Amazon*

production statistics are scarce in the 19th century, but the end date is more decisive. Although the economy of the Amazon was dominated by rubber production into the 1920s, 1913 was the first year in which the quantity of wild rubber harvested in the Amazon was less than that grown on rubber plantations in Malaya.¹⁴ During this boom period, practically all of the rubber produced for industrial purposes was harvested from wild rubber-trees by roaming tappers. Because of the peculiarity of the method required to harvest wild rubber as well as the remoteness of the rubber trees from industrial infrastructure, both the means of labor control and the patterns of capital investment differed from those seen in the industrialized US.

Two of the elements that differentiate the Amazonian rubber economy vis-a-vis the industrialized economies of the North are a lack of a stable and waged labor force, and a lack of capital investment aimed at increasing the productivity of labor. Each of these contributed to the absence of stable long-term industry in the Amazon - its underdevelopment. In theorizing the unique characteristics of industry in the Amazon, Stephen Bunker coined the phrase 'mode of extraction' to indicate an economic logic different from the 'mode of production' which prevailed in the economies of the US and Europe. For Bunker, "[t]he crucial difference between production and extraction is that the dynamics of scale in extractive economies function inversely to the dynamics of scale in the productive economies to which world trade connects them."¹⁵ In other words, whereas in productive economies greater output is brought about by an increase in the productivity of labor through its combination with an investment in technology, in extractive economies a greater yield is effected by an increase in the amount of labor employed to exploit

Rubber Boom, 1850-1920 (Stanford University Press, 1983).

¹⁴ P. W. Barker and E. G. Holt, *Rubber Statistics 1900-1937: Production, Absorption, Stocks and Prices*, no. 181, Trade Promotion Series (U. S. Department of Commerce, 1938).

¹⁵ Stephen G. Bunker, *Underdeveloping the Amazon: Extraction, Unequal Exchange, and the Failure of the Modern State* (University of Illinois Press, 1985). 25

each unit of a dwindling natural resource. A greater extractive output corresponds to a fall in the productivity of labor. Because of this, "when extractive systems respond to increased external demand, they tend to impoverish themselves (1) by depleting non-self-renewing resources or (2) by exploiting self-renewing resources beyond their capacities for regeneration."¹⁶ The extreme exploitation of resources, and especially the high labor inputs, raise the cost of extraction so high as to stimulate cultivation or synthetic production of the natural resource elsewhere.

This model perhaps is a better fit for the Amazon's other industries than for rubber, as it is technically renewable. There are several types of plants that produce quality rubber in commercial quantities. The two most prominent in the Americas are *Castilla Elastica* which grows as far north as Mexico, and *Hevea Brasiliensis*, which grows only in the Amazon. Of these, *Castilla* was the first to be harvested. It provided the rubber used by Mesoamerican peoples and was the dominant source of rubber through the first decades of the Amazon Rubber Boom. In order to extract natural latex from *Castilla*, harvesters would usually cut down the entire tree and manually extract the sap from the bark as the trunk decays. By contrast, *Hevea Brasiliensis* has a capillary system that bears rubber near the bark. This system can be exposed through shallow cuts, and the tree can be repeatedly and productively tapped.¹⁷ Indeed, Bunker notes that *Hevea* "will yield indefinitely if properly scarred and tapped."¹⁸

Both sources of rubber are renewable, but it is only the first which fits into Bunker's schema for impoverishment by over-exploitation. *Castilla* was more prominent in the upper reaches of the Amazon, and sources indicate that by the start of the twentieth century it was

¹⁶ Bunker, *Underdeveloping the Amazon*. 25

¹⁷ O. F. Cook, "Rubber Production from Castilla and Hevea," *Science* 85, no. 2208 (1937): 406–7, <https://doi.org/10.1126/science.85.2208.406>.

¹⁸ Bunker, *Underdeveloping the Amazon*. 66

sparse from over-harvesting.¹⁹ Production from *Hevea*, which accounted for the vast majority of Amazonian rubber, was limited not by over-harvesting but by *under-harvesting* owing to a dearth of labor. In one of the earliest works analyzing the Amazonian rubber industry Charles Akers claims that the natural stock of rubber trees "for practical purposes may be considered inexhaustible; but the supply of latex for the manufacture of rubber for export is controlled by the lack of any large laboring population."²⁰ In the 90s, Coomes and Barham reaffirm this analysis, stating that "the scarce factors of production in the wild rubber industry were labor and capital, in contrast to rubber trees which were relatively abundant."²¹

That the development of a region so rich in natural resources could be stymied by a scarcity of labor and capital in an era of unprecedented advances in infrastructural advancement and labor mobility is baffling. Indeed it would be incorrect to assert that capital and labor did not flow into the Amazon. There were concerted efforts by Brazil to import labor, both from abroad and from other regions inside the country, into the Amazon, many of them quite successful.²² Furthermore, the urban centers of the Amazon saw massive expansion with Manaus boasting an "excellent system of waterworks, an efficient garbage collection and disposal system, electricity, telephone service, handsome public buildings, and comfortable private residences" - all indications of significant capital investment in the region.²³ But the influx of people and cash did

¹⁹ Cook, "Rubber Production from Castilla and Hevea."

²⁰ Charles Edmond Akers, *The Rubber Industry in Brazil and the Orient* (Methuen & Co. Ltd., 1914). 8

²¹ Oliver T. Coomes and Bradford L. Barham, "The Amazon Rubber Boom: Labor Control, Resistance, and Failed Plantation Development Revisited," *Hispanic American Historical Review* (Durham) 74, no. 2 (1994): 231–57.

²² For an example of failed foreign settlements in the Amazon, see Lawrence F. Hill, "Confederate Exiles to Brazil," *The Hispanic American Historical Review* 7, no. 2 (1927): 192–210. The more successful system of internal migration will be dealt with below.

²³ E. Bradford Burns, "Manaus, 1910: Portrait of a Boom Town," *Journal of Inter-American Studies* 7, no. 3 (1965). 401.

not alleviate the problems of labor and *productive* capital investment. This in part because of the ways in which both money and labor were employed in the Amazon. The labor situation will be dealt with in following sections, but it is instructive to treat here the growth of the rubber boom towns.

During the rubber period, Manaus and Belem grew enormously. Between 1872 and 1920, Manaus grew from 29,000 inhabitants to 75,000, whereas the population of Belem increased from 61,000 to over 230,000 in the same period.²⁴ This rapid expansion in population was accompanied by the aforementioned infrastructural projects, but also more by conspicuous displays of wealth, such as the notorious Amazonas Theater in Manaus. This theater, which was completed in 1896, was made almost entirely from imported materials:

For its construction not only architects, builders, painters, decorator and sculptors were brought from Europe, but also various materials such as Carrara marbles, Murano chandeliers, pieces of worked iron from England, French tiles and European wood for the floor. In the internal structure, the steel dome was manufactured in Belgium.... The dome roof is composed of 36 thousand pieces of enameled ceramic pottery and glazed tiles, coming from Alsace.²⁵

In addition, there was a robust market for commodities from Europe and the United States, as advertisements listed for sale such items as "Smith and Wesson revolvers, Collins machetes, Victor gramophones, Walk-over shoes, Parfum Lubin, Omega watches, Black and White whiskey, Booth gin, Scandinavian butter, Fussell condensed milk, Otto motors, Liqueurs Bardinet, Sun typewriters, Deutz automobiles, Underwood typewriters, White Label whiskey, etc." There were also French restaurants, sports clubs, and other trappings of European life, but it

²⁴ IBGE, *Censo Demográfico*, 1872; and IBGE, *Censo Demográfico*, 1920.

²⁵ Sampaio Maria S.. M., "Amazonas Theater Architectural Construction and Restorations History," paper presented at 12th International Conference on Structural Analysis of Historical Constructions, 2020, https://www.scipedia.com/public/M. Sampaio_2021a.2.

is specifically the importation of commodities from the core industrial countries that is significant for this study.

Manaus is held up as a prototypical boom town, where quick fortunes were made in the day by the exploitation of land and people and then lost at night in the bordellos and bars. The conspicuous consumption on display in Manaus is sometimes considered a weakness, which ultimately undermined the possibility for stable development - why was this money not spent on infrastructure, or on rationalizing production?²⁶ This local focus misses the crucial role that such consumption played on a global scale. Commodity imports are not unidirectional: while commodities flowed into Brazil, the surplus from the rubber industry flowed outward to the United States and Europe. Capitalism is not merely a system of commodity production but of commodity consumption, and the new markets for finished goods created in the boom towns of the periphery realize the value produced in the core can be realized.

In short, the Amazon region benefited the industrial Global North in two ways - both in the production of cheap raw materials, and in the consumption of finished goods. Both roles undermined the possibility of long-term stable production, as the labor turnover rate (as well as the death rate) in harvesting was high, and the surplus earned was more often spent on conspicuous displays of wealth than invested into increasing the productivity of labor. The following sections will examine the labor relations and capital investment dynamics in Amazonian Rubber during and following the boom, highlighting the Amazon's failure to transition from what Bunker calls an extractive economy to a productive one.

²⁶ See the section *Weaknesses in the Brazilian Industry* in Randolph R. Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945," *The Business History Review* 51, no. 3 (1977): 341-66.

The 'Free' Laborer

The procurement of labor for commodity rubber production in the Amazon differed in key respects from the process of proletarianization seen in industrial companies, especially in its failure to establish a body of free wage-laborers. But under capitalism, free labor takes on a second, ironic meaning. The proletarian is "free in the double sense that as a free individual he can dispose of his labour-power as his own commodity, and that, on the other hand, he has no other commodity for sale, i.e. he is rid of them, he is free of all the objects needed for the realization of his labour-power."²⁷ In other words, workers under capitalism are 'freed' from the ability to reproduce themselves by the dissolution of their traditional social ties and community relationships. This 'freedom' comes about as a result of a historical process of dispossession whereby workers are cut loose from the land (or from other social structures which sustain them) and forced into a market for labor. Though not formally wage-workers, the rubber tappers in the Amazon were free in this second, ironic sense.

In the nineteenth century, several major events undermined the social and economic relations that structured Brazilian society, and left hundreds of thousands of workers without their established means of survival. Chief among these were the abolition of slavery and the *Grand Seca* that devastated the northeast. These two events each set loose some portion of the population which had previously been bound in a customary social hierarchy. Unlike their fellow workers of the industrialized Global North, however, neither the climate refugees from *Grand Seca* nor the formerly enslaved Africans became wage workers. They were rather recruited to work under conditions of debt peonage as *seringueiros*, or rubber workers, in the Amazon where

²⁷ Marx, *Capital*, vol. 1. 272-273.

market penetration was shallow and the flow of those commodities necessary for survival was tightly controlled by a group of middlemen-traders, or *aviadores*.

In the rubber fields, they joined the existing population of peoples Indigenous to the Amazon whose traditional ways of life had been similarly disrupted. Through hundreds of years of colonization, during which colonists massacred them, displaced them, and pressed them into involuntary servitude, Indigenous communities had been cut off from the land. By being compelled to labor not for their own subsistence but for a series of colonial masters, from Jesuit missionaries to state agents and settlers, these communities were forced to become dependent on a flow of commodities from outside the Amazon as their metabolism with the land was broken. This section will trace how each of these three groups - Indigenous peoples, climate refugees from the *sertão*, and formerly enslaved Africans - were 'freed' from the social structures that maintained them and were recruited into the rubber trade.

The Indigenous Peoples of the Amazon

When rubber found a market on a global scale, the Indigenous peoples of the Amazon were among the first to be pressed into the production of rubber as a commodity. This was not a new dynamic, as colonial powers (and later the Latin American states) had exploited the inhabitants of the Amazon since the arrival of Europeans to South America at the turn of the 16th century. Indeed "exploited" is too gentle of a word, as the European powers and their Latin American successors committed genocide against the Indigenous people. Massacres, enslavement, forced displacement, and ecocide depopulated the Amazon by 80%, from around 5 million in 1500 to around a million four hundred years later, falling further to around 200,000 in

1980.²⁸ Rubber barons themselves participated in the genocide of Indigenous Amazonians, such as in the Putumayo region of the upper Amazon where state presence was more limited. But in much of the lower Amazon the demographic collapse of Indigenous peoples undermined the nascent rubber industry.

The combination of declining population and an ecology unfit for labor-intensive plantation development led early European settlers to organize "their economic penetration of the region during the seventeenth and eighteenth centuries around highly mobile groups of Indian gatherers who dispersed throughout an area to extract natural substances such as turtle oil, spices, hardwoods, vegetable oils, and cacao beans."²⁹ The collectors were sometimes paid a wage, but more often given tools, cloth, or trinkets in exchange for the goods they gathered. This differed dramatically from the economic foundation of the rest of Brazil which developed plantations on the basis of the labor of enslaved Africans. However, the system of voluntary exchange with the Indigenous population was not sufficient to meet the demands of the Europeans, and so the colonists began "a series of slaving expeditions in the major areas of Indian settlement in order to assure themselves of an adequate labor force."³⁰ Indigenous communities resisted these raids, but their resistance was often used as a pretext for more violent retaliation. The force of the colonists was overwhelming. The slaving raids accelerated the collapse of Indigenous communities and by the time that the rubber industry began to grow in the nineteenth century, the labor requirements were in considerable excess of the population.

²⁸ Chris C. Park, *Tropical Rainforests* (Routledge, 1992). 108.

²⁹ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 10

³⁰ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 11

During the 17th and 18th centuries, there was another group influential in the capture and exploitation of Indigenous labor in the Amazon: the Jesuits. Often portrayed as benevolent towards the population of the Amazon, the Jesuits were in fact instrumental in justifying a system of forced labor, relying on the colonial principles of both Just War and *resgate* (rescue) to determine who could be legally enslaved.³¹ The Portuguese crown invested in founding mission villages in the region, which would be responsible for both the spiritual well-being of the nearby communities and for establishing a system of control over Indigenous labor. The majority of the surrounding Indigenous groups would be pressed into service at the mission itself such as tending to foodstuffs and "conduct[ing] other Indians down from their jungle homes to designated mission villages."³² The Indigenous communities, as well as their relations of reproduction, were destroyed, and the people were forced to labor for the reproduction of the missions themselves - producing food and procuring more labor. Those people not pressed into service at the missions were made available as laborers first the government and finally to private settlers.

The Jesuits and settlers clashed over control of the Indigenous population of the Amazon, with the missions periodically disallowing colonists access to their labor, allegedly due to the colonists' mistreatment of the workers. In 1759 the Portuguese sided with the colonists and expelled the Jesuits from Brazil (and other areas of the Portuguese Empire), but "the collapse of the Amazonian mission system did little to resolve the settlers' economic troubles, since the

³¹ A. J. R. Russell-Wood, "Introduction: New Directions in Bandeirismo Studies In Colonial Brazil," *The Americas* 61, no. 3 (2005): 353–71.

³² Colin M. MacLachlin, "The Indian Labor Structure in the Portuguese Amazon, 1700-1800," in *Colonial Roots of Modern Brazil: Papers of the Newberry Library Conference*, ed. Dauril Alden (University of California Press, 1973). 201

fundamental problem was the prolonged and precipitous decline of the Indigenous population."³³ Access to labor was to be a major issue for extractivist ventures in the Amazon, and with a dwindling local population, it would have to come from somewhere else. The major sources would be enslaved or formerly-enslaved Africans, and climate refugees from the drought-stricken region of Northeast Brazil.

The Abolition of Slavery in Brazil

Shortly before the Jesuits were removed from Brazil, the Portuguese Crown shifted its sights from Indigenous to enslaved African labor in the Amazon. In 1755, "all forms of Indigenous slavery were abolished" and the Grão-Pará and Maranhão General Trading Company was established "to promote the importation of enslaved Africans to the Amazon."³⁴ By the end of the 18th century, the Company had imported almost 20,000 enslaved people into the Amazon region from other parts of Brazil. Although this number was too small to fill all of the labor needs of colonists in the region, it had a significant impact on the labor relations in the Amazon and the demographic composition. During the 18th and 19th century, the distinction between such racial categories as "Indian" and "Black" became permeable, as did the distinction between free and unfree labor. Today, a majority of the residents of the Amazon (at least 65% in each Amazonian state) identify as 'Black.'³⁵

³³ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 12

³⁴ Adalberto Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon," trans. Bryan Pitts, *International Review of Social History* 62 (2017). 26.

³⁵ David Pereira Junior, "AMAZON REGION AND BLACKNESS: NOTES TO UNDERSTAND BLACK INVISIBILITY," *Journal of the Research and Training Center*, 2025, 117–39.

During the Atlantic slave trade Brazil was the largest importer of enslaved Africans, serving as the final destination for over 5 million persons.³⁶ They were employed first in sugar cane plantations, and later in mining and coffee operations. Mortality in these fields was very high - on the sugar plantations, the death rate of the enslaved exceeded their birth rate.³⁷ In theory it was possible for enslaved people to purchase their freedom, and at the death of an enslaver there was a "generalized practice of manumissions, which originated an enormous population of ex-slaves."³⁸ However, the plight of those released from slavery was so unstable that "[n]ot only did the freedman have less security than the slave and generally become the dependent of his exmaster, but he was often used in the most dangerous work where the masters feared to trust their slaves."³⁹

For individuals of either African or Indigenous ancestry in the Amazon - whether enslaved or not - labor was subject to various types of coercion. As they could still be compelled to work for the state or private citizens, the freedom that had been granted to the Indigenous population in 1755 was, "not the freedom to come and go as one pleased, but was rather more akin to a limited concession that could be curtailed and conditioned at any time."⁴⁰ In 1798 a law was passed modifying the post-slavery system of labor for Indigenous people that "added a fundamental innovation to previous Indigenous policy by imposing compulsory service on the

³⁶ For a survey of the early critical historiography of Brazilian Slavery, see Richard Graham, "Brazilian Slavery Re-Examined: A Review Article," *Journal of Social History* 3, no. 4 (1970): 431–53. For a survey of more recent literature, see Rafael Marquese and Ricardo Salles, "Slavery in Nineteenth-Century Brazil: History and Historiography," in Dale Tomich, ed., *Slavery and Historical Capitalism during the Nineteenth Century* (Lexington Books, 2017).

³⁷ Kit Sims Taylor, "The Economics of Sugar and Slavery in Northeastern Brazil," *Agricultural History* 44, no. 3 (1970): 267–80, JSTOR.

³⁸ Marquese and Salles, "Slavery in Nineteenth-Century Brazil: History and Historiography." 145.

³⁹ Graham, "Brazilian Slavery Re-Examined: A Review Article." 452-453

⁴⁰ Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon." 27.

entire free, non-white population, including blacks, *mamelucos*, and *cafuzos*."⁴¹ The liberty of free Africans in the Amazon thus became likewise subject to the labor needs of both the state and colonists.

Naturally, both the Indigenous peoples and the Africans resisted the oppressive system of racialized labor imposed on them. Between 1835 and 1841 a civil war was waged across northern Brazil called the *Cabanagem*, the longest and largest rebellion in the history of Imperial Brazil.⁴² The elites in Brazil perceived this war as a general uprising of the *gente de cor*, or people of color. The *Cabanagem* left as many as 30,000 dead and resulted in "the widespread disorganization of the rural labor force" as "enslaved Africans and quasi-enslaved Indians joined the guerrilla bands which roamed the region."⁴³

In the aftermath of the *Cabanagem* the Brazilian state doubled down on attempts to pacify racialized labor in the Amazon valley. In the 1838 Pará created a labor corps, the *Corpo de Trabalhadores*, to press the population of color into service for a nominal wage. These corps engaged in the forced resettlement of the Indigenous population to towns where labor needs were higher, and along with the legal framework that established them "represented a virtually unlimited expansion of the right of any civilian, military, or state-sanctioned private authority to exploit the involuntary labor of Indians, blacks, and *mestiços*."⁴⁴ It is interesting to note that these

⁴¹ Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon." 29.

⁴² David Cleary, "'Lost Altogether to the Civilised World': Race and the Cabanagem in Northern Brazil, 1750 to 1850," *Comparative Studies in Society and History* 40, no. 1 (1998): 109–35.

⁴³ Weinstein, "The Persistence of Precapitalist Relations of Production in a Tropical Export Economy: The Amazon Rubber Trade, 1850-1920." 59.

⁴⁴ Paz, "Free and Unfree Labor in the Nineteenth-Century Brazilian Amazon." 31.

labor corps considered rubber collectors to be vagrants, as they "lived in complete independence," and would conscript them to work in more settled industries.⁴⁵

However, this particular system of involuntary servitude was short lived, and the workers corps were eventually abandoned in the state of Pará in the late 1850s. During this time, Brazil was entered its abolition period, but more importantly rubber began at this period to dominate the Amazon and its need for 'independent' workers trumped the concerns of the less lucrative Amazonian agrarian industries. However, the abolition of the workers corps did not mean the freedom of labor, especially for racialized workers and the formerly enslaved.

Although it persisted in modified forms of coerced labor, legal slavery was formally abolished in all of Brazil in 1888. As a legal institution it had been crumbling for decades. As mentioned earlier, the enslavement of Indigenous people was outlawed in 1755, and in 1871 Brazil passed the Rio Branco Law, or the Law of the Free Womb, which granted free status to all children born to enslaved women. In March 1884, four years before the nationwide decree, slavery was ended in the state of Ceará, whose economy was more dependent on the labor of peasants than of enslaved Africans.⁴⁶ Four months later, Amazonas, which had about 2% of the enslaved population of its neighboring Amazonian state Pará, followed suit.⁴⁷

The abolition of slavery in Brazil was slow and incomplete. Brazil was the last country in the Americas to grant freedom to the victims of the Atlantic slave trade, and they did so without

⁴⁵ Weinstein, "The Persistence of Precapitalist Relations of Production in a Tropical Export Economy: The Amazon Rubber Trade, 1850-1920." 59.

⁴⁶ Erick Matias, "March 25, 1884: Ceará Abolishes Slavery Before the Golden Law and Enters Brazilian History," *O Maringá*, March 25, 2026, <https://omaringa.com.br/en/coluna/datas-historicas/25-de-marco-de-1884-ceara-abole-a-escravidao-antes-da-lei-aurea-e-entra-para-a-historia-do-brasil/>.

⁴⁷ Gabriel Abreu and Bruno Pacheco, "137 Years Ago, AM Signed the Slave Liberation Charter: 'We Are Still the Protagonists of Our Own Freedom,'" trans. Bruno Sena, *Revista Cenarium*, July 12, 2021, <https://agenciacenarium.com.br/137-years-ago-am-signed-the-slave-liberation-charter-we-are-still-the-protagonists-of-our-own-freedom/?lang=en>.

compensation. Rather than providing the newly freed people with the means to participate fully in civic and economic life, the liberation of Africans in Brazil "resulted in either (1) the marginalization of former slaves from capitalist production or (2) the incorporation of former slaves into new, race-based regimes of coerced labor."⁴⁸

Common to all of the racialized labor regimes in Brazil - enslaved, coerced, or formally free but precarious - was the expectation that people of color in the Amazon were obligated to provide labor to meet the needs of capital and the state. Just as importantly, through the forced resettlement of Indigenous and enslaved people as well as the economic legacy of slavery, the ability of Brazil's racialized underclass to sustain itself by its own labor had been thoroughly undermined. For both Indigenous and African communities, "[t]he abolition of slavery in Brazil was, in fact, premised on the creation of social conditions that assured the former slave population's uninterrupted servitude, guaranteed by the heavy hand of extramarket structures and institutions."⁴⁹ During the rubber trade, both of these populations were incorporated into the growing class of rubber tappers and subject to new forms of both economic and extra-economic compulsion. The growth of the rubber industry was so great, however, that the Black and Indigenous residents of the Amazon could not meet the demand for labor. The majority of rubber workers would have to be pulled from elsewhere in Brazil - the peasant population of the northeast.

⁴⁸ David Baranov, *The Abolition of Slavery in Brazil: The "Liberation" of Africans Through the Emancipation of Capital*, Contributions in Latin American Studies 17 (Greenwood Press, 2000). 7.

⁴⁹ Baranov, *The Abolition of Slavery in Brazil*. 168.

The Grand Seca

The northeast of Brazil is dominated by the *sertão*, a semi-arid and sparsely populated region visited periodically by great droughts. The turn of the century saw a series of especially bad droughts, the worst of which was the *Grand Seca* of 1877. This drought and those which followed saw a fundamental reconfiguration of the *sertão*'s social fabric. Between 1877 and 1919, two million people in the northeast died from the droughts and their effects. In addition to this, tens of thousands of the inhabitants of the *sertão* left their homes to look for work and subsistence in other parts of Brazil, principally working as rubber harvesters in the Amazon. The already precarious conditions of life and labor in the northeast were so decisively undermined that "the infamous drought of 1877 ... definitively ended slavery in Ceará, an event which anticipated the emancipation of all slaves throughout Brazil in 1888."⁵⁰

Mike Davis argues that drought "always has a manmade dimension and is never simply a natural disaster."⁵¹ Indeed, although the droughts in the *sertão* are linked to the semi-regular occurrence of the El Niño in the Pacific, their human cost is a result not merely of climatic caprice but is inflamed by various human factors, from the development of infrastructural projects (or lack thereof) to the prevailing property relations in affected regions. In the nineteenth century, Brazil was a major producer of both coffee and sugar, but these were mainly limited to the south of the country. In Ceará, the state most affected by the *Grand Seca*, the population relied on cattle and cotton produced for export. Although better suited to the arid climate of Ceará than other industries, cattle raising cannot be said to have thrived there. Davis notes that

⁵⁰ Reuben H. Brooks, "HUMAN RESPONSE TO RECURRENT DROUGHT IN NORTHEASTERN BRAZIL" *The Professional Geographer* 23, no. 1 (1971): 40–44, <https://doi.org/10.1111/j.0033-0124.1971.00040.x>. 42

⁵¹ Mike Davis, *Late Victorian Holocausts: El Niño Famines and the Making of the Third World* (2001; Verso, 2017).
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cattle were "forced out of the *zona de mata* by the sugar boom," and settled in the *sertão* as a sort of last resort. Whereas in much of the world a hectare of ranch land can support 1-2 cattle, in northeast Brazil, a "typical ranch of 1,000 hectares, for example, might sustain only 50 scrawny cattle; and even the biggest *fazendas* (10,000 hectares or more) rarely pastured herds larger than 1,000."⁵²

Cotton began to eclipse cattle in the eighteenth century as droughts weakened cattle herds, especially during the American Civil War which disrupted the global cotton market. Large plantations combined cotton and cattle, but they crucially also relied on subsistence agriculture to maintain the working population. In the absence of a robust commodity market for the means of subsistence, the large *fazendas* had to be able to maintain not just the workers, but their families as well. Well into the 20th century, the population of the *sertão* "employs the same food technology as its Indian predecessors, subsisting primarily on manioc and maize and secondarily on sweet potatoes, beans, kidney beans, pumpkins, peanuts, pineapple, pepper, gourds, and tobacco."⁵³

The right to cultivate subsistence crops was a custom granted to the *moradores* working the *fazendas* as a part of a system of mutual obligation between the owners and the workers who lived on and worked their land. While the owners could dictate land use, demand labor for upkeep around the *fazenda*, fire unruly workers, and had a right to any cotton produced by the *moradores*, they had several obligations to the *moradores* as well. *Moradores* and their families were entitled to housing, hunting rights, and could fish in the dammed ponds, although they did

⁵² Davis, *Late Victorian Holocausts*. 405

⁵³ Bernard J. Siegel, "Migration Dynamics in the Interior of Ceará, Brazil," *Southwestern Journal of Anthropology* 27, no. 3 (1971). 235

not have "an automatic right to the use of this water for irrigation purposes."⁵⁴ The workers also did not have the right to ask a bank for a loan, and had to go through the owners.

Crucially, however, unlike peasants in feudal societies, the *moradores* and their families were not legally bound to the land. Mobility was a constant part of life in the *sertão*. Bachelors were the most mobile and could attach themselves to *moradore* families, but once a nuclear family was established they were generally expected to "settle into a tenant relation with an owner."⁵⁵ Even then, however, families moved between *fazendas* for a variety of reasons, from poor housing to personal quarrels with the owners. Bernard Siegel, in a study of migration in Ceará, notes that the region was home to an 'endemic semi-nomadic pattern,' and that "migration from one plantation to another is almost incessant, with an average annual turnover ... of 8 percent."⁵⁶

This feature of life in the *sertão*, the constant recourse to migration as a solution for environmental or labor difficulties, can help us contextualize the "manmade dimension" of the droughts in the late 19th and early 20th centuries. State measures such as an investment in infrastructure could have mitigated the effects of the great droughts, but during the nineteenth century, they were only sporadically applied. Mike Davis writes that "the modern history of the *sertão* is striking for the absence of any significant state developmental role until the 1960s."⁵⁷ Projects such as road building or the construction of dams were occasionally implemented during drought years but then abandoned in years of plenty. While they provided necessary income that

⁵⁴ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil." 251

⁵⁵ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil." 253

⁵⁶ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil." 243

⁵⁷ Davis, *Late Victorian Holocausts*. 398

allowed some individuals to survive the worst years, they amounted to a stop-gap rather than a robust or long-term solution.⁵⁸

At the coming of the unusually severe 1877 drought, then, the government's minimal investment in public works and relief shipments were woefully insufficient. People started to leave the place of hunger, and the regional cities swelled with refugees: in just two years Fortaleza and Aracati, two cities on the coast, swelled from 25,000 and 5,000 residents respectively, to 114,000 and 60,000. This did not alleviate the need for food, however, and so "[a] government pressed on all sides by masses they could not feed decided that the best alternative was to ship thousands of the drought fugitives to the Amazon where they could start life over."⁵⁹

Why the Amazon? Rubber was booming during the worst years of drought, but it was not the only industry recruiting workers. Coffee, grown on plantations in the southeast of Brazil, was similarly booming and needed workers, but they refused to recruit the climate refugees from the *sertão*. Davis notes that "southeastern coffee planters, for their part, wanted only 'white' overseas laborers after Emancipation," and even in the drought stricken regions of the northeast, "[w]hile state authorities were roadblocking *retirantes*' route to the cities and forcibly interning them by the thousands in camps, they continued efforts to lure European immigrants with expensive subsidies."⁶⁰ He describes this policy as part of an obsession among Brazil's elites with "Whitening" and "de-Africanizing" Brazil, which led to the creation of "the mother of all dual labor markets." The racial division of Brazilian society was a fundamental structure of the rubber

⁵⁸ Siegel, "Migration Dynamics in the Interior of Ceará, Brazil."

⁵⁹ Brooks, "HUMAN RESPONSE TO RECURRENT DROUGHT IN NORTHEASTERN BRAZIL." 43

⁶⁰ Davis, *Late Victorian Holocausts*. 403

boom, in which racialized groups were pushed away from the flourishing towns of the coasts and the rich plantations of the south and into the Amazon where free labor was at best a formality.

Rubber Tappers

The rubber tappers of the Amazon have occasioned much scrutiny. In the first part of the twentieth century, attention was paid to the inefficiency of the wild tapping system and the abject working conditions of the tappers themselves.⁶¹ Early on, the rubber trade was placed in a global context, as scholars, diplomats, and industrialists like tried to understand the different trajectories of the South American and Asian rubber industries, as well as how attention from the countries of North America and Europe could shape their futures. In the last 40 years, scholarship on the workers of the Amazonian rubber trade has shifted towards micro-economic and local considerations, and a re-evaluation of the infamously brutal working conditions. Scholars such as Barbara Weinstein have demonstrated that within the Amazonian system of labor, "the *seringueiro* [tapper] was not necessarily as helpless or dehumanized as one might believe from the standard accounts. On the contrary, the tapper showed himself ready to resist innovations which threatened to infringe further on his autonomy."⁶² We should not take this too far, however, for although it is true that rubber tappers had substantial freedoms when compared to those living under slavery or the compulsory labor schemes of the nineteenth century, they were far from autonomous producers. Rather they labored in a system that offered them superficial legal labor rights while trapping them in a cycle of debt bondage. Despite this, due to the isolated nature of rubber tapping they had significant leverage over key aspects of production that they could

⁶¹ For the first see Akers, *The Rubber Industry in Brazil and the Orient*; for the second see W. E. Hardenburg, *The Putumayo: The Devil's Paradise*, ed. C. R. Enock (T. Fisher Unwin, 1912).

⁶² Barbara Weinstein, "Capital Penetration and Problems of Labor Control in the Amazon Rubber Trade," *Radical History Review* 27 (1983). 132.

exercise to their advantage. This limited power enjoyed by the tappers played a crucial role in shaping the Amazonian rubber industry during and after the boom.

For Barbara Weinstein, the key to understanding the nature of the Amazon rubber trade, and especially its unique characteristics when compared to other Latin American export booms, lies in an analysis of the productive relations internal to the rubber industry. Writing in the 1980s, she criticizes the work of dependency theorists in Latin America for "relegat[ing] such issues as internal class formation to a position of secondary importance," arguing that we must "look first at the relations of production and exchange that defined the regional economy, and the forces inhibiting any attempts to transform those relations."⁶³ She characterizes her approach as "the exact opposite point of view" of dependentistas who prioritize "production for the world market, rather than the mode of production as the crucial variable." In her analysis, the local productive relations served as a bulwark against the intrusion of capital, as "[e]ven the most heavily financed attempts to transform rubber production and place it on a more 'capitalist' footing had to contend with internal resistance from both the rubber tappers who produced the export commodity and the domestic middlemen who marketed it locally."⁶⁴ Her empirical work into the local conditions of production is invaluable, but her analysis is limited by an insistence on 1) the analytical separability of the local mode of production and the contours of global capital flows, and 2) the primacy of the first over the second. In this section I insist on the co-constitution of local and global manifestations of capitalism - and the inextricability of production and exchange. The local relations of property and the conditions of class struggle in the Amazon did not serve as a bulwark against capital's intrusion, but were themselves fundamentally

⁶³ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 2

⁶⁴ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 3

transformed to serve capital's need not merely for the production of commodities, but their consumption as well.

Central to Weinstein's argument is an insistence on the continued presence of "pre-capitalist relations of production."⁶⁵ Despite the transformation of the Amazon by the rise of the automotive industry's need for rubber, in her view "the Amazon's increased integration into the world capitalist system did not generate a genuine transition to the capitalist mode of production."⁶⁶ Rather, the rubber industry continued to be dominated by the collecting expeditions that had defined the colonial era. Because of the dispersed nature of resources in the Amazon, their exploitation depended on a similarly dispersed labor force that collected the desired goods on long expeditions into the jungle. The rise of rubber both intensified and reinforced these patterns of production.

Rubber tappers belonged to a complicated hierarchy of tappers (*seringueiros*), their bosses (*patrões*), landlords (*seringalistas*), middlemen and traders (*aviadores*), and exporters. The tappers would be responsible for several looping trails that connected one or two hundred rubber trees. They would collect rubber from these trees, process it, and deliver it on a regular basis to either the local landowner or the trader who managed the local market. The trader is the key figure in the structure of Amazonian rubber production, as one (or several) stood as intermediaries between the rubber collector and the outside market. The trader stationed in the interior served two functions: they collected the rubber from the workers, but they also "operated a wet and dry goods store in one of the tiny interior towns and acted as an agent for a large commercial house, which supplied him with the wares he sold on credit, and accepted as payment the rubber

⁶⁵ Weinstein, "Precapitalist Relations."

⁶⁶ Weinstein, "Capital Penetration." 121.

he received either from the patrão [boss] or directly from the tapper."⁶⁷ The trader would generally advance to the tapper the tools necessary to tap rubber, as well as foodstuffs and necessary domestic goods - all at an inflated price.

Thus would form the notorious debt relation that bound the tappers to their positions, and forced them into an endless cycle of work to pay off an ever-increasing debt. The provision of goods and tools was essential, especially for laborers from outside the Amazon who were unaccustomed to the pattern of life on rubber expeditions. For those familiar with the terrain, the jungle provided such foodstuffs as Brazil nuts or cacao beans which could be eaten or sold, but even Indigenous communities had been disrupted to the point that they became reliant on goods provided by the traders. Weinstein says of a particular community in the Amazon called the Mundurucú that "[p]articipation in the trade did eventually destroy much of the tribal culture, but this was due largely to the corrosive effects of the merchandise the Mundurucú received in exchange for their rubber. Increasingly dependent on commodities that they themselves could not produce, members of the tribe spent more and more time on the rubber trails."⁶⁸ The commodities themselves were not locally or even nationally produced, but were largely imported. Resor notes that "the seringueiro subsisted largely on imported canned food, and most of his tools were also imports."⁶⁹ Through the debt contract, the traders connected the rubber tappers to the international market in two distinct ways, ensuring that they produced cheap rubber for international industry, and holding them as a captive market for the commodities produced by international canning industries and tool manufacturers.

⁶⁷ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 18. This description of the rubber trade's hierarchy is drawn from pages 15-31.

⁶⁸ Weinstein, "Precapitalist Relations." 64-65

⁶⁹ Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945." 350.

Weinstein downplays the coercive nature of this debt, however, even saying that the Mundurucú people were 'coaxed' and not 'coerced.' This distinction is problematic as even in the absence of violence, debt or the threat of starvation are more than enough to coerce the hungry to work. As Jairus Banaji asks: "When is a contract 'voluntary'? The answer is, probably never."⁷⁰ Throughout her descriptions of the rubber trade Weinstein is ambivalent about the forces that compel the tappers to work, describing the relationship between tapper and trader as both "not necessarily coercive in any obvious sense," but also liable to "deteriorate into a coercive relationship, with the tapper spending more and more time in the rubber fields at the behest of his merchant *patrão* (boss)."⁷¹

Her ambivalence seems tied to her insistence on characterizing the rubber trade as pre-capitalist, a claim which relies on the fact that the industry "did not generate a transition to a mode of production in which the producers were completely separated from productive resources or compensated by wages."⁷² In other words, in parts of the Amazon the debt relation provided some limited freedoms to the tappers, especially some control over the means of production. She argues that "in a region as vast as the Amazon, escape to another zone of rubber extractions particularly during the early years of the rubber era, seems to have been relatively easy, thereby limiting the effectiveness of debt as a means of control."⁷³ That the rubber fields were sparsely populated and remote also meant that the laborers retained control over their own working conditions. This modicum of control over the conditions of production mean that the tappers

⁷⁰ Jairus Banaji, *Theory as History: Essays on Modes of Production and Exploitation* (Brill Academic Publishers, 2010). 131.

⁷¹ Weinstein, "Precapitalist Relations." 57-58

⁷² Weinstein, "Precapitalist Relations." 56

⁷³ Weinstein, "Precapitalist Relations." 61

were able to resist proletarianization and the consequent loss of control over their working conditions. At the same time, however, the isolation of the worker undermined the possibility of collective action on the part of the tapper and, as the rubber gatherers penetrated more deeply into the jungle and farther from cities, it increased the tapper's dependence on the trader for essential food, tools, and other commodities.

Coomes and Barham challenge Weinstein's view that the social structure of the Amazon rubber industry was anachronistic. Wage labor did not fail to develop because the laborers, through their resistance to proletarianization, halted the transition to capitalism, and the unique structure of Amazon rubber production was not stalled at an inefficient stage of development. Rather, the debt relation that structured labor in the Amazon was a stable dynamic with its own coherent logic that was suitable to the production of rubber as a commodity in the Amazon. Coomes and Barham's analysis differs from previous analyses in two ways. First, they assert that local actors in the industry retained a significant amount of surplus, and second, that the "much-criticized debt-merchandise contract provided the basis for a stable and relatively efficient relationship among the trader, the patron, and the tapper."⁷⁴

These two points have significant consequences. The first would indicate that the failure of the rubber industry in the Amazon to survive the rise of Asian plantations was not due to a lack of local surplus that could be re-invested in production. Some of this surplus was spent in the conspicuous consumption that characterized the boom cities, and some was invested in the extensive expansion of production by opening new rubber trails deeper in the Amazon (rather than intensive expansion with investment into labor-saving technologies). But Coomes and Barham also make the case that tappers themselves retained significant surplus: according to

⁷⁴ Coomes and Barham, "Amazon Rubber Boom Revisited," 241.

public land records in Iquitos, rubber workers and their families owned a fair amount of property in the city. Because of the unsupervised nature of rubber collection, tappers could also retain surplus through illicit trades. After providing enough rubber to their contracted trader to service their debt, they could sell any remainder on the side to roaming traders who were looking to purchase any surplus rubber. All told, the returns on rubber tapping were great enough to push that the daily wage in the urban areas of the Amazon succeeded that in other parts of Brazil by a factor of four or five. These city wages were "necessary not only to attract trained labour into the region (e.g. professionals, administrators, etc.), but also to hold untrained labour out of rubber extraction to work the ports and service industries of the cities."⁷⁵

The second assertion counters the characterization of Amazonian production as backwards or inefficient. On the contrary, Coomes and Barham convincingly argue that the structure of the debt relations "ensured the lowest-cost, most sustainable delivery of wild rubber to lenders and estate owners in the decentralized Amazonian environment."⁷⁶ In their view, the debt relation was not a one way siphoning of surplus, but a "mutual commitment of credit for rubber" by which "patron and tapper met their respective needs and ensured that wild rubber would be procured at low cost per unit capital and per tree."⁷⁷ They go further than Weinstein in stating that not only did the tapper retain some leverage in the relationship with the trader, but that the "[t]he debt-merchandise contract filled the common need of trader, patron, and tapper for a stable, long-term relationship in which the trader and patron minimized transaction costs and

⁷⁵ Bradford Barham and Oliver Coomes, "Wild Rubber: Industrial Organisation and the Microeconomics of Extraction During the Amazon Rubber Boom (1860-1920)," *Journal of Latin American Studies* 26, no. 1 (1994). 68.

⁷⁶ Coomes and Barham, "Amazon Rubber Boom Revisited." 245.

⁷⁷ Barham and Coomes, "Wild Rubber." 63.

risks, and the tapper obtained some insurance and access to credit in a risky work environment."⁷⁸ Wage labor, which requires the close monitoring of workers, would have been too costly in the wild rubber scenario where labor productivity (the amount of rubber collected per unit-time) was low, and piece-rate labor would run into "the problem of provisioning the tappers for a season of work."⁷⁹ The debt relation resolved both of these problems. Monitoring laborers was unnecessary because the debt had to be repaid and rubber procured in order to access the commodities necessary for survival in the Amazon. Likewise, the cost of provisioning was offloaded onto the tapper as debt. This debt served as a sort of insurance both for the tapper and their trader. On the one hand, the exorbitant prices charged by the trader dispersed some of the risk bound up in advancing provisions, while at the same time ensuring that the trader was invested in the continued well-being of the tapper. If the loan was to be recovered, the tapper had to be provided with the means of survival.

The analyses of Weinstein and of Coomes and Barham are incomplete, especially on the question of coercion in the relationship between tapper and trader. Coomes and Barham's picture of the debt-merchandise relation as a symbiotic one falls flat in view of the high mortality the tappers faced. Meanwhile Weinstein's distinction between coercion and 'coaxing' is vague and unconvincing. Labor conditions in the Amazon, especially for the migrants from the northeast, were brutal as laborers faced disease, starvation, sweltering weather, and an unwelcoming terrain. Each section of the rubber workers - the Indigenous Amazonians, the peasants from Ceará, and the formerly enslaved Africans - had been cut off from their previous methods of survival. They may not have been forced to work under direct threat of violence, but the

⁷⁸ Coomes and Barham, "Amazon Rubber Boom Revisited." 245

⁷⁹ Coomes and Barham, "Amazon Rubber Boom Revisited." 246

possibility of death hung over their heads all the same. Their situation is precisely that described by Marx in Capital Volume 1:

[T]hey are compelled to sell themselves voluntarily.... The silent compulsion of economic relations sets the seal on the domination of the capitalist over the worker. Direct extra-economic force is still of course used, but only in exceptional cases. In the ordinary run of things, the worker can be left to the 'natural laws of production', i.e. it is possible to rely on his dependence on capital, which springs from the conditions of production themselves, and is guaranteed in perpetuity by them.⁸⁰

The tappers had precisely the freedom of the proletariat: the freedom to work or starve.

The fact remains, however, that the rubber tappers were not, strictly speaking, wage workers. If wage labor was to be established in the rubber industry, it would need to concentrate both labor and the rubber trees in a smaller geographic area where labor could be supervised and its productivity could be increased. Several attempts were made to establish plantations in the Amazon, especially after the boom as production in the region lagged behind the Asian plantations, but they were largely unsuccessful. The next section will look at the most infamous of these attempts: Henry Ford's Brazilian plantations of Fordlandia and Belterra.

Fordism and the Failure of Plantation Rubber

Almost all rubber produced in the Amazon, both during the boom and in the years after, was harvested from wild rubber trees. Plantations were rare, and to the extent they existed they were small projects "generally regarded as casual experiments by their owners, and were promptly abandoned when problems arose."⁸¹ The main exception to this was Henry Ford's attempt to produce plantation on a large scale in the 1920s and 1930s. His two plantations, at Fordlandia and Belterra, though unsuccessful and eventually abandoned were anything but

⁸⁰ Karl Marx, *Capital: A Critique of Political Economy*, vol. 1, trans. Ben Fowkes (Pelican Books, 1976).

⁸¹ Weinstein, *The Amazon Rubber Boom, 1850-1920*. 33

casual. Over the course of two decades, Henry Ford spent millions of dollars in a Sisyphean struggle to produce rubber in Brazil using management principles perfected in his Michigan factories.⁸² But every time success seemed to be within reach, catastrophe struck and the boulder tumbled down the hill again. His project was stymied by labor unrest and ecological disaster, and in the end his attempt to transplant the relations of production and reproduction that he pioneered in America to the Amazon was a total failure.

However it was, for us, an instructive failure. Henry Ford, fresh off his success in reconfiguring not just the factory floor, but the private lives of his workers, correctly perceived that there was a strong connection between the manner in which workers lived and the manner in which they produced. His mistake was in thinking that this connection was absolute rather than modulated by such factors as ecology, culture, and politics. As we have noted, Henry Ford's reconfiguration of patterns of production was tied up with a project of Americanism - an attempt to elaborate a new kind of man that Gramsci notes was uniquely possible in the United States.⁸³ Existing cultures and communities in the American Midwest had been wiped out through centuries of displacement, resettlement, and genocide and the slate of the US American working classes had been blanked and prepared as a receptacle for Henry Ford's American ideals. In Brazil, just as in Europe, cultural patterns and social structures persisted - they had been changed by the arrival of capital, but not erased. Thus, while correct in his intuition that capitalists in the Amazon would need to pay attention to questions of education, food, child-rearing, demography and recreation, Ford was wrong to import US American answers wholesale.

⁸² Greg Grandin, *Fordlandia: The Rise and Fall of Henry Ford's Forgotten Jungle City* (Metropolitan Books, 2009).

⁸³ Antonio Gramsci, "Americanism and Fordism," in Antonio Gramsci, *Selections from the Prison Notebooks*, trans. Hoare Quintin and Geoffrey Nowell Smith (International Publishers Co, 1971).

The factors contributing to Fordlandia's collapse can be grouped roughly into two categories: ecological difficulties exacerbated by Ford's implementation of factory-based regimentation in the Amazonian context, and labor troubles arising likewise from Ford's strict regimentation of the daily lives of his workers according to his experiences in Michigan. This first difficulty - that of the hostility of the Amazonian ecology to the establishment of rubber plantations - did not exclusively affect Fordlandia, but undermined the fundamentals of plantation economics. In the Amazon, rubber trees grow alongside their predators: a fungus that causes South American leaf blight, and a whole host of insects that feed on various parts of the rubber trees. In the wild, rubber trees grow far enough apart that the spread of these predators is limited, and affected trees are less likely to spread their parasites to other hosts. In plantations, however, the close planting of trees create a prime breeding ground for the enemies of the rubber plants.

It was assumed that such horticultural problems would be easily defeated. In the trade journals of the 1920s, proud rubber men declared that "Rubber culture has now been reduced to a science. Tropical diseases have been conquered through proper sanitation. When Mr. Ford enters on his task, therefore, of turning Brazilian jungles into rubber plantations, he will be equipped by modern science to overcome automatically many of the unfavorable elements which brought disappointment and failure to his predecessors."⁸⁴ But Ford's factory science was not up to the task of rationalizing Amazonian ecology. Greg Grandin details the methods used to combat insect infestations and fungal infections implemented by the staff at Fordlandia and Belterra, which were so labor intensive as to seriously undermine the profitability of plantation rubber in South America. These included manual culling of insects from the plants by the men and women

⁸⁴ "Labor Is Seen as Principal Obstacle to Ford in Brazil," *The Rubber Age* (New York) 22, no. 3 (1927). 125

of the plantations and regular inspections of the plants for fungus. Eventually Ford's scientists created a system of grafting that combined "durable roots, a high-yield trunk, and a full, verdant crown of blight- and bug-resistant leaves" from three different strains of rubber trees, but "[t]he procedure was time-consuming and costly and entailed building bulky scaffolding in the field to perform the operation and to support the graft."⁸⁵ Even if the management of labor had presented no difficulties, these measures put Amazonian plantations at a significant disadvantage against Asian plantations where the rubber pests were absent.

The management of labor in the Amazon in fact presented quite a few difficulties for Henry Ford. In an essay in the collection *Imperial Debris*, Grandin describes the current state of Fordlandia where the "hospital has collapsed, the jungle has reclaimed the golf course and tennis courts, and bats have taken up residence in houses where American managers once lived...."⁸⁶ He notes with great irony the similarity between the ruins of Ford's city in the jungle and the urban decay of Detroit, or of Ford's abandoned mining and logging towns in Michigan's upper peninsula. But whereas in Michigan the amenities provided by the company - the high pay, the parks, the healthcare - were a crucial element in the decades-long success of Ford's social experiment, in the Amazon their restrictive nature was more readily apparent. The rubber workers rejected the claustrophobic, managed life that Ford offered them at nearly every turn.

At first, Ford's social policies were welcomed. A hagiographic profile of Henry Ford published in a Brazilian paper in 1928 called Ford a "creative spirit" who was "renewing the face of the planet."⁸⁷ He was expected to bring benefit not only to the captains of industry, but to the

⁸⁵ Grandin, *Fordlandia*. 330

⁸⁶ Greg Grandin, "Empire's Ruins: Detroit to the Amazon," in Ann Laura Stoler, ed., *Imperial Debris: On Ruins and Ruination* (Duke University Press, 2013). 110

⁸⁷ José Carvalho, "Henry Ford," *O Paiz*, June 14, 1928. "Mas o que é certo é que Henry Ford é um espirito verdadeiramente creador e que está renovando a face do planeta."

whole of society, and workers would flood his plantation to receive the benefits of working for Ford: higher salaries, quality housing, good schools, a state-of-the-art hospital, and ample recreational facilities. Workers did flood the plantation, but found Fordlandia to be far from a worker's paradise.

Other than some settlements and holdings across the river, Fordlandia was an isolated city. To attract stable workers with families, the town had to provide everything its residents might need. Fordlandia was equipped with radio, a film projector, parks and other perks, but they were all provided on Ford's terms in a way as to promote Ford's family values. In practice, almost every benefit offered by Ford was accompanied by a corresponding reduction in freedom. Housing was one of the most important improvements lauded by Fordlandia's boosters in the press, but it was slow to come, and initially only the Americans in the town lived in permanent housing. These Michigan-style homes were high quality, but were built from materials shipped in from the US, and were designed for a colder climate. Henry Ford evidently detested the look of the thatched roofs and mud walls on the homes workers had built for themselves, considering them unbecoming of his workers. By the early 1930s as the construction of housing expanded, the residents of Fordlandia were encouraged to abandon their well ventilated housing for the stifling heat of metal-roofed houses, described as "galvanized iron bake ovens."⁸⁸ In a program similar to the Five Dollar Day, which increased the pay of workers who met Ford's lifestyle requirements, the extra rental cost of the new houses was "not actually paid by the laborer inasmuch as his salary [was] increased by that amount in the event that he live[d] in one of these houses."⁸⁹ Despite this, many of the married workers eligible for the 'modern' houses "prefer to

⁸⁸ Grandin, *Fordlandia*. 274

⁸⁹ A. Ogden Pierrot, "A Visit to Fordlandia," *The Rubber Age* (New York) 31, no. 1 (1932): 19–22.

remain in their huts by the river - palm thatched shelters with dirt floors - in preference to the frame houses with screened windows, running water, bathrooms and electric lights."

The food and recreation offered by Ford were likewise rejected by Brazilian workers. Drinking was banned in Fordlandia but the settlements across the river were outside his jurisdiction and they grew into "'filthy small cafes, restaurants, meat and fruit shops,' gambling houses, and thatched bordellos established by local merchants and staffed mostly with women from Brazil's poor northeast."⁹⁰ Rather than spend time listening to music or visiting the parks of Fordlandia, the workers preferred to wind down off premises across the river. When they ate on site, the workers were provided food in an eating hall, but they "grumbled about being fed a diet set by Henry Ford, consisting of oatmeal and canned peaches imported from Michigan for breakfast and unpolished rice and whole wheat bread for dinner."⁹¹ Adding to their frustration, the cost of meals were deducted automatically from the worker's paychecks. In December 1930, the workers rioted in the cafeteria, and the frustrations of the compound's restrictions boiled over as they ransacked the town. In response the company fired all but a few hundred men, and they were forced to renew their search for a suitable mass of workers free from "shiftless and vicious types" and "communistic elements."⁹²

Even the hospital - maybe the most successful of Fordlandia's social improvements - was a source of conflict. Boosters in trade publications such as *India Rubber World* and *The Rubber Age* lauded the eradication of diseases endemic to the Amazon, even going so far as to claim that Ford's settlement "is probably as free from malaria and other tropical scourges as any American

⁹⁰ Grandin, *Fordlandia*. 157

⁹¹ Grandin, *Fordlandia*. 228

⁹² Pierrot, "A Visit to Fordlandia." 20

city."⁹³ Indeed the hospital built at Fordlandia provided care not only to employees and their families, but to anybody from the neighboring districts. The introduction of health measures were largely successful, but residents bucked against the injunction to give birth at the hospital, rather than at home with a midwife. Ford's doctors also insisted on providing mothers and new children with powdered milk - which meant that "utensils had to be provided as well, since most workers didn't own dishes, or 'even a spoon,' to prepare the powdered milk, using instead their fingers to mix the powder in empty cans."⁹⁴

Common to all of the restrictions Ford placed on his workers was the substitution not merely of American culture for Amazonian practices, but the substitution of American commodities for local materials. For Henry Ford the ideal worker is one who not only produces commodities, but consumes them as well. In the early years of the plantation, the establishment of a cash system with nowhere to spend it undermined Ford's dream of a modern (and teetotaling) Midwestern city in the jungle. To rectify this, cash wages had to be combined with a robust domestic cash market. Ford attempted to manage the whole of the circuit of commodity production and consumption on site - by 1932, Ford's Amazonian cities were homes to rows of shops. Central to Fordlandia's propaganda was the image of domestic commodity consumption:

The housewives who for the first time in their lives know the joys of shopping are to be seen moving from store to store and finally emerging with market baskets of conventional design filled with groceries. This is a considerable transformation from the erstwhile occupation of these same women of gathering a few bananas from the small plot behind the hut on the edge of the jungle and serving them with whatever fish the male members of the family may have caught....⁹⁵

⁹³ "Ford Rubber Plantation," *India Rubber World* (New York) 85, no. 5 (1932). 48

⁹⁴ Grandin, *Fordlandia*. 291

⁹⁵ Pierrot, "A Visit to Fordlandia." 22

But these small glimpses of Midwestern life being lived in the Amazon were not generalized to a large Amazonian rubber working class. They were a mirage, carefully reproduced as propaganda for the trade journals of the rubber industry. This was a facsimile of Midwestern industrial life, and it was not attractive to workers in Brazil. They chafed at the closely managed routines of Fordlandia, and although they came to Fordlandia in large quantities, they mostly did not stay long enough to see the rubber crop through: "A fair number disappeared after their first pay day with more money than they had ever seen at one time, and after more than two years have failed to put in another appearance."⁹⁶

By the early 1940s, some 15 years after Ford purchased land in Brazil neither Fordlandia nor its sister plantation Belterra were producing rubber in commercial quantities. Pest and disease management was expensive, which along with the higher cost of labor in the Amazon made competing with cheap Asian rubber difficult. The biggest issue, however, was labor recruitment. Despite the amenities and higher wages offered at the plantation, Ford was unable to attract sufficient labor to Brazil. In an article examining the difficulties facing Fordlandia and Belterra in 1942, Joseph A. Russell notes several factors exacerbating the labor supply problems. Prominent among them were the "disinclination to work on the part of the natives who are without employment," and the "bad reputation among Brazilian labor" of the rubber region of the Amazon.⁹⁷

⁹⁶ Pierrot, "A Visit to Fordlandia." 21.

⁹⁷ Joseph A. Russell, "Fordlandia and Belterra, Rubber Plantations on the Tapajos River, Brazil," *Economic Geography* 18, no. 2 (1942). 140.

The Second Rubber Boom: Rubber Soldiers

Periodically, after the boom years, discussion would arise of a rebirth of Brazilian rubber. In 1916 Joseph F. Woodroffe published *The Rubber Industry of the Amazon and How Its Supremacy Can Be Maintained*, a lengthy book which set the conventions of the genre: despite the rise of plantation rubber in Asia, with the right tweaks to both production and the reproductive conditions of the workers, Brazilian rubber could still be profitable.⁹⁸ His suggestions primarily centered on developing a stable working population, to be achieved through the importation of laborers from Asia, as well as replacing the dominance of rubber collection with an agricultural system that combined rubber with crop and livestock cultivation. In his vision, the imported labor would "clear and settle on the lands available up the Amazon, and whilst forming home-steads and small farms, to tap and cure rubber to obtain 'pocket-money,' whilst their women and children do the lighter work on the home farms."⁹⁹ A population of small proprietors would be significant change from either plantation development or the existing system of debt-bound tappers, but Woodroffe was insistent that the self-sufficiency of tappers would improve the long-term stability of the rubber industry in Brazil.

Aware that the revival of Brazilian rubber could be seen as a challenge to British dominance in the rubber trade, Woodroffe invited both former president Roosevelt and Lord Bryce in England to comment for his book on the relevance of the Monroe doctrine to British investment in Brazil. Both declined to say anything of substance on the matter, and indeed in later discussions Brazilian rubber was presented as a crucial matter of national interest for the United States. After all, in the 1920s, "[w]hile British companies in British colonies produced 67

⁹⁸ Joseph F. Woodroffe, *The Rubber Industry of the Amazon and How Its Supremacy Can Be Maintained* (T. Fisher Unwin, 1916).

⁹⁹ Woodroffe, *The Rubber Industry of the Amazon and How Its Supremacy Can Be Maintained*. ix.

per cent of the world's rubber, most of their exports went to the United States, which consumed over 70 per cent of the world's production."¹⁰⁰ This situation came to a head in early 1942 when Japan captured Singapore from the British and cut off the Allied powers from their supply of natural rubber. Almost overnight, the United States was deprived "of 97% of its supply of a key material for which it had effectively no domestic sourcing."¹⁰¹ Shortly thereafter US chauvinists began to extol the virtues of the long-neglected Amazon as "[t]he loss of Asiatic sources of supply has, of course, turned the attention of the United States to the potentialities of Brazil's rich valley" - this time, of course, the lesson was learned, the mistakes of the past would not be repeated, and "the new phase of the trade [would] follow along economic and constructive lines."¹⁰²

In reality, no such thing happened - rather than investing meaningfully in a stable rubber industry, Brazil fell back on old patterns of extraction. In the 1940s Brazil made one last push to harvest rubber from the Amazon, but the lesson it had learned was less about the investment in stable industry than the disposability of labor. Surveys of the Amazon indicated that the region could supply 100,000 tons of wild rubber per year, but this output "implied, according to the survey's optimistic calculation, 100,000 tappers, to be drawn mostly probably from the arid and impoverished northeast, as in the days of the boom."¹⁰³ The previous system of tapping remained mostly intact, and attempts to bypass it by offering tappers higher prices were countered by the *seringalistas* raising the prices of the provisions that tappers relied on - ensuring that they were

¹⁰⁰ Randolph R. Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945," *The Business History Review* 51, no. 3 (1977). 361.

¹⁰¹ Alexander J. Field, "The World War II US Rubber Famine," *Business History Review* 99, no. 3 (2025).365.

¹⁰² John Melby, "Rubber River: An Account of the Rise and Collapse of the Amazon Boom," *The Hispanic American Historical Review* 22, no. 3 (1942). 468.

¹⁰³ Warren Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History* (Cambridge University Press, 1987). 93.

able to maintain profits. Further, along the rubber chain traders were hoarding rubber, waiting for prices to rise further, and production was much lower than expected. The Brazilian government decided to abandon their attempts at control, "turn the entire system 'back to the Brazilians'" and focus their efforts on recruiting more labor.¹⁰⁴

Labor, long the primary obstacle to the viability of Amazonian rubber, was not easy to obtain. "The Brazilian government in February 1943 turned the recruitment of tappers from outside the region into a quasimilitary operation, contracting them for two years, paying advances and promising family allowances, and furnishing transportation."¹⁰⁵ As previously, the benefits were exaggerated, medical services were inadequate, and advances went unpaid. The rubber tappers brought into the region had to battle disease, malnutrition, and mistreatment, with minimal protection from the state. The labor conditions of these 'rubber soldiers' were so bad that as early as 1946, Time Magazine reported that "[o]f the 18,000 men who went into Amazonia, only a few were ever seen again."¹⁰⁶ Sources disagree on the exact number of people conscripted to gather rubber during the war with more recent estimates exceeding 55,000, as many as half of whom died in the jungle.¹⁰⁷ Amazon exports to the US quadrupled during the war period, reaching just over 20,000 long tons in 1944 - just over a fifth of total US rubber imports, and less than 2% of the total preblockade imports of over a million long tonnes.¹⁰⁸ Compared with the US

¹⁰⁴ Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History*. 95

¹⁰⁵ Dean, *Brazil and the Struggle for Rubber: A Study in Environmental History*. 96

¹⁰⁶ "BRAZIL: Lost Army," *Time Magazine*, July 22, 1946.

¹⁰⁷ Gilson Laone Pereira, "'SOLDADOS DA BORRACHA' – ESQUECIDOS OU NÃO LEMBRADOS?," *Revista Margens Interdisciplinar* 8, no. 11 (2014).

¹⁰⁸ *Report on the Rubber Program, 1940 - 1945* (Rubber Reserve Company, 1945).

Government's expectations of importing 40,000 long tons in 1944, the Amazon's second rubber boom was a resounding failure.¹⁰⁹

Conclusion

At no point in the history of Amazon rubber - from the heyday of the rubber boom to the failed plantation experiments to the wartime conscription of rubber soldiers - was a body of free wage laborers established. Despite this, the key categories of life under capitalism that seemed so important for Henry Ford to grab hold of played a crucial role in structuring the lives of the rubber tappers. Every aspect of the lives of workers were bent to the needs of capital accumulation, including the manner in which they reproduced themselves while not working, and the commodities they consumed.

Capital is sometimes seen as a homogenizing force, which by collapsing all labor into abstract labor-time evacuates it of its specificity. The role of wild rubber in the Amazon points to a different interpretation. The exchange relation posits the commensurability of various types of concrete labor, but it does not collapse them. The need of capital for various specific concrete labors flourishes, conditioned by differences the processes of production, differences in political and cultural structures, and differences in the ecological conditions of extraction. Perhaps more to the point, the individuals who provide this concrete labor to capital serve another role as well: they are a crucial market for the commodities they produce. As consumers they are differentiated as well: the wage worker in the automotive factory purchased quite different goods than the

¹⁰⁹ Edward D. McLaughlin, "Report by the Consul at Para (McLaughlin) on Changes Brought About in the Amazon Rubber Program as a Result of Recent Conferences in Washington," Foreign Relations of the United States, Diplomatic Papers, US Department of State, 1943.

rubber tapper in the Amazon, but they both contributed to the essential final stage of commodity production: consumption.

The rise and fall of rubber production provides key insight into the categories of Fordism. Henry Ford recognized that the private lives of his workers were in a real sense internal to the production process. For automotive production to succeed as more than a niche luxury good, cars would have to be produced more cheaply, and the working classes would have to serve as a portion of the automotive market. This, combined with homeownership and an increase in domestic commodities reduced the flexibility of factory workers and tied them to their towns and jobs. Likewise, in the Amazon, in order for workers to be able to produce rubber cheaply, they would have to be severed from their previous means of subsistence and posited as commodity consumers. The substitution of consumption for subsistence production served the double purpose of tethering the workers to their employment and providing a market to the producers of canned food and tools abroad.

The debt relation was the dominant labor dynamic in the Amazonian rubber trade, but it was not the only one. In the upper reaches of the Amazon, state presence was minimal and the extraction of quality rubber became more difficult. Under these conditions, quite a different labor dynamic developed to meet the rubber needs of the automotive industry. In the Putumayo region of what was then Peru, but is now Colombia, the agents of the Peruvian Amazon Company subjected the Indigenous inhabitants to a system of slavery and genocide. In the context of different ecological and productive relations, the needs of capital shifted, and human life became comparatively less valuable - less worth reproducing. The next chapter will take a closer look at this particular case.

Chapter 4: The Putumayo Atrocities and the Reproduction of Genocide

Although most of the latex produced during the period of wild rubber was tapped by workers bound by debt rather than violence, there existed pockets of violence in the rubber industry so brutal that they seem to defy analysis. Contemporary interpreters resorted to metaphor, calling the rubber produced in these regions "red rubber," or "the devil's milk," and naming the part of the Amazon where rubber workers were enslaved "the devil's paradise." The most notorious of these regions was in Africa, which produced about half as much rubber as the Amazon, in the Congo Free State. King Leopold II of Belgium held the Congo Free State as absolute monarch and he exploited the population and resources with a rare brutality. As a result of his exploitation of the native population to produce ivory and rubber "between 1880 and 1920, the population of the Congo was cut 'by at least a half'," or some 10 million people.¹ When Roger Casement traveled to the Congo in 1903 in his role as British consul, he was convinced that "only a humanitarian crusade could abolish the evils of the Leopoldian regime,"² and he became famous for his advocacy against the rubber atrocities.

Although the scope of the rubber atrocities in the Congo were enormous, and their situation important in the history of rubber, my project focuses on a less well known series of rubber atrocities in the western Amazon around the Putumayo River, which were also exposed by Roger Casement. In this remote region of the Amazon, the Peruvian Amazon Company captured

¹Adam Hochschild, *King Leopold's Ghost* (Houghton Mifflin, 1998), 233.

²William Roger Louis, "Roger Casement and the Congo," *The Journal of African History* 5, no. 1 (1964), 120.

and enslaved the indigenous population and established what has been called a "regime of terror." They killed tens of thousands of the region's inhabitants through forced labor, starvation, and massacres, depopulating the region by 90%. I have chosen to highlight this case over the Congo atrocities primarily because it belongs to the same system of wild rubber collection as existed in the Brazilian Amazon, and so its points of difference are instructive as to the underlying dynamics of wild rubber collection in the most productive rubber field on earth: the Amazon. The questions raised in this chapter about forced labor for the market and the reproduction of genocide provide a fertile grounds for future research into the Congo as well.

The relationship between the Putumayo rubber atrocities and the capitalist world system is not a simple one, and scholars often consider the operations in the Putumayo antithetical to capitalism. Margarita Serje says that in the Putumayo, "credit and debt, usually considered to be one of the engines of capitalism, were here at the core of a system that was not only opposed to the market logic, but also to the possibility of developing a modern local economy."³ On the contrary, I argue that the Putumayo was dominated by the market. On the one hand, the products produced in the Putumayo were commodities - both rubber and enslaved people were 'produced' for the global and regional market. On the other hand, the Putumayo was - in much the same way as the lower Amazon - a consumer of commodities produced in the core. Serje betrays a teleological conception of capitalism by referring to some ideal of a 'modern local economy'. The Putumayo genocide was not a pre-modern aberration in global capitalism, but a core element. And it was not merely modern, but constitutive of modernity.

³Margarita Serje, "The Peruvian Amazon Co.: Credit and Debt in the Putumayo 'Wild Rubber' Business," *Enterprise & Society* 22, no. 2 (2021), 478.

To make this case, I will look at several aspects of the rubber industry in the Putumayo. I will first briefly examine relationship between the local Peruvian economy and the world market. I will then look at the operations of the primary corporation in the region, the Peruvian Amazon Company, as well as its actions against the indigenous inhabitants of the Putumayo. I should flag here that although I will not dwell on violent description, any discussion of genocide necessarily contains some graphic elements. I will then move on to an analysis of the racial hierarchy of labor in the Peruvian Amazon Company, paying special attention to the importation of Black laborers from Barbados to serve as overseers for the enslaved natives. Lastly I will look at the ways in which the social structure of genocide was reproduced, the necessarily short-term nature of this reproduction, and the way in which the Peruvian Amazon Company contributed to the larger dynamic of capitalist reproduction on a global scale.

Global Integration

The development of the rubber industry in the Amazon which was stimulated by the growing automotive industry had a profound transformation on Amazonian communities. As the region was reoriented towards commodity production for the world market, the dissolution of traditional patterns of communal life was accelerated, domestic commodity consumption was increased, and massive cities emerged, built using materials from Europe and North America. In other words, the growth of the rubber stimulated by the automotive industry led to the increased integration of the economy of the Brazilian Amazon into the world economy. This integration was, however, not limited to the Brazilian Amazon - the Peruvian Amazon was also pulled into international trade, and the local consumption of international goods increased significantly.

The western Amazon was less productive than the Brazilian Amazon for several reasons: it was further from the coast, it had less infrastructural development, and the quality of the rubber was lower. In 1901-1902 approximately 5% of the rubber produced by the Amazon valley passed through Iquitos, the Peruvian port city on the Amazon River - or 1.4 million⁴ kilograms out of 29 million.⁵ Given that Peru currently covers 13% of the Amazon, and claimed an even larger portion a hundred years ago, it is clear that the region, while still significant for the global market, was less productive than the eastern Amazon. The Peruvian region's marginal position in the Amazon rubber industry had several important implications for the relations of production that took hold there. The main characteristics of Amazonian production were present: the debt-relation and the increase in commodity consumption. However, these characteristics were warped by the remoteness of the region and the lower productivity: commodity consumption was more limited than in the rest of the Amazon, and more importantly the debt relation was combined with a genocidal violence exercised by the rubber corporations against indigenous workers. The remainder of this section will contextualize the importation of commodities to the Peruvian rubber fields, and the following sections will deal with the genocidal violence that was employed there.

The purchase and consumption by peripheral regions of goods produced in core regions served to extend the valorization of production in industrial centers. This is not a novel phenomenon. Marx writes of the agricultural revolution in Europe that "[t]he expropriation and eviction of a part of the agricultural population not only set free for industrial capital the

⁴Rupert J. M. Medd and Hélène Guyot, "Eyewitness Accounts during the Putumayo Rubber Boom: Manuel Antonio Mesones Muro—the 'Madman of the Marañón River,' Carlos Oyague y Calderón—the State Engineer, and Roger Casement—'Not of the Real World' Humanitarian," *Journeys* (New York, USA) 20, no. 2 (2019), 64.

⁵P. W. Barker and E. G. Holt, *Rubber Statistics 1900-1937: Production, Absorption, Stocks and Prices*, no. 181, Trade Promotion Series (U. S. Department of Commerce, 1938).

workers, their means of subsistence and the materials of their labour; it also created the home market."⁶ The means of subsistence that were formerly provided outside the market by the reproductive labor of the family and community have been turned into commodities; they "are now transformed into articles of manufacture, the markets for which are found precisely in the country districts."⁷ Marx is speaking of the division between the town and country side in Europe, but the dynamic holds true on the global scale: the peripheral countries which provided raw materials to the manufacturing companies of the capitalist core, themselves served as markets for the core's finished goods.

During the period of the rubber boom, Peru increased its importation of US-produced goods, especially consumer good such as tinned foodstuffs. Historian Geoff Bertram, in a study of Peruvian imports, classifies the use cases of imported goods in several categories. Consumer goods imports, such as food, clothing, medicine, and other commodities, fluctuated between 30% and 40% of imports between 1891 and 1930. The value of this category rose from about 5,000 soles in the 1890s to consistently over 50,000 in the 1920s.⁸ This was by far the largest absolute growth of any import category during these years. Other categories, such as manufacturing inputs and fuel fell as a proportion of total imports, while agricultural inputs, such as livestock feed grew slightly. The biggest proportional growth in imports was capital goods such as agricultural equipment and machinery - from less than 5% to over 15%.

During this period, the United States grew to be Peru's greatest trading partner. In the 1890s, the US accounted for barely 10% of imports to Peru and most years that decade received

⁶Karl Marx, *Capital: A Critique of Political Economy*, trans. Ben Fowkes, vol. 1 (Pelican Books, 1976), 910.

⁷Marx, *Capital*, vol. 1, 911.

⁸Geoff Bertram, "End-Use Classification of Peruvian Imports, 1891-1972," St Antonys College Oxford, 1977.

an even smaller proportion of its exports. By the mid 1910s, the US accounted for over 50% of Peru's trading, both imports and exports.⁹ The US was displacing the United Kingdom, which fell from over a third of imports and half of exports to about 20% of each over the same period. Some of this change was the result of trade disruptions during the first world war, but the trend of declining trade with Britain and increasing trade with the US had been well established prior to the war.

Despite the remoteness of the Peruvian Amazon, the commodities imported from the US made it to the workers in the rubber fields - especially foodstuffs. Lists of goods for sale at company stores around the Putumayo river include things like tinned milk, lard, and meat¹⁰ - all things being produced for export in spades in the US. Centered in Chicago, the US meatpacking industry was highly rationalized - the automation of Chicago slaughterhouses had even provided inspiration for the automotive assembly line. This industry was growing rapidly during the late 19th and early 20th centuries, and in 1909 over 88 million animals were slaughtered for food, mostly hogs and cows but also sheep.¹¹ This massive operation fed the meatpacking business, and in 1910 exports were in excess of \$130 million.¹² This canned, salted, pickled, or otherwise cured meat was precisely the type of food that sustained rubber workers in the Putumayo.

⁹Dirección de Estadística del Ministerio de Fomento, *Extracto Estadístico correspondiente al año 1918* (Lima, Peru, 1919), 57.

¹⁰Roger Casement, *Correspondence Respecting the Treatment of British Colonial Subjects and Native Indians Employed in the Collection of Rubber in the Putumayo District : July 1912*, Miscellaneous ; No. 8 (1912) (His Majesty's Stationery Office, 1912), 22.

¹¹US Census Bureau, *Slaughtering and Meatpacking* (1910), 345.

¹²US Census Bureau, *Slaughtering and Meatpacking*, 329.

The Putumayo Genocide

The food shipped in from the US was generally very expensive, of low quality, and it replaced the more nutritious local diet. Indeed, the effects of incorporation into the global market were uniformly detrimental to the people who lived in the western Amazon, undermining indigenous communities, and introducing severe forms of exploitation and violence. This is true throughout the Amazon, but was especially severe in the western region. In addition to the debt bondage that prevailed in Brazil, in the western Amazon indigenous peoples were enslaved and forced to work for rubber companies. The best documented case of Amazonian rubber slavery was in the Putumayo region of Peru (now Colombia). There rubber companies enslaved the native inhabitants with methods so brutal that tens of thousands died in a rubber genocide. The system that prevailed in Putumayo was a brutal one, but it should not be thought of as 'pre-capitalist' or 'non-capitalist.' Rather it was an integral part of the global system of capitalist production. The brutality of the rubber companies was a response to the demands of the global market, and the problems of commodity production in such a difficult and remote region determined the living and laboring conditions of the rubber workers. This section will examine these conditions, and will look at how a rigid racial labor hierarchy was established and maintained with the importation of Black workers from Barbados. Family and gender dynamics played an essential role in the consolidation of power, and they were explicitly manipulated by the white company men to subdue both the native workers and the Barbadian overseers. The reproductive scheme they established was essential in producing the cheapest possible labor, combining force, debt, and family obligations to compel the workers.

The Putumayo River is a tributary of the Amazon river which forms the current border between Colombia and Peru. The river basin is the traditional home of several indigenous peoples: the Huitoto were the most numerous, but there were other groups as well, such as the Bora and Muinane people. In the late 19th and early 20th centuries, the land north of the river was claimed by both Peru and Colombia, each of whom sent settlers into the area to establish land claims and harvest its rubber. Rubber was booming, and there was certainly a fortune to be made in its exploitation, but the rubber of the Putumayo region was of an inferior quality. In addition to the well-known *Hevea* trees, which can be tapped, the region was home to the lower quality *Castilloa Elastica*, which was often cut down in the process of harvesting its natural latex. Evidently harvesting was so severe that a contemporary observer in 1907 noted that Colombian harvesters complained there was very little of this second type left.¹³ The poor quality of the region's rubber required production costs to be kept low in order that the product might be viable on the world market. Both Colombian and Peruvian companies operating in the region relied heavily on the local indigenous population to perform the actual work of harvesting the latex.

The working conditions were brutal, beyond imagining. Throughout the Putumayo basin, indigenous people were enslaved, compelled to labor as rubber tappers, and massacred in great numbers. In what was known at the time as the Putumayo atrocities, but which is now called the Putumayo Genocide, tens of thousands of the indigenous inhabitants of the region were killed. One of the groups targeted, the Muinane people, were reduced from 2000 to just 20 as a result of the massacres. In 1993, their descendants in the Muinane census still numbered less than 150.¹⁴ Exact figures are disputed, but Roger Casement gives a conservative estimate of 30,000 total

¹³W. E. Hardenburg, *The Putumayo: The Devil's Paradise*, ed. C. R. Enock (T. Fisher Unwin, 1912), 94-5.

¹⁴Juan Alvaro Echeverri, "The Putumayo Indians and the Rubber Boom," *Irish Journal of Anthropology* 14, no. 2 (2011): 13-18.

victims; the Putumayo region was depopulated by 90% during the rubber boom. In addition to outright massacres, thousands were killed by starvation, overwork, or were forcibly removed from their home region.

The major perpetrator of this genocide was a single company, the J. C. Arana and Hermanos Company which was later renamed the Peruvian Amazon Company (PAC) after receiving a significant investment of British capital in 1907. The origins of the company were relayed by W. E. Hardenburg, an adventurer and engineer who traveled to the Putumayo in search of a rubber fortune. He experienced the brutality of the PAC first hand, and was the first to spread awareness of the atrocities in English. According to Hardenburg, the company was helmed by Julio César Arana, who had originally come to Iquitos as a hat salesman. Arana began to work with small Colombian rubber interests in the late 1890s, and in 1903 started his own rubber company along with his brothers-in-law. He soon bought out his Colombian partners and attempted to push out the remainder of the Colombian competitors. Hardenburg recalls, based on his conversations with the Colombians, and on his later first hand experience, that Arana and his company were ruthless - when dealing with former partners, Arana "[took] advantage of their ignorance and stupidity to rob them scandalously."¹⁵ When cunning was insufficient, Arana "forced José Cabrera, owner of Nueva Granada, to sell out to them at an insignificant price by threats of killing him, by shooting at him from ambush, by forcibly taking away his Indians [sic], and by the other methods for which this company is known."¹⁶

There is something significant in Hardenburg using the phrase "taking away *his* Indians" worth returning to, but for now it's enough to note that over time Arana's methods became more

¹⁵Hardenburg, *The Devil's Paradise*, 200.

¹⁶Hardenburg, *The Devil's Paradise*, 200.

and more terrible. While Hardenburg was visiting Colombian rubber estates in the Putumayo he heard stories of Arana's men attacking his white competitors, murdering their workers, enslaving their children and raping their wives. The PAC conducted raids on independent estates to frighten them out of the region and to steal their rubber stores. In some cases they appear to have been accompanied by Peruvian soldiers, evidently there to enforce Peru's claim to the region.

Hardenburg and his traveling companions were captured in one of these raids and held for some days at one of Arana's estates where it became evident that Arana's violence was not aimed only at competitors, but was directed against the local indigenous population as well. Whereas the Colombian rubber harvesters had paid the native workers - underpaid, to be sure, and often in goods rather than direct wages - the agents of the Peruvian Amazon Company compelled local indigenous groups to collect rubber by force. Slavery is the obvious term for this, but what was described by Hardenburg and then corroborated by the testimonies Casement collected later went far beyond a system of forcibly extracting labor from unfree subjects. It was a regime of terror. The indigenous Huitoto were forced to collect rubber and were provided only the barest amount of food. If they failed to collect enough rubber they were punished severely, and not only the men, but the women and children all bore the scars of the lash, which came to be known as the "marca de Arana."¹⁷ When they grew too old or sick to work they were killed and fed to dogs. Beyond this, the company officers kept harems of girls as young as nine whom they raped and killed at will; they used children as target practice, or lit them on fire for amusement.

In 1910, after Hardenburg's initial account had been published in the British newspaper *Truth*, Roger Casement was appointed by the UK Foreign Office to travel to the Putumayo to investigate several related matters: "whether any British subjects have suffered or are in distress,

¹⁷Hardenburg, *The Devil's Paradise*, 179-80.

and if so from what causes, and whether they stand in need of relief," as well as "the methods of rubber collection and the treatment of the employ  s of the Company."¹⁸ He was accompanied by a separate committee appointed by the Peruvian Amazon Company itself, which was tasked first with reporting "on the possibilities of commercial development of the company's properties" and secondly "to enquire into the present relations between the native employ  s and the agents of the company."¹⁹ The British subjects whose welfare formed the primary object of Casement's investigation were men and women from Barbados, recruited in 1904-5 and "shipped for Peru at the request of Se  or Abel Alarco, at that time a partner of the firm in question," some of whom stayed with the company long after their initial two year contracts had expired.²⁰ Their testimonies provided the greatest part of the evidence submitted by Casement to the Foreign Office.

Casement arrived in Iquitos on August 31st of 1910, and left for the Putumayo along with the commission from the PAC on September 14th. Casement stayed in the Amazon, visiting various stations within the Putumayo, interviewing company employees and observing the operations until December. His two page preliminary report notes that the charges brought by the men from Barbados against the company "were of the most atrocious kind, and added to the accumulating weight of evidence that we had gathered from station to station, and the condition of the Indian population as we had opportunity to observe it in passing, they left no doubt in our minds that the worst charges against the company's agents were true."²¹ In the testimonies, however, the men from Barbados do not figure as mere witnesses to the company's white agents'

¹⁸Casement, *Correspondence*, 1.

¹⁹Casement, *Correspondence*, 1.

²⁰Casement, *Correspondence*, 7.

²¹Casement, *Correspondence*, 3.

atrocities against the native population. Rather they testify in part against themselves, revealing a subjectivity in which they are both the recipients and dealers of terror - active, if often reluctant, participants in the construction of a racialized labor hierarchy.

Although the men from Barbados were initially told that they would be contracted as laborers, there were signs that this was false even before they arrived in Peru. When they arrived in Manaus on the way up the Amazon river they were warned "by outsiders that in the countries to which they were going they would not be employed as labourers, but would be armed and used to force the Indians to work for their employers."²² Several of the men protested and asked for the British vice-consul to intervene and release them from their contracts. Their protests were ignored, however, and they "had to be taken on board the river steamer waiting to convey them to the Putumayo under police supervision."²³ Their suspicions were confirmed upon arrival in Putumayo, as the first group to arrive was immediately armed with rifles and ammunition and sent into the jungle on a raid against the indigenous Andokes people. This raid was in part an attempt to secure labor for the collection of rubber, but it served another purpose as well. The Andokes had previously attacked a group of Colombian settlers who had come to enslave them, and had stolen their rifles. The Barbadians were sent out "on what were termed punitive expeditions" to kill or capture the responsible native people and to recover the lost rifles.

This was to become a pattern, in which the hired Barbadians would be placed between the indigenous people of the Putumayo and their white Colombian or Peruvian oppressors. They were given the jobs that were considered too dangerous or unpleasant for the whites. The rubber which was collected by the indigenous workers had to be carried on foot through difficult forest

²²Casement, *Correspondence*, 11.

²³Casement, *Correspondence*, 11.

paths obstructed by deep mud, rivers, and fallen trees. The journey from the furthest rubber station to the river was 70 miles, and took four or five days to traverse. Casement notes that "The Indians were not supplied with food for this journey. They were guarded by armed men both going and returning, and Barbados men frequently were employed for this work, just as they were used, in the first instance, in forcing the Indians to collect the rubber in the forest and bring it into Matanzas."²⁴ This was the general pattern that the Barbadians' labor took: they were engaged both in capturing and disciplining native labor. They held the guns that separated indigenous families and forced them to work past the point of starvation. By the time Casement arrived, only 20 of more than 100 Barbadians remained in the company's employ, but "with the exception of three men at La Chorrera itself, whose duties were those of ordinary labour, all the men ... were employed in guarding or coercing, or in actively maltreating, Indians to force them to work and bring in india-rubber to the various sections."²⁵

Although the men from Barbados were brought in as slave-drivers, but they were also themselves driven. In addition to outlining the debt they incurred and the conditions of starvation they faced, Casement recounts at length four cases in which the Barbadians were tortured and mistreated by the white company agents. Every station that the company owned had a set of stocks that had been built to punish the native workers who drew the ire of company agents. Although the stocks were much too small for the men from Barbados, they were often used as punishment for them too. In at least one case, a Barbadian man named Edward Crichlow was chained up in the stocks that he himself had built. In other instances, the Barbados men were lifted off the ground and suspended by their arms which had been tied behind their backs - in this

²⁴Casement, *Correspondence*, 11-12.

²⁵Casement, *Correspondence*, 12.

posture they were beaten with tapir-hide whips or with machetes. In most cases these punishments were meted out after a minor quarrel with a white employee. It should be noted that the punishments given to the Barbadians, while extreme, were moderate versions of those given to the native population. Where the Barbadians would be held in stocks for a few hours to a few days, a native man would be held for weeks. Where the Barbadians were suspended and beaten, a native man would be hanged until dead.

The relationship between the Barbadians and the indigenous people enslaved by the company in a sense mirrors the racial stratification in Michigan and the rest of the Amazon. Although the stratification was enforced much more violently in the Putumayo, in every region there has been a racialized underclass managed or overseen by a separate racial group(s). Previous chapters have demonstrated that in the factories of Michigan, only a portion of the workforce was paid well enough to consume commodities on an enlarged scale: the workforce was stratified along racial lines into well-paid white wage-workers and under-paid Black and immigrant wage-workers. The poorer workers still consumed commodities, but to a lesser extent, and their consumption was supplemented by informal and reproductive labor, such as family gardens. In Putumayo a stratification was enforced, but between under-paid Black wage-workers and enslaved natives. This stratification manifested through the differential application of violence, but also through differential access to family, food, and other necessities.

Like the white company men, although to a lesser extent, the Barbadian men also took part in the spoils of slavery. Casement reports that "every so-called civilised employé receives from the agent of the company, on arrival, and Indian woman to be his temporary wife."²⁶ According to Casement, "[w]ith the Barbados men it was no doubt a more or less voluntary

²⁶Casement, *Correspondence*, 16.

contract on each side," but I hesitate to speak of volition in this context. Regardless of their wishes, the women were considered "the direct gift or loan" of the company, and when a worker was reassigned to a different station, there was no guarantee that they would be able to bring their wife or children with them. One Barbadian, Joshua Dyll, evidently had 9 successive "wives" - a different indigenous woman was given to him every time he changed post.

The company's interest in this system is clear. Casement notes that "[m]ost of the twenty men I actually encountered in the services of the company were in debt; that is to say, owing to their purchases from the company's stores of things they needed from time to time, either for their own wants or those of their Indian wives and possibly children, they had been debited with amounts which often considerably exceeded the wages credited to them."²⁷ Although the contract between the Barbadians and the company stipulated that in addition to salary, the workers would receive free medicine and food, in reality the men were forced to purchase these at inflated cost from the company stores. Furthermore, once the men were given wives they became responsible also for purchasing their food and medicine, with one official claiming that "their indebtedness in these accounts was 'for medicines required for the men's Indian wives.'"²⁸ These nominal familial ties served to put the workers in the company's debt, which both disallowed the Barbados men from leaving, and cheapened their labor, as the profits from the company store far outweighed their salaries.

Thus the large majority of the salaries of the Barbadians were eaten up by debts - especially for food. Although there was some crop cultivation at the stations, it seems to have not been very extensive. Despite the long odes to the fertility of the Amazon found in the accounts of

²⁷Casement, *Correspondence*, 16.

²⁸Casement, *Correspondence*, 19.

Hardenburg and Enock, much of the land around the stations was cleared and then left empty. Casement claimed that at one station with 200 cleared acres, "scarcely 1 acre is under any form of cultivation."²⁹ As a result, "[f]ood which might easily be raised locally is brought literally from thousands of miles away at great expense, and often in insufficient quantity."³⁰ Although the terms of the contract stipulated that food would be provided to the Barbadians, in reality they often had to purchase this expensive food from the company, contributing to their great debt. Casement lists the account of Edward Crichlow, which included cassava meal, rice, biscuits, flour, condensed milk, cocoa and milk, preserved meat, sardines, salmon, lard, sugar, tea, and cod roes. All of these except the cassava, the rice, the flour and the tea were in tins. In addition to these foodstuffs were pots, pans, tobacco, soap, and other small items, along with a large quantity of cotton goods.³¹

Casement seemed baffled by the substitution of imported goods for locally grown food, and it is on some level baffling. The immediate decision to undermine local agriculture and rely on tinned food was certainly not made in the name of profit or efficiency. However, there are several possible explanations. Destroying the ability of the native workers to cultivate their own food put them more thoroughly at the mercy of the company. Starvation was a major element of the terror imposed on the Putumayans, and was crucial in the destruction of native communities. Once captured, native women and children seemed to be more prized as sexual objects - both for the enjoyment of company men and as debt-anchors for the Barbadian workers - than as

²⁹Casement, *Correspondence*, 18.

³⁰Casement, *Correspondence*, 18.

³¹Casement, *Correspondence*, 22.

agricultural workers. As for the Barbadians, reliance on food purchased from the station stores was a way for the company to claw back their wages and then some.

Food was also a central point of contention between the Barbadians and the indigenous workers. When the food provided or sold by the company was insufficient, employees "were often forced to steal food from the Indians, or to go out into the forest and the surrounding plantations and look for it and rob it."³² In some cases they had to forage their own food and survive on "a wild herb called by the Indians 'airambo'."³³ On the long marches through the jungle, "[n]o food is to be obtained ... except from the few Indians who may be dwelling in the neighborhood, and these poor people now have little enough for themselves."³⁴ Casement recounts the story of one Barbadian at a station with scarce food supplies. He applied to be transferred and "was glad to be sent to a sub-station where more food was obtainable, chiefly from the Indians, and where his duties mainly consisted in flogging these same Indians."

As enslaved people, the native Putumayans were almost entirely left out of the wage-relation and thus excluded from purchasing commodities. Their communities and existing agricultural practices had been destroyed and so they were often entirely at the mercy of their enslavers for sustenance. Although there was sometimes a nominal system of barter whereby natives traded some rubber for sundry goods from the colonists, this served only to establish a debt against the natives. This debt could never be paid off, and was merely a veneer which covered the system of violent coercion. The economic compulsion of the wage was replaced by the direct compulsion of the whip and so labor was cheapened, but at the cost of removing from

³²Casement, *Correspondence*, 12.

³³Casement, *Correspondence*, 20.

³⁴Casement, *Correspondence*, 11-12.

capital's ambit a potential market. Further, the genocidal viciousness of the slave drivers undermined both the short term and long term reproduction of labor, meaning that labor renewal was largely effected by constant slaving raids.

The system of slavery on the Putumayo destroyed the generational reproduction of native labor, and pushed the daily reproduction to a minimum. Families were separated, with men forced to collect rubber and women and children being taken by company men, enslaved and raped. For the working men, Hardenburg states that their food was "given to them once a day, at noon; it consisted of a handful of *fariña* and a tin of sardines - when there were any - for each group of four Indians, nothing more. And this was to sustain them for twenty-four hours, sixteen of which were spent at the hardest kind of labour."³⁵ On the long marches through the forest, the workers were given no food at all. When the workers inevitably fell ill, they were not given medicine, but were "exposed to the burning rays of the vertical sun and the cold rains and heavy dews of early morning until death released them from their sufferings. Then their companions carried their cold corpses ... to the river."³⁶ The renewal of the labor force depended entirely on a steady stream of newly enslaved workers from a population that was, like the rubber trees, fast dwindling.

Although the native Putumayans were not allowed to labor to meet their own needs, they were compelled to do reproductive labor for the company. In addition to collecting rubber, both men and women were forced to work to provide food and shelter for their enslavers. Casement notes that "[w]hether it was food-stuffs that were required, labour for building houses or planting cassava, sugar-cane, maize, or other produce for the needs of each station, or the collection of

³⁵Hardenburg, *The Devil's Paradise*, 180.

³⁶Hardenburg, *The Devil's Paradise*, 180.

india-rubber, the Indians had to satisfy all the demands of the so-called commercial establishment which had planted itself in their midsts."³⁷ Indeed all of the houses at the rubber stations were built by the natives:

The white settlers in the forest from the first compelled the Indians to build houses for them. The plan of the house would be the work of the white man, but the labour involved and all the materials would be supplied by the neighbouring Indian tribe or tribes he had reduced to work for him. All the houses that I visited outside the chief station of La Chorrera in which the company's agents lived, and where their goods were stored, were and are so constructed by the surrounding Indians, acting under the direct supervision of the agent and his white or half-caste employés. This labour of the Indians goes unremunerated. Not only do they build the houses and the stores for the white men, but they have to keep them in repair and supply labour for this purpose whenever called upon.³⁸

These houses were built on huge clearings of up to 200 acres, the clearing and planting of which also fell to the Putumayans.

Reproduction of Genocide

What was the purpose of this system of brutality? Contemporary accounts are shot through with attempts to make sense of something so offensive to sense and reason. It seems to go directly against the considerations operating in both Michigan factories and the rubber fields of the lower Amazon: maintaining an active and renewable labor pool, and stimulating the consumption of commodities as well as their production. Genocide is an eradication of a people, not their reproduction, and the native people enslaved by the company were not significant consumers of food commodities, as they were barely given enough food to survive. My framework, however, emphasizes not only the reproduction of people or workers, but also the reproduction of the relations of production. From this perspective, the genocide contributed not

³⁷Casement, *Correspondence*, 9.

³⁸Casement, *Correspondence*, 17.

only to short term profitability, but to the transfer of property from indigenous communities to profit-seeking enterprises, the maintenance of a racially stratified labor force, and even some small amount of commodity circulation by the Barbadian wage workers - although this commodity consumption was still much smaller in extent than in the Brazilian Amazon.

As a social system, the Putumayo genocide was relatively short lived. It might seem odd to speak of the reproduction of a system that cannibalized itself in just a few short years, but the concept of social reproduction includes not only the generational, but also the daily scale. In the Putumayo, the long-term stable reproduction of the system of rubber collection in was sacrificed in order to intensify the short term collection of rubber. This is true both of the consumption of the rubber commodity and the consumption of labor. But in other ways, the Putumayo genocide contributed to the long-term reproduction of the capitalist world-economy itself. This is clear in the way that the destruction of local food production opened up the Putumayo as a market for non-perishable goods, especially tinned meats from abroad. There are several key aspects of social reproduction to be examined here: the establishment and reproduction of property relations, and the perpetuation of the conditions of commodity production.

Land

Industrial production requires raw material inputs, and as production expands to serve the end of capitalist accumulation, the pool of available resources must expand also. This is partly a matter of finding the resources and developing the necessary infrastructure to exploit them, but it is also a matter of social construction. In order to make profit, capital must exploit natural resources as cheaply as possible: as Jason Moore writes, "[t]he condition of the rise of

capitalism, in other words, was the creation of Cheap Nature."³⁹ In the case of the Putumayo the natural world had to be made cheap, and capital employed several techniques to do so, such as violent dispossession of the native population and fraudulent accounting practices.

The first person to expose the Putumayo genocide to the English speaking world, Walter E. Hardenburg, also wrote at length about natural wealth of the Putumayo region. Hardenburg was an engineer who traveled to the Putumayo to look for business opportunities, and when he published his travel accounts in book form, they were accompanied by an introduction written by a friend and fellow engineer C. Reginald Enock. Their writings express outrage at the genocide of the indigenous Putumayans, but alongside this is a certain instrumental view of nature. In their view, the Amazon basin itself was an untamed and untapped source of wealth which only needed to be subject to the rationalizing discipline of capital. As they described the natural space of the Putumayo basin and its inhabitants, they did so as surveyors and engineers, cataloging every natural and human element in terms of their potential to produce profit. Enock writes: "In the Peruvian Montaña, in its upper regions, Nature has been lavish of her products and opportunities. The rancher who should take up his abode there, with a small amount of capital, can rapidly acquire estates and wealth.... It is sufficient to cut down and burn the brush and scratch the soil and sow any seed, to recover return of a hundred for one."⁴⁰ And a page later: "The whole Amazon Valley, when it shall have been opened up, will prove to be one of the most valuable parts of the earth's surface."⁴¹ What's odd about these quotes is that they are sandwiched between the diatribes against the "savages" of North and South America and the extended musings on the

³⁹Jason W. Moore, "The Rise of Cheap Nature," in *Anthropocene or Capitalocene? Nature, History, and the Crisis of Capitalism* (PM Press, 2016), 99.

⁴⁰Hardenburg, *The Devil's Paradise*, 36.

⁴¹Hardenburg, *The Devil's Paradise*, 37.

racial mixing of Spanish and Latin American races. It's written in a scattered way, as if Enock couldn't decide between the relative importance of accumulation and expounding his racial theories.

Hardenburg's account is somewhat more focused on profit, as his journey was planned explicitly as an expedition to find potential areas of investment. He wanted to take advantage of the rubber boom, but kept a keen eye on any potentially profitable ventures. He similarly discussed the suitability of the region for agriculture, noting that "[a] part of the valley is low and swampy, but the rest is good, rich soil, quite suitable for agricultural purposes, and covered with a thick, short grass. Although all the encircling mountains are clad with forests, the valley is, at present, cleared and ready for cultivation."⁴² As he traveled down the river, he took samples of rocks that might indicate the presence of valuable minerals.

It was rubber, of course, which stood out as the most profitable venture in the Putumayo. However, the latex producing trees were in general much more difficult to manage than those in the lower Amazon. As mentioned in the previous chapter, there are several species of latex producing plants:⁴³ *Guayule* which can grow in the US and which produces hypoallergenic rubber;⁴⁴ *Landolphia Owariensis*, an "fragile and easily killed" African vine which served as the principle source of rubber collected during the Congo Genocide;⁴⁵ *Cryptostegia*, which was native to Madagascar but was prohibitively labor intensive to tap;⁴⁶ and literally hundreds more,

⁴²Hardenburg, *The Devil's Paradise*, 55.

⁴³For a much more extensive list and analysis of the many latex producing plants, see Harold Brown, *Rubber: Its Sources, Cultivation and Preparation* (J. Murray, 1914).

⁴⁴Katrina Cornish et al., "Guayule Latex Provides a Solution for the Critical Demands of the Non-Allergenic Medical Products Market," *Agro Food Industry Hi Tech* 12, no. 6 (2001): 27–32.

⁴⁵Robert Harms, "The End of Red Rubber: A Reassessment," *The Journal of African History* 16, no. 1 (1975): 75.

⁴⁶SR Hoover et al., "Cryptostegia Leaf Rubber.," *Industrial & Engineering Chemistry* 37, no. 9 (1945): 803–9.

including the common dandelion. Among the world's latex producing plants *Hevea Brasiliensis* is by far the most productive and most widely processed, but even in its native Amazon it competed with the *Castilla* plant. In fact, *Castilla* processing predated *Hevea* among indigenous peoples, and was initially much more common with several unique advantages. The most important of these was that *Castilla* trees have a much higher short term production, as the "free flow of latex from the tapping cuts ... gave more rubber with less labor than was possible with *Hevea*."⁴⁷ Unfortunately, "they will not produce such yields if tapped more frequently than once or twice per year,"⁴⁸ and in South America it was harvested "by felling and circling the trunks,"⁴⁹ destroying the trees.

In the 19th century, *Hevea* latex was more common in the eastern Amazon, and *Castilla* in the west. However because the harvesting of *Castilla* latex often destroyed the tree, it began to die out in the wild by the end of the nineteenth century. When Hardenburg traveled through Colombia and Peru in 1907 and 1908 he was told that "the whole region of the Upper Putumayo had once abounded in *caucho negro*," the local name for the rubber produced by *Castilla* tree, "but that at the present date very little remained, owing to the fierce onslaughts of the *caucheros* many years ago."⁵⁰ Thus there had been a shift to *Hevea*, but there are several ways in which *Hevea* production in the Putumayo region was inferior to that in the lower Amazon.

In the first place, there are several species in the genus *Hevea*, not all of which are as productive as the *Hevea brasiliensis* which grows in the east. The species that grow in the

⁴⁷EM Blair and TF Ford, "Castilla as a Western Hemisphere Rubber.," *Industrial & Engineering Chemistry* 37, no. 8 (1945): 760.

⁴⁸Ernest P. Imle, "Hevea Rubber: Past and Future," *Economic Botany* 32, no. 3 (1978): 264–77, JSTOR.

⁴⁹O. F. Cook, "Rubber Production from Castilla and Hevea," *Science* 85, no. 2208 (1937), 406.

⁵⁰Hardenburg, *The Devil's Paradise*, 95.

Putumayo are *Hevea guianensis* and *Hevea benthamiana*, which are the "most productive and highest quality varieties within the category of weak rubber," but which in the rest of the Amazon are less exploited than *Hevea Brasiliensis* "due to their low production and the occasionally sticky character of their latex which makes their rubber undesirable on the market."⁵¹ From July 1901 to June 1902, 1.4 million kilograms of rubber passed through Iquitos, of which only a third was classified as high quality,⁵² compared to about half in Brazil.⁵³ During the final period of Amazonian dominance on the world rubber market, roughly 1900 to 1914, fine rubber averaged a 36% premium over lower quality rubber, peaking at more than double the value in 1910.⁵⁴

The increased difficulty of harvesting rubber in the Putumayo vis-a-vis the rest of the Amazon appears to have been a compounding factor. In describing the harvesting patterns of tappers in Bolivia, contemporary descriptions describe tappers operating one or two looping trails that stayed close to the vicinity of their domicile. The work was organized so as "to make the *estradas* [paths] each of a convenient length for one man to tap all the trees included in it, and carry the rubber milk back to the hut and cure it, in one day."⁵⁵ Tappers would alternate between trails on subsequent days and carry their rubber to their *aviadores* at the end of the week

⁵¹"La **Hevea guianensis** y su variedad **lutea** y la **Hevea benthamiana** son las tres especies y variedades más productivas y de mejor calidad entre las denominadas siringas fracas o jebes débiles, es decir, gomas de Hevea que tienden a perder mas peso que las siringas verdaderas y que, por lo tanto, tienen menor valor en el mercado. Además, son de extracción más difícil. Con las demás especies y variedades de jebes débiles sólo se realiza su explotación en casos excepcionales, debido a su producción muy baja y, en algunos casos, a las características pegajosas del látex que hace la goma indeseable en el mercado (siringaranas)." Camilo Domínguez and Augusto Gómez, *La Economía Extractiva En La Amazonia Colombiana: 1850 - 1930* (Tropenbos Colombia, 1990), 93.

⁵²Rupert J. M. Medd and Hélène Guyot, "Eyewitness Accounts during the Putumayo Rubber Boom: Manuel Antonio Mesones Muro—the 'Madman of the Marañon River,' Carlos Oyague y Calderón—the State Engineer, and Roger Casement—'Not of the Real World' Humanitarian," *Journeys* (New York, USA) 20, no. 2 (2019), 64.

⁵³Randolph R. Resor, "Rubber in Brazil: Dominance and Collapse, 1876-1945," *The Business History Review* 51, no. 3 (1977), 353

⁵⁴P. W. Barker and E. G. Holt, *Rubber Statistics 1900-1937: Production, Absorption, Stocks and Prices*, no. 181, Trade Promotion Series (U. S. Department of Commerce, 1938), 43.

⁵⁵"The Rubber Situation in Bolivia," *India Rubber World* (New York) 27, no. 1 (1902), 15.

or month. This description differs sharply from that of the Putumayo gathering expeditions, in which enslaved workers would be forced to carry at gunpoint the rubber they had gathered over months sometimes up to 70 miles over hard terrain.⁵⁶ This is indicative not only of the hard conditions of labor, but also of the more dispersed nature of rubber and the lack of infrastructure in the Putumayo which made tapping more labor intensive.

All of this indicates that if the Putumayo was going to compete with the rest of the Amazon, it would not be in the realm of quality. The Putumayo could produce rubber, but given its lower quality and consequently its lower price, to be profitable it would have to drive down the costs of production. This is evident in the methods of labor discipline employed by the PAC: the forced labor, the sub-starvation-level provisions, and the debt relation all served to depress the cost of labor and therefore the cost of producing rubber.

Recent scholarship into the corporate structure and accounting practices of the PAC has deepened our understanding of the methods used to conduct profitable business in the unique economic considerations that differentiated the region from the rest of the Amazon. Although the debt-bondage relation that characterized Amazonian wild rubber production was operative in the Putumayo, it served a somewhat different purpose. Whereas in the lower Amazon, the debt relation allowed workers some level of autonomy, as has been argued convincingly by Barbara Weinstein, in the Putumayo it served to mask the extreme levels of unfreedom that actually prevailed. This was especially important for Arana's solicitation of capital investment in Britain, as the debt relation served to add a veneer of legality to a system of enslavement that otherwise would have deterred British capital investment. As the Putumayo region was contested territory between Colombia and Peru, land deeds were scarce, and the PAC was required to establish the

⁵⁶Casement, *Correspondence*, 11-12.

legitimacy of its claim to the land it was exploiting. The debt relation and control over the indigenous population served as a crucial element in their assertion of land rights.

Stephen Walker's recent article "The Peruvian Amazon Company: An Accounting Perspective" makes the case that the debt-bondage relation is one of many accounting practices which should be "no longer perceived as a purely technical endeavor that is isolated from the organizational and social context in which it is practiced. Accounting information is now comprehended as socially constructed, as value laden, and the production of accounts is recognized as a highly political process."⁵⁷ There were several anomalies in the PAC's accounting practices which contributed both to the extension of control over the native population, and to disguising the violent nature of this control. Among these anomalies were the valuation of land not based on ownership rights but on access to the land's inhabitants, capital outlays for *Gastos de Conquistacion* (expenses of conquest), and the charging of arbitrary debts to the Indigenous workers and treating these debts as company assets.

As Hardenburg had witnessed firsthand, the PAC came into possession of many of its estates through the "systematic persecution of ... Colombian establishments ... with the object of making them abandon the region and then taking possession of their properties."⁵⁸ Casement confirms this point, but notes that the Colombians had also been settlers who exploited the native population, and he frequently mentions them as parties to raids and massacres of the indigenous people. The Colombians' purpose in dealing with any Indigenous community was the same as the PAC: "enslaving that tribe and forcing it to work rubber for them."⁵⁹ Although "[i]n some cases

⁵⁷Stephen P. Walker, "The Peruvian Amazon Company: An Accounting Perspective," *Enterprise & Society* 25, no. 1 (2024), 282.

⁵⁸Hardenburg, *The Devil's Paradise*, 201.

⁵⁹Casement, *Correspondence*, 11.

these Colombian settlers appear to have held concessions from their government," in the contested region of the Putumayo the legality of land titles was shaky. At any rate by the time the estates passed into the hands of the Peruvians, the majority of their estates did not have legal titles, and "[t]he PAC's chairman conceded that the company held only squatter's rights."⁶⁰ The PAC's rights to the estates had been asserted by extra-legal means and persisted through the overwhelming use of force against anyone who might challenge them - native or Colombian. Far more important than a legal claim to title over the Putumayo was the ability to effectively exploit the land's inhabitants. Whereas land assets are often tallied on a per-acre basis, "PAC accounting practices suggested that valuation was primarily understood to be a function of the presence of exploitable labor."⁶¹ In a rain forest spanning half a continent that was full of rubber, it was primarily the presence of ready workers that would not have to be imported that made the Putumayo unique.

Workers would not have to be imported, but the native Putumayans did not accept enslavement quietly. In fact, their resistance cost the company significantly. In order to subdue the indigenous population, the PAC brought in Black Barbadian migrants to serve as overseers, and laid out significant capital on gastos de conquistacion. These armaments, though not representing a large portion of the company's assets, were significant enough to appear on a 1909 financial audit of the PAC, where they were lumped into a category of "office furniture, armaments, and moveable plant."⁶² The head of the PAC, J.C. Arana, argued that "conquest referred to attracting Indians, a process of converting uncivilized people to a relationship of

⁶⁰Walker, "The Peruvian Amazon Company: An Accounting Perspective," 289.

⁶¹Walker, "The Peruvian Amazon Company: An Accounting Perspective," 292.

⁶²Walker, "The Peruvian Amazon Company: An Accounting Perspective," 295.

exchange," but a later investigative committee "affirmed that the *Gastos de Conquistacion* represented the cost of the manhunts and slave raids, which 'were regarded as preliminary expenses' of starting a section for collecting rubber."⁶³

The appearance of a large amount of munitions on a list of company assets offers us a glimpse into the violence that underpinned the operations of the PAC. It is a sort of economic residue of non-economic compulsion, and it demonstrates a tension at the heart of the wild rubber industry. In order to cheaply extract rubber from the Putumayo region, the company employed genocidal violence against the local population. But in order to participate in the global economy, all assets had to be accounted for and framed as legitimate - and so the guns were lumped in with office furniture. The contradiction between the violence of the gun and the quieter compulsion of the balance sheet is a fundamental dynamic of global capitalism that plays out in the subjugation of peripheral regions and the transformation of people into workers.

Labor Discipline

Labor scarcity was reported to be an additional pressure on rubber production in the Putumayo, but early diagnoses of labor scarcity in the Putumayo are contested. A contemporary newspaper article mentioned that as "labour and women are both scarce, and as there is a strong demand for one and the other, bands of armed men are constantly organised for sudden descents upon groups or communities of the savages,"⁶⁴ but more recent analysis indicates that, at least relative to the lower Amazon, the Putumayo had a "a strategic advantage constituted by the large Indian population that had not been decimated by the advances of slave traffickers and Luso-

⁶³Walker, "The Peruvian Amazon Company: An Accounting Perspective," 294.

⁶⁴Article from *El Comercio*, 1906 as cited in Hardenburg, *The Devil's Paradise*, 24.

Brazilian settlers of the seventeenth and eighteenth centuries."⁶⁵ Insofar as labor was imported, it was not employed in rubber gathering, but in overseeing and disciplining enslaved laborers. Indeed the brutality of the Putumayo Genocide cannot be entirely explained either as a response to difficult conditions of production or the scarcity of labor.

Casement makes the argument in his report that the PAC terrorized and enslaved the Huitoto people as a method of extracting their labor: "The object of the "civilised" intruders, in the first instance, was not to annihilate the Indians, but to "conquistar," *i.e.*, to subjugate them, and put them to what was termed civilised, or at any rate profitable, occupation to their subduers."⁶⁶ The Barbadians were brought in to offset some of the dangers and difficulties of the slaving raids, and overseeing the native workers on their long marches through the forest. Due to the expense of transporting the rubber from the Putumayo to market, the PAC combined debt peonage with extreme violence to maintain a docile and compliant labor force. However, there are several issues with a purely economic explanation. As Michael Taussig claims, the view that torture was employed for the purposes of labor discipline, while not incorrect, can only be a partial explanation, as "terror and torture do not derive only from market pressure (which we can regard here as a trigger) but also from the process of cultural construction of evil as well."⁶⁷ Indeed, the violence employed by the PAC was not only directed at the native workers, but at their entire communities and even the salaried Barbadian employees. More significantly, the brutality of the PAC undermined the workforce by starving, maiming, and ultimately slaughtering thousands of would-be rubber tappers.

⁶⁵Serje, "The Peruvian Amazon Co.: Credit and Debt in the Putumayo 'Wild Rubber' Business," 483.

⁶⁶Casement, *Correspondence*, 10.

⁶⁷Michael Taussig, "Culture of Terror--Space of Death. Roger Casement's Putumayo Report and the Explanation of Torture," *Comparative Studies in Society and History* 26, no. 3 (1984), 479.

Given the apparent insensibility of the genocide from an economic standpoint, it is easy to see why scholars have considered it a pre-capitalist relic. Seen within the proper context, however, it is clear that the brutality of the PAC was an element of the ongoing process of racial stratification that is a constituent component of the capitalist world. The horror of the Putumayo genocide can here be placed within the much longer project of what Michael Taussig calls the "construction of colonial reality that occurred in the New World... where the Indian and the African became subject to an initially far smaller number of Christians."⁶⁸ The brutality of the PAC's methods were crucial in the elaboration, not only of a new kind of man, but of new kinds of family and a new kind of society built on a rigid racial hierarchy. Terror played a fundamental role in this elaboration: terror "which as well as being a physiological state is also a social fact and a cultural construction whose baroque dimensions allow it to serve as the mediator *par excellence* of colonial hegemony."⁶⁹

The actions of the Peruvian Amazon Company can be seen as a continuation of colonial policy against indigenous peoples in the Americas. The discussion of how to frame their actions is in a sense a continuation of the Valladolid debate: an attempt by European conquering powers to, as Jason Moore says "justify and enable the profit-driven conquest, appropriation, and exploitation of humans and the rest of nature so as to sustain the endless accumulation of capital."⁷⁰ Part of the justification came from framing resistance by the indigenous Putumayans as evidence of their brutality and backwardness: Taussig notes that propaganda of the "savagery

⁶⁸Taussig, "Culture of Terror," 468.

⁶⁹Taussig, "Culture of Terror," 468.

⁷⁰Jason W. Moore, "Our Capitalogenic World: Climate Crises, Class Politics, and the Civilizing Project," *Annales Universitatis Paedagogicae Cracoviensis. Studia Poetica* 11 (2023), 98.

of the wild Indians occupied a key role in the propaganda of the rubber company."⁷¹ The genocide was indeed a Civilising Project by a "civilising company" in that it was an attempt to impose a civilizational order that not only extracted cheap labor from its racialized subjects, but that actively devalued both indigenous populations and their ways of life, as well as the natural world.

In his introductory remarks to the republication of Peruvian Judge Valcarcel's *El Proceso del Putumayo*, Alberto Chirif outlines a long lineage of resistance by Indigenous Amazonians to European colonization. He identifies three successive periods in which different justifications were claimed for the domination of the indigenous Americans.⁷² Initially, the indigenous peoples were subjugated and enslaved by Europeans on the grounds that they were no human, but after the papal bull of 1537 which declared their humanity, the justification shifted. The natives peoples of the Americas were now considered human but atheists, and the west was justified in taking responsibility for their salvation. Finally, with the development of secularism in the 19th century, the theory of evolution was taken up by the advocates of western civilization who framed indigenous peoples as savage enemies of civilization. However, as Chirif notes, each of these successive justifications for conquest were in reality cumulative - they built on each other to further the civilizing project.⁷³ In the development of this logic, any action that the indigenous peoples took was framed negatively. If they fought back they were considered as either subhuman or heathen savages, and if they didn't then they had "weakness of character and

⁷¹Taussig, "Culture of Terror," 484.

⁷²Carlos A. Valcárcel, *El Proceso Del Putumayo y Sus Secretos Inauditos* (CETA, 2004), 28.

⁷³Valcárcel, *El Proceso Del Putumayo y Sus Secretos Inauditos*, 28: "Las tres imágenes históricas elaboradas por los occidentales para justificar su dominio frente a los indígenas (no-humanos, ateos y salvajes), en realidad no son sucesivas sino acumulativas."

docility of temperament." Moments of resistance especially were incorporated into the civilizing narrative as justifying the brutal methods of subjugation.

In the case of Peru, the state was interested in incorporating the resource-rich Amazon into the national economy, as it was being under-utilized by its inhabitants.⁷⁴ In the 1860s, Peru began to assert its presence in the rain forest, and in 1866 sent a steamship along the Putumayo to determine its navigability. In August the steamship was attacked by the region's inhabitants, and two men were killed. In response, a second team was sent out with the express purpose of demonstrating the superiority of the Peruvians. This demonstration took the form of a massacre: an unrecorded number of dead men, along with three women and fourteen children taken prisoner. The place was renamed by the Peruvians the "port of Punishment."⁷⁵

This pattern of indigenous resistance to Peruvian incursion being met with overwhelming violence was continued in the Putumayo. As noted previously, the first task of the Barbadians after they arrived in the Putumayo was to conduct a series of retaliatory raids on the Andokes people. These raids were characterized in two ways - on the one hand they were intended to "open up what were styled trade relations" with the Andokes people, but at the same time they were explicitly "termed punitive expeditions," payback for the theft of rifles and the killing of civilized Colombians.⁷⁶ The resistance of the Andokes people was used as justification for slaving expeditions: the phrase "trade relations" was a euphemism for the attempt to "capture Indians and compel them to come in and work for Señor Sanchez."⁷⁷ The primary norm of civilized society, trade, was synonymous with the enslavement of the uncivilized.

⁷⁴Valcárcel, *El Proceso Del Putumayo y Sus Secretos Inauditos*, 23.

⁷⁵Valcárcel, *El Proceso Del Putumayo y Sus Secretos Inauditos*, 21-22.

⁷⁶Casement, *Correspondence*, 11.

⁷⁷Casement, *Correspondence*, 11.

Like the Andokes people, the Huitoto also attempted to steal rifles from their enslavers. Casement's report narrates the story of an unnamed 'capitán' of the Huitoto who was being held prisoner at the Matanzas station. After another native worker managed to multiple times escape capture, this capitán sent his son on an errand to bring him a rifle that he had secreted away in his house in the forest. The son was caught smuggling the rifle into the station and was hanged. The father was subsequently also hanged, but not before being beaten with machetes. One of the Barbadian workers, an Augustus Walcott, expressed some mild sympathy with the natives, remarking that "this was not the way to punish people" and that "it was a brutal act."⁷⁸ Walcott was threatened with punishment for making this comment, and for an evidently unrelated incident was himself later beaten with machetes and hanged - but by his hands rather than his neck, and so he survived.

This expression of sympathy between the Barbadians and the Putumayo natives was not isolated, but such expressions were overwhelmed by conflict between them. Of course the Barbadians were mostly employed in violently forcing the natives to work, and the fact that they seem to have been compelled to do so by the violence visited on their own heads does not negate their brutality towards the native workers. They were therefore occasionally the targets of acts of resistance. Although rifles were not easy to come by for the native peoples, they had other methods of resisting enslavement and death. Some simply ran away and evaded capture, but they also enacted positive means of self defense, attacking and wounding the company's employees. Casement notes that the Andokes and Boras peoples were sometimes successful in killing their would-be capturers, and he says that one Barbadian man, Clifford Quintin, had been wounded in one of his feet when they met. "This wound has been caused by a splinter of wood, probably

⁷⁸Casement, *Correspondence*, 14.

poisoned, placed in the ground by the Indians in order to obstruct the approach of their dwelling places, in the hope of thereby hindering the armed raids made upon them to compel them to work rubber."⁷⁹ The employment of such means of defense around their homes was not merely retaliatory violence, but was quite literally a defense of their homes, and communities - their patterns of life.

But the patterns of life among the native communities were precisely the target of colonial violence - it was the 'savagery' that would be wiped out by civilization. Taussig notes that the practices and culture of the native Putumayans were interpreted very differently by the Peruvians and Casement. Casement emphasizes the irrationality of fearing the Putumayans, indicating that their communities were fractured and they were both too poorly armed and too docile to provide much resistance. But among Peruvian and Colombian settlers, the natives were seen as almost mythically dangerous. When the Colombian slavers were killed by the people they tried to enslave, Casement's informant says that the only reason they did not eat them was "that Indians 'had a repugnance to eating white men, whom they hated too much.'"⁸⁰ Even the lack of cannibalism in this instance was somehow twisted into evidence of savagery.

Thus resistance to colonial expansion was turned into fodder for the intensification of colonial activity. It was portrayed as uncivilized 'savagery' or '*malas acciones*' and used to justify a far greater degree of brutality. This pattern, in which resistance to colonial violence is metabolized by the colonial beast and revisited on the indigenous population, is accompanied by a great mythmaking. Taussig notes two related themes around which these myths coalesce: "the

⁷⁹Casement, *Correspondence*, 13.

⁸⁰Taussig, "Culture of Terror," 489; and Casement, *Correspondence*, 30.

horrors of the jungle, and the horrors of savagery."⁸¹ The Amazon jungle was a place of horrors which degrades those who enter, and the people who have made it their home are, naturally horrific themselves. In the context of Peru's project of civilizing the jungle, the native population took on a "powerfully ambiguous image, a seesawing, bifocalized, and hazy composite of the animal and the human. In their human or human-like form, the wild Indians could all the better reflect back to the colonists the vast and baroque projections of human wildness that the colonists needed to establish their reality as civilized (not to mention business-like) people."⁸² In their hybrid identity, the native peoples of the Amazon served a dual purpose for the rubber companies: they were both the engines of civilization as laborers, and the object of civilization as savages in want of pacification.

For the companies and state actors engaged in the project of civilizing the jungle, the acts of resistance by the indigenous Amazonians were crucial, in that they provided justification for the project itself. The fact of their pacification and employment as laborers meanwhile - which should be considered not a natural state *à la* Casement, but a state created and enforced by unimaginable violence - provided proof of the project's success. Both the resistance of the Putumayo peoples and the violent reprisal were elements in the project of genocide-as-accumulation. They figured into its creation as well as into its reproduction and maintenance.

Finally, it is important to note that even the liberal humanism which opposed the genocide, such as expressed in Hardenburg's travelogue, was still fully committed to the construction of colonial hegemony. This comes through clearly in certain passages both of his book, and in the introduction to his book written by a friend and fellow engineer Charles R.

⁸¹Taussig, "Culture of Terror," 482.

⁸²Taussig, "Culture of Terror," 483-4.

Enock. Both authors object to the brutality of the PAC's actions against the indigenous Putumayans, but remain committed to a racial hierarchy that positions non white people as in need of the edifying power of labor discipline. Enock in particular views peoples of the Putumayo as "worthy of preservation" in so far as they are docile and free from immorality. He projects an explicit framework of racial classification in which some peoples are more valued than others. He initially rejects the prevailing racialization of the indigenous Putumayans, saying that "[t]hese people, although known as *infieles* and *salvages* - that is, 'un-faithed' and 'savage' - cannot be described as savages in the ordinary sense of the term. They have nothing in common with the bloody savages of Africa and other parts of the world."⁸³

Even Enock's praise is built on the devaluation of other races, but faint as it is he cannot hold onto it for long. A mere two pages later he attempts to describe the various peoples in Peru, saying "[t]he highland and coast people were those who formed the population under the Inca government, and under whose control they had reached a high degree of aboriginal civilisation; whilst the indigenes of the forests were more or less roving bands of savages, dwelling on the river banks, without other forms of government than that of the *curacas*, or petty chiefs of families or tribe." As evidence of the virtue of the people in the coastal region, Enock says they live under an intelligible structure, in which "every village contains its *iglesia* and village priest."⁸⁴ In perhaps the most bizarre expression of his racial calculus, Enock develops an explanation of why the Peruvian 'Christians' would commit such terrible crimes, and why the English stockholders should to some extent be exudes. To quote somewhat at length:

Apart from topographical considerations, the sinister occurrences on the Putumayo are, to some extent, the result of a sinister human element - the Spanish and

⁸³Hardenburg, *The Devil's Paradise*, 16.

⁸⁴Hardenburg, *The Devil's Paradise*, 19.

Portuguese character. The remarkable trait of callousness to human suffering which the Iberian people of Portugal and Spain - themselves a mixture of Moor, Goth, Semite, Vandal, and other peoples - introduced into the Latin American race is here shown in its intensity, and is augmented by a further Spanish quality. The Spaniard often regards the Indians as *animals*. Other European people may have abused the Indians of America, but none have that peculiar Spanish attitude towards them of frankly considering them as non-human.[...]There is yet a further trait of the Latin American which to the Anglo-Saxon mind is almost inexplicable. This is the pleasure in the torture of the Indian as a *diversion*, not merely as a vengeance or 'punishment.'⁸⁵

The explanation? That the Peruvians carry bad Iberian blood, blood of a lower class. Enock's analysis exists within a rigid racial hierarchy - even his criticism of ethnic cleansing reinforces colonial hierarchy. This is a criticism not *of*, but *internal to* the construction of colonial hegemony.

Both the genocide itself and the critical viewpoint embodied by Hardenburg and Enock shared a common goal: the pacification of the indigenous population and their transformation into a docile class of workers, and the production of rubber for the world market. Although Enock and Hardenburg balked at the violence of the genocide, the worldview they shared is precisely that which created it. The genocide cannot be reduced either to an attempt at labor discipline nor to simple bloodlust, because they cannot be extricated. The brutality of the individual perpetrators of the genocide may have been motivated by a capricious love for violence, but it took place in and was conditioned by the context of accumulation. The extermination of the Putumayo peoples was instrumentalized by the forces of capital as a crucial element of the world-system.

⁸⁵Hardenburg, *The Devil's Paradise*, 37.

Conclusion

By the time Casement had arrived in Peru, both the population of the indigenous residents of the Putumayo and the rubber output of the region had fallen considerably: from about 50,000 inhabitants in 1906 to "not more than 8,000 in 1911," and from 645,000 kilograms of rubber to 236,000 in the same period. Despite this, Casement's notes that the individuals exploiting the Putumayo seemed unconcerned: "The disappearance of the London company as an active factor restricts the necessities of profit finding to a narrower field. While it needed a heavy output of hundreds of tons of rubber in 1906 upon which to found an appeal to British investors or speculators in 1907 ... a diminished yield of only 200 tons will still maintain many families and individuals on the spot."⁸⁶

According to the correspondence included in Casement's consular report, by July of 1911 the Peruvian government had issued more than 200 arrest warrants for the perpetrators of the Putumayo genocide. However, "[m]any [had] escaped to Brazil. Only accessories remain[ed] in the territory."⁸⁷ Many of the warrants were voided by Peruvian courts or tied up in bureaucratic technicalities in an attempt to bury the matter.⁸⁸ In 1913 the Peruvian Amazon Company was dissolved, but Arana maintained his estates and continued exploiting their rubber and residents.⁸⁹ Over 6,000 inhabitants of the Putumayo - evidently most of the remaining residents, if Casement's numbers were correct - were forced to leave their home territory in the late 1920s by

⁸⁶Casement, *Correspondence*, 157.

⁸⁷Casement, *Correspondence*, 147.

⁸⁸Pilar Jordán, "A Propósito de Redes Sociales, Económicas y Políticas En La Iquitos de Inicios Del Siglo XX," *Boletín Americanista*, ISSN 0520-4100, N^o. 56, 2006, Pags. 103-118, January 2006, 112.

⁸⁹J. Goodman, *The Devil and Mr. Casement: One Man's Battle for Human Rights in South America's Heart of Darkness* (Farrar, Straus and Giroux, 2010), 239.

two of Arana's managers. Half of them died on the march.⁹⁰ In the 1920s Arana became a Peruvian senator. In 1927 the territorial conflict over the Putumayo was finally settled between Peru and Colombia in favor of Colombia. Arana resisted the treaty, but managed to ensure that he would be compensated for his property. A Colombian bank bought Arana's rights, but he died before receiving the full payment. In the 1980s, a Colombian bank paid his heirs \$160,000.⁹¹ Taussig approvingly cites Edmund Morel's claim that the intervention spearheaded by Casement was one of the "only two [occasions] in which British diplomacy rose above the commonplace" and helped bring an end to the genocide. I'm not so sure. The rubber genocide in the Putumayo did not end because of external pressure, but because Arana ran out of people to kill.

The genocide cannot be reduced to its economic dimension. Taussig refers to market pressure as a 'trigger' - not the only cause of the genocide, just its immediate context. But although the market may not have demanded the specific torture of the Huitoto people, it did demand a radical reconfiguration of life in the Putumayo. Meeting the market demand for rubber in previously unimagined quantities required a logic of extraction to be imposed on the land, and labor discipline to be imposed on the workers. New forms of social and capital reproduction developed in response to the rise of industry in the Global North - both its need for raw material inputs, and its need for an increased market for consumer goods. The structure of the global market made way for this reconfiguration to be borne out by especially bloodthirsty actors in the remote region of the Putumayo.

The atrocities discussed here are more violent and more shocking than the debt bondage of the lower Amazon and the racial stratification in Detroit Michigan, but they are part of the

⁹⁰J.V.S. Román et al., *Perfiles Históricos de La Amazonía Peruana* (Centro de Estudios Teológicos de la Amazonía, 1994), 192-3.

⁹¹Echeverri, "The Putumayo Indians and the Rubber Boom," 17.

same system. In each location, capital operated within the prevailing political, social, and natural constraints to establish a compliant workforce that could cheaply produce automobiles or their constituent raw materials. The following chapter will look at another such location: South and Southeast Asia, where rubber was first grown productively on plantations. In sparsely populated Colonial Malaya, British owned and run plantations imported workers from Southern India. In doing so they created and maintained new forms of racial stratification and labor discipline, transforming the local patterns of social reproduction toward the production of profit.

Chapter 5: Plantation Rubber and Working Families in British Malaya

The rise of the automotive industry in the early twentieth century created a demand for rubber that could not be met by the Amazon. In 1913 Asian rubber plantations surpassed Amazonian production for the first time, and by the next year accounted for over half of all global natural rubber production. This was not so much due to a sudden fall in Amazonian rubber but to an explosion of production on Southeast Asian plantations that would soon make peak Amazonian production look like little more than a rounding error. In 1912, the Amazon reached its peak production of 41,619 long tons, while Southeast Asia produced a little over 33 thousand. Just over 20 years later, in 1934, Amazonian production had fallen to less than 9,000 tons while Southeast Asia produced over a million: more than 98% of global production that year.¹ How did a sparsely populated region thousands of miles from the native home of *Hevea Brasiliensis* come to so thoroughly dominate its cultivation? This chapter will explore the development of South Asian rubber in plantations of British Malaya, which accounted for between 40 and 50 percent of global production in the 1920s and 1930s. Although the Dutch East Indies were also major producers of plantation rubber in the twentieth century, they lagged behind Malaya for many years. This chapter will focus on Malaya over Dutch East Indies for this reason, and in order to spend more time on its unique system of labor recruitment and discipline.

¹P. W. Barker and E. G. Holt, "Rubber Statistics 1900-1937: Production, Absorption, Stocks and Prices," Trade Promotion Series (Washington: U. S. Department of Commerce, 1938).

The key to the rise of Asian rubber was its plantation system. South Asia imported rubber plants that had been carefully cultivated in botanical gardens, and that were free of the South American Leaf Blight. This, combined with a favorable climate, an experienced class of planters and the consolidation of large plantations created the conditions for the rapid expansion of rubber cultivation, but the dominant factor was a massive system of exploitation of immigrant workers. Between 1840 and 1940, approximately 4 million people, traveled from India to Malaya.² Most of this migration happened after rubber plantations were established at the turn of the century - on average almost 100,000 Indians migrated annually by the end of the 1910s.³ Initially migrant workers were indentured - unfree for the length of their contract - but due to international pressure, indenture for Indian workers in Malaya was abolished in 1910. These workers, without whom the rapid expansion of rubber production would not have been possible, suddenly found themselves in the middle of a global struggle over the rights of workers and the status of free labor. Indian nationalists and international trade unionists insisted on legal rights of free labor, and so planters and colonial authorities were forced to try and resolve the contradiction between free labor and their desire to compel migrants to work under brutal, but profitable, conditions.

This contradiction was resolved, for a time, through the exploitation of family and community bonds via what was called the *kangany* system. This article will explore how family connections among workers, and between workers and overseers, were used to create a compliant workforce, while maintaining the legal fiction of free labor. I draw on colonial reports of the administrators of British Malaya, as well as the opinions of anti-indenture activists expressed in newspapers and trade union publications to demonstrate how the category of the

²Amrith, *Crossing the Bay of Bengal*, 104.

³Colonial Office Great Britain, *Annual Report on the Social and Economic Progress of the People of the Straits Settlements* (H.M. Stationary Office, 1918).

free wage laborer was transformed in the context of South Asian rubber plantations by the exploitation of family relations, and how this propped up the global rubber industry.

The Arrival of Rubber in British Malaya

Rubber is a naturally occurring compound that can be found in thousands of plants around the globe, such as the common dandelion.⁴ Very few of these plants, however, produce rubber in commercially viable quantities or forms. The vast majority of naturally produced rubber is collected from the *Hevea Brasiliensis* plant, which grows natively in the Amazon. As early as the 1870s, seeds of the rubber plant were smuggled out of South America to Great Britain where they were studied at Kew Gardens.⁵ The imperial countries of the West knew that wild-collected rubber from South America and Africa would not be able to meet industry's rising demand, and so they began to experiment with rubber plantations in other parts of the world - most prominently Malaya. This section traces how, over the course of the last decades of the nineteenth century and the first half of the twentieth, Malaya was transformed from a predominantly peasant society into a highly specialized producer of plantation rubber.

In the nineteenth century, European states established a system of botanical research as a component of imperial governance. The system was centered on botanical gardens, such as Kew Gardens near London, which collected and studied specimens from around the world. The purpose of this was not merely the advancement of science, as "[i]n 1879, Kew acquired the vegetable collections of the India Museum and established a museum of economic botany with samples and illustrations of every possible use for vegetable products."⁶ Daniel Headricks notes

⁴John Tully, *The Devils Milk* (NYU Press, 2011), 21.

⁵Tully, *The Devils Milk*, 185.

⁶Daniel R. Headrick, *The Tentacles of Progress: Technology Transfer in the Age of Imperialism, 1850-1940* (Oxford University Press, 1988), 213.

that Kew soon became an imperial institution, and "[b]y the turn of the century, alumni of Kew were serving as directors of botanical gardens, forest departments, and agricultural research stations throughout the empire."⁷ Kew was instrumental in researching crops and establishing the colonial plantations that dominated the "fertile, well-watered, but vacant lands, such as existed in Ceylon, Malaya, Sumatra, Assam, and Indochina."⁸

Vacant is not the right word as each of these places was populated by indigenous peoples - if sparsely in comparison to today. It is true, however, that the new plantations required more labor than could be extracted from the local population. Ceylon and Malaya especially required the large scale importation of migrant laborers to run the coffee (Ceylon) and rubber (Malaya) plantations and extract the tin (also Malaya) that were so important to the British Empire. The Empire tried to put a positive spin on this, downplaying the human cost of these developments. In the 1926 *Handbook to British Malaya*, prepared by the colonial apparatus of the Malayan Civil Service, the contention is made that "[t]rade, and not imperialistic aspiration, was the motive which brought the British to Malaya; the Spaniards and the Portuguese were the missionaries and conquerors." Although the trading posts and factories were to be transformed into forts, "[t]he consideration with which the natives were treated when our trade depended on their goodwill was continued as a matter of principle when the trading station became a colony."⁹ This description would surely come as a surprise to the predominantly Indian laborers on Malayan rubber estates whose exploitation at the hands of the British attracted the ire of groups ranging from local social activists to nationalists and reformers in India to communists in Britain

⁷Headrick, *The Tentacles of Progress*, 214.

⁸Headrick, *The Tentacles of Progress*, 210-211.

⁹"Handbook to British Malaya" (London: Malay States Information Agency, 1926), 40.

itself. It does, however, indicate a desire on the part of the British colonizers to present at least the image of a free and equal partnership between themselves and the peoples they colonized in Malaya.

The first British interests in Malaya were the Straits Settlements: Singapore, Penang, Malacca, and Dinding. These were primarily port cities, but they included some rural holdings that would become important agricultural producers. Throughout the British occupation of Malaya, the Straits Settlements served as the primary points of dis/embarkation for laborers coming to find work, or leaving for home at the end of their contracts. They were governed as part of British India until 1867 when they were constituted as a separate colony. At the end of the nineteenth century, the four Federated Malay States, Perak, Selangor, Negri Sembilan and Pahang, came under the administration of the British, and over the next few years, the Unfederated Malay States of Johor, Kedah, Kelantan, Perlis, and Terengganu also accepted British 'advisors'.

During this period of consolidation, the primary economic activity was tin mining, largely run on Chinese labor and backed by Chinese capital. Historian Jomo K. S. notes that tin production just from the state of Perak "rose from 650 tons in 1874 to 18,810 tons in 1893, while the other States produced 18,930 tons."¹⁰ This precipitous increase in production, driven by the development of British tin-plating, presented a problem for the largely peasant population of Malaya: "As there was no Malay proletariat who could be set to work on the mines, the existing Malay rulers agreed to arrangements from which they obtained large commissions from Chinese miners."¹¹ By 1900, the Chinese population of the Federated Malay States was about 300,000,

¹⁰Jomo Kwame Sundaram, *A Question of Class: Capital, the State, and Uneven Development in Malaya* (Singapore: Oxford University Press, 1986), 159.

¹¹Jomo Kwame Sundaram, *A Question of Class*, 157.

about 80% of which was employed in tin mines. This pattern of importing foreign workers to fulfill a labor demand that the indigenous population would not or could not meet would continue with the subsequent rise of Malayan agriculture.

Despite the dominance of tin, British authorities tried to encourage agricultural production. J Norman Parmer quotes the Secretary of State for the Colonies as writing in 1895 that the development of Malayan agriculture was the "point of greatest importance" if the Malay States were to remain profitable "when their mineral resources in the course of years show signs of depletion."¹² Parmer notes that miners "felt strongly that the government was being biased and ungrateful. They correctly pointed out that the planters were being subsidized with the revenues which the mines produced."¹³ The most important crop throughout most of the nineteenth century was sugar which was planted in estates that were, like the tin mines, mostly owned by Chinese capital. According to Parmer, the British were at first reluctant to invest capital in Malayan plantations until "a leaf disease and bad times"¹⁴ ruined coffee crops in Ceylon, forcing planters to look elsewhere to grow the crop. Coffee moved to Malay in the 1880s, and "[g]ood coffee prices in 1894-1895 furnished an impetus to the extension of coffee planting."¹⁵ By the 1896, there were 47,000 acres in over 70 coffee plantations backed by British capital in Malaya, rivaling the economic output of the Chinese sugar estates. Almost immediately, however, the coffee price fell, and planters began to look again for alternative crops to maintain the profitability of their plantations. The most profitable of these was to be rubber.

¹²Joseph Chamberlain quoted in J. Norman Parmer, *Colonial Labor Policy and Administration: A History of Labor in the Rubber Plantation Industry in Malaya*, Monographs of the Association for Asian Studies, IX (Locust Valley, NY: J. J. Augustin Incorporated Publisher, 1960), 6.

¹³Parmer, *Colonial Labor Policy*, 6.

¹⁴Parmer, *Colonial Labor Policy*, 7.

¹⁵Parmer, *Colonial Labor Policy*, 7.

Small rubber plantations had been established by the British in Assam in 1873, although they were unsuccessful and failed within a few years. Despite the cutting edge botanical research at Kew, transfers were a risky business. In 1876, Henry Wickham famously smuggled 70,000 rubber seeds out of Brazil, but only 2,700 germinated. Of these 2,000 made it to Ceylon, where they were distributed among other research stations in Asia. Most of these transfers also came to naught, but "[t]he one transfer that mattered was the shipment of twenty-two seedlings from Ceylon to Singapore in 1877. Nine were planted in the garden of Sir Hugh Low, the British resident at Kuala Kangsar in Perak, while the rest remained in the Singapore Botanic Garden. These twenty-two seedlings are the ancestors of almost all the rubber trees in Southeast Asia today."¹⁶ The Dutch East Indies would later become rubber power in their own right, but they lagged behind Malaya for many decades, not catching up in production until the late 1920s.¹⁷

For many years after its first planting, rubber remained a marginal crop in Malaya, mostly grown in state research gardens. It wasn't until the collapse of coffee prices in the 1890s that Malayan planters began to plant it in earnest. Rising rubber prices driven by the automotive industry's increasing demand for rubber as well as governmental incentives encouraged planters to convert more and more of their plantation acreage to rubber. The labor force on rubber estates grew to over 100,000 workers by 1910, a need that could be met neither by Chinese immigrants who were mostly employed in the still-productive tin mines, nor by the indigenous Malayan population. The next section will focus on the consolidation of smaller holdings into large rubber plantations, and the recruitment of Indian labor to work them.

¹⁶Headrick, *The Tentacles of Progress*, 244.

¹⁷Headrick, *The Tentacles of Progress*, 247.

The Recruitment of Foreign Workers to Malayan Estates

Prior to the rubber boom of 1910, the majority of the small amount of rubber produced in Malaya was grown by planters who owned their own farms. They had begun planting rubber alongside other crops in the 1890s to ward against falling coffee prices, but soon began to convert most of their acreage to rubber. The boom of 1910 saw an influx of foreign capital, and many local planters began to sell their farms to European planters. Periodic slumps in the 1920s and 1930s squeezed the remaining small producers and furthered the consolidation of large rubber estates in the hands of European owners. By 1940, 2.1 million acres of rubber had been planted in estates ranging from 100 acres to over 10,000. Of this 2.1 million acres, 75% was owned by European capital, and a 20% by Chinese and Indian planters. In addition to this, a further 1.3 million acres of rubber was planted on small and medium sized farms of less than 100 acres each.¹⁸

The labor required to cultivate this crop was massive, though the plantation labor force fluctuated over the decades. From 58,073 laborers in 1907, the employment on large estates rose to the twentieth century high of 332,300 in 1946. In the interim there were local peaks of 237,128 and 258,780 in 1919 and 1929 respectively, each of which was followed by a dramatic crash in which the labor pool contracted by a third to a half (down to 156,341 in 1921 and 123,924 in 1933).¹⁹ The growth and contraction of plantation labor was facilitated by large scale migration from other regions of Asia. Even before rubber cultivation, the labor demand on Malayan plantations was too great to be satisfied by the local Malayan population. Although migrants and slaves had been traveling to Malaya for centuries - Sunil Amrith states that the Bay

¹⁸Parmer, *Colonial Labor Policy*, 8-9.

¹⁹From a table compiled in Jomo Kwame Sundaram, *A Question of Class*, 181.

of Bengal was a "maritime highway between India and China"²⁰ - the demands of rubber production transformed the patterns of human mobility across the bay. As labor requirements and migration increased, largely comprising Tamil workers from Madras, so too did the administrative apparatus that governed migration. Over the course of the early twentieth century, the governments of India and Malaya, Malayan small-holders and European planters all formed various associations and committees to try to direct the flow of workers to their own benefit. Through their negotiations, a complex set of labor and citizenship categories were developed. Amrith says that the increased governance of Tamil migration "linked the two coasts of the Bay of Bengal more closely together, and at the same time introduced new distinctions between different kinds of migration - free and unfree; 'assisted' and 'unassisted'; Chinese and Tamil - and between migrants and indigenes in Malaya."²¹

In the 1890s, prior to the large scale cultivation of rubber, already proprietary planters and estate managers were forming committees to address the labor shortage in Malaya. These committees would merge into the Planters Association of Malaya or PAM in 1907, and their primary interest was to petition the government to develop a migration apparatus that would secure a stable pool of migrant labor. Parmer notes that in many cases the planters were competing for labor with the government itself: "When European employers experienced difficulty in obtaining labor, they usually turned to the government asking that it assume some responsibility for the labor supply. Employers' demands were given added impetus by the tendency of plantation laborers to seek employment with the government which sometimes

²⁰Amrith, *Crossing the Bay of Bengal*, 1.

²¹Sunil S. Amrith, "Indians Overseas? Governing Tamil Migration to Malaya 1870-1941," *Past & Present* 208, no. 1 (August 2010), 232.

offered better wages."²² The government was slow to respond to employer demands, but due to pressure from PAM and other groups it slowly began to take measures to control the inflow of labor.

During the late nineteenth and early twentieth centuries, there were several mechanisms by which Indians migrated to Malaya. There has always been some amount of unassisted free labor migration: workers who pay for their own journey and are in theory free to enter contracts with any employer. However, in the early decades of British rule, most workers who came to Malaya were indentured. Their transit was paid for by their future employers, the cost of which was then docked from their pay. In the late nineteenth century, the conditions of indentured labor were quite poor, and a Labor Commission was established in 1890 to conduct an inquiry into high mortality rates among indentured Indian laborers. The Commission recommended that the government take on responsibility for overseeing labor recruitment and for improving labor conditions.

The government of Malaya was reluctant to assume total responsibility for procuring labor on behalf of the planters, but being itself a major employer of Indian immigrant labor it saw the need for a steady and regulated inflow of migrant workers. In 1907 the Malayan government established the Indian Immigration Committee and the Tamil Immigration Fund to supervise the recruitment of workers from south India, as well as to pay for any associated costs.²³ The fund was started with initial loans from the government, but thereafter drew its money from assessments on any employers of Indian labor. As the government was a major employer of Indian labor it paid into the fund at the same rate as planters. The primary purpose

²²Parmer, *Colonial Labor Policy*, 18.

²³Parmer, *Colonial Labor Policy*, 39.

of the Committee and its fund was to secure Indian labor at a fixed low cost and to discourage planters from poaching others' workers through high wages. The Indian Immigration Committee operated from its conception in 1907 until 1938 when India would outlaw all forms of assisted labor migration to Malaya.²⁴

At the same time that the Tamil Immigration Fund was established in Malaya, an anti-indenture movement was growing in India among various interests. The opposition to indenture came from two different sectors of Indian society, and they opposed indenture on rather different grounds. The local planters and ryots in Madras argued that the practice of indentured labor migration had a detrimental affect on the local labor market. They claimed that landless laborers were systematically dodging debts to their landlords by moving abroad, and they complained that returned coolies were in such ill health that they were no longer suited to agricultural labor. They appealed to the government of Madras to regulate indentured migration, but this was met with indifference. Although officials confirmed that many emigrants were leaving behind debts, they blamed the ryots for unfair labor practices which drove the workers to leave, suggesting that "that the Government could not do anything to protect the ryot until he realised that if he wanted to keep his labourers he must pay them better."²⁵ The Raj, caring little for the profits of local Tamil planters, claimed that the system of emigration was for the workers own good, a 'free' labor market providing workers with the opportunity to move to wherever labor conditions suited them.

The planters were joined in denouncing emigration by Indian nationalists, although their complaints against indenture had less to do with promoting a domestic labor market, and more to

²⁴Parmer, *Colonial Labor Policy*, 57.

²⁵M. M. Mani, "Abolition of Indentured Labour Emigration to the British and French Colonies: Role of the People, the Press, and the Planters of Tamilnadu," *Proceedings of the Indian History Conference* 1957 (1996), 698.

do with bolstering a nascent republicanism that was concerned with the treatment of Indian citizens abroad. M.M. Mani notes that partisans of Indian independence, such as Gopalakrishna Gokhale, Mahatma Gandhi, and C.F. Andrews "wrote in the press and spoke on public platforms requesting the Government to abolish the system of Indentured labour emigration."²⁶ Prominent among the newspapers carrying nationalist opinions were *The Hindu*, published in English in Tamil Nadu, and *New India*, a daily published by home rule advocate Annie Besant. The arguments made in these pages were often moral, emphasizing the evils of indenture as a national injury. Mani writes of *New India*, that "the paper inferred that the self respect of India was hurt by that 'refined and civilized form of slavery'. Commenting on emigration Mahatma Gandhi said that emigration did not do any good to the country from which people emigrated. The returned emigrants were morally corrupt."²⁷ A major aim of the nationalist propaganda was to counter the Raj's contention that it maintained the system of indentured emigration to improve the lots of workers, and to demonstrate that under British rule, Indians were being systematically mistreated.

Ultimately, the Raj gave in to the combined pressure of the nationalists and the ryots. In 1910 indentured labor contracts were banned for migrants traveling to Malaya, although in-place contracts were allowed to run their course. In 1917, labor recruiting was temporarily paused due to the War, and a few years later the 1922 Emigration Act formally banned all indentured emigration from India to other colonies.

The 1910 abolition of indentured labor to Malaya only applied to Indian labor, and Chinese and Javanese workers were still brought in on indenture contracts. However, most of the

²⁶Mani, "Abolition of Indentured Labour Emigration," 696.

²⁷Mani, "Abolition of Indentured Labour Emigration," 704.

workers on rubber plantations were Tamils, and so the Tamil Immigration Fund had to find alternate ways of ensuring the continued influx of ready workers. They had two important considerations: They had to counter widespread reporting in India of the deplorable working conditions on Malayan plantations, and they had to find some way of compelling workers to stay on their plantations while still upholding the image of 'free labor'. Planters in Malaya were compelled to improve the living and working conditions on their estates - but only up to a point. In the legislation passed by the Malayan government, the primary reason given for the concern with labor conditions was not the well-being of the worker, but the detrimental effect that poor conditions had on recruitment.

Along with the difficulties of attracting labor, planters in Malaya found domestic reasons to pressure the Malayan government to regulate the supply of labor. This pressure came in part from planters who wanted a regulated system of labor importation to reduce poaching of 'free' laborers by other estates. The colonial government itself also seemed anxious to create a more settled labor force, a concern that was tied up with a disdain for the Malayan-owned small holdings. In the 1923 *Annual Report on the Social and Economic Progress of the People of the Straits Settlements*, the colonial secretary complains of outbreaks of Mouldy Rot in Malacca which was spreading through the Malayan owned small holdings. He notes that the disease was absent on larger estates in part because "they have settled labor forces and do not employ coolies indiscriminately picked up, there being ordinarily therefore no contact with infected trees...."²⁸ In years during which the price of rubber fell, the smaller holdings could not afford prophylactic measures against disease, and so the Rubber Research Institute and the Departments of Co-

²⁸Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1923), 29.

operation and Agriculture developed a "Rural Lecture Caravan" which would tour villages in Malaya to give Malay-language lectures "on agricultural and co-operative subjects, illustrated or followed by the lantern slides and moving pictures in the evening." At least part of the goal seems to have been to discourage small holders from planting rubber altogether, as the section of the report on the Rural Lecture Caravan ends simply: "Good audiences were attracted and a number of enquiries for selected padi seed resulted."²⁹

Throughout the annual reports, a similar complaint recurs: that periods of high rubber prices had caused "land owners to plant up unsuitable land with rubber, so that in parts of Malacca and Province Wellesley low-lying land with a high water table, suitable for padi or certain annual crops, is often found to contain stunted and unhealthy rubber trees with an inadequate root system... finally it has caused public interest to be centred on rubber cultivation and has led to the neglect of all other crops giving reliable but smaller profits."³⁰ And then again in 1924: "as a general rule it can be stated that small holders have more frequently than large holders selected unsuitable (often swampy) land for rubber cultivation during the days of high prices."³¹ The implication of these concerns is clear. In the eyes of the colonial secretary, not only were Asian workers vectors of disease who need to be corralled into their worksites, but even as administrators Malaysians were incompetent and needed to be educated at a great expense to the colonial apparatus. Malaysian planters should not share in the profits of export production, but should focus on producing domestic goods like rice.

²⁹Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1931), 24.

³⁰Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1922), 32.

³¹Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1924), 28.

The annual reports of the colonial authorities in Malaya are shot through with a tension between the desire to affirm the formal equality of all British subjects and the desire to enrich the white planters at the expense of the non white populations - both indigenous and migrant. There was an urgent need for a system of labor hierarchy that would appease critics of bonded labor from India, but that would still meet the considerable need for a compliant workforce. This tension is expressed quite aptly in a footnote to the 1917 annual report:

"Assisted immigrants" are of two classes: - (i) the independent labourer who gets a free ticket from the Emigration Agent in India upon proof that he is a bona fide labourer, and who is free to work where he pleases; (ii) the Kangany labourer, who is recruited in India by a licensed Kangany, and who goes to a particular place of employment, though he is free, upon arrival there, to give a month's notice of his intention to leave. There is no indentured Tamil labour in the Colony.³²

Laborers of both types had their transit costs covered by the Tamil Immigration Fund, and both were formally free to work for whomever they desired, but there were significant differences between them. It is odd that despite the significant costs of recruitment, workers produced by the kangany system were preferred to so-called independent laborers. This is partly due to the specifics of kangany recruitment: a former estate worker who wishes to become a recruiter first applies for a license and then "proceeds to his own village where he informs his friends and relations of the conditions of labour on his estate."³³ The kangany would then often travel back to Malaya with his recruits and serve as their overseer on the plantations. In his role as both overseer and recruiter, the kangany was able to exercise his authority as both job-giver and relative to ensure the compliance of the workforce.

³²Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1917), 28.

³³Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1931), 25-27.

Parmer makes an interesting claim about the effect that the kangany system had on the individual worker: he says that in the system of labor migration, "[e]xploitation was practiced - not by the planter but by the professional recruiter. The government, rather than trying to abolish or reform the recruiter, had propped up the system by law."³⁴ Although I doubt that exploitation was "not [practiced] by the planter" - indeed there is much evidence to the contrary - this quote indicates that the colonial government was interested in, and had some success in, maintaining a system of exploitation and at the same time shifting that blame away from the planters and the colonial apparatus and onto an indigenous population - the kangany recruiters.

The following section will focus on the kangany system of recruitment. I will trace the development of the kangany into a hybrid role of recruiter and overseer, and its parallel with other overseer roles in Indian labor. Especially important is the fact that kanganies generally recruited from their own villages, and among their own kin. My discussion will demonstrate how the kangany system allowed the colonial state to resolve, for a time, the contradiction between bonded and free labor by exploiting existing family and community ties in the villages of the Madras Presidency.

The Kangany System of Recruitment and Labor Discipline

The word kangany is of Tamil origin, indicating "'overseer" (kan means "eye" in Tamil); in the Tamil Bible, kangany is the term for a bishop, the overseer of the congregation."³⁵ The kangany, however, forms part of a larger structure of labor recruitment and discipline throughout south and south-east Asia, called by different names in various places: *kangany* among Tamil populations, especially in Malaya or Ceylon; *maistry* among Indian workers in Burma, and

³⁴Parmer, *Colonial Labor Policy*, 25.

³⁵Amrith, *Crossing the Bay of Bengal*, 115.

sardar in the region around Calcutta. Dipesh Chakrabarty discusses the *sardar* at length in *Rethinking Working Class History*, and although this role differed from the *kangany* in many ways, each being the product of a specific industry and local social context, they shared a common, and for our purposes crucial, feature: the exploitation of family and community bonds in service of labor recruitment and discipline.

For Chakrabarty, writing about the role of culture in the labor history of Bengali jute mills, the figure of the *sardar* represents a challenge to the dominant western historiography of capitalism. In his reading, Marx situated the working classes in the context of bourgeois social relations, most important of which were the category of citizenship, formal equality before the law, and the freedom of the contract. These categories of bourgeois equality are limited and "necessarily inimical to the construction of any socialist, communist, or communitarian ethic or order Yet Marxism shares in some of the premises of bourgeois egalitarianism (while being critical of it) and the task for classical Marxism was always one of transcending (in the Hegelian sense of the world) its shortcomings."³⁶ In this reading, citizenship, equality, and contract-freedom are the constituent components of capitalism that serve as the grounds of contestation in the struggle to supersede capitalism. The *sardar* system however, as a prevailing structure of the Bengali working class, is based on a rather different social order: hierarchy rather than equality, and labor discipline based on family and community relations and their expectations rather than on the impersonal compulsion of the contract wage. For Chakrabarty, this radically different class structure requires a rethinking of the categories of capitalism and a more careful consideration of the role of 'culture' in the formation of the working classes.

³⁶Dipesh Chakrabarty, *Rethinking Working-Class History: Bengal, 1890-1940* (Princeton, NJ: Princeton University Press, 1989), xiii-xiv.

Chakrabarty's intervention into the relationship between capitalism and bourgeois social relations is crucial, but his framing often falls short of his own insight. Rather than emphasizing the flexibility of capitalism and its ability to subsume diverse social structures in service of accumulation, he insists that the sardar system is 'precapitalist' or 'prebourgeois', at times indirectly reinforcing an orthodox, stagist conception of the history of capitalism. It is worth quoting Chakrabarty's discussion of the authority of the sardar over his workers at length:

The respect that the sardar commanded had roots in certain other aspects of the culture to which both the sardar and the worker belonged. It derived from a precapitalist culture with a strong emphasis on religion, community, kinship, language, and other similar loyalties. It seems significant, for instance, that all the words used by the workers (and others) to describe *sardari* extortions - payments summed up in the legalistic expression 'corruption' or 'abuses' - were words of precapitalist, pre-British origin: *dastoori*, *bakshish*, *batta*, *parbani*, *salami*, and so on.... Besides, the *sardar's* mode of operation had certain crucial precapitalist elements. He usually recruited on the basis of the often overlapping networks of community, village, and kin.... Much of the *sardar's* social control of the workforce lay in the relationships of community, kinship, religion, and so on, and in the ideas and norms associated with them.³⁷

Workers under a sardar were compelled to work not by the impersonal domination of the wage that we see in traditional Marxist historiographies - the freedom to work or to starve - but by the personal domination of a family or community member who often employed naked violence to compel his workers to labor. The division of personal and impersonal domination into 'pre-capitalist' and 'capitalist' forms of compulsion is common, and indeed the rapid expansion of the impersonal compulsion of the wage is a fundamentally capitalist development. But to assign all direct or personal forms of cultural and hierarchical domination to a category of 'pre-capitalism' is to underestimate the flexibility of capitalism in subsuming and transforming diverse social systems in the service of accumulation.

³⁷Chakrabarty, *Rethinking Working-Class History*, 112.

In the case of the kangany system on rubber plantations, the so-called 'pre-capitalist' pattern of domination through 'networks of community, village, and kin' were in fact constituent of global capitalist accumulation. They were not elements of an alternative path to capitalism specific to South Asia, but were crucial to the operation of capital as a global system, and in particular to the development of a global automotive industry that required large inputs of rubber. The 'strong emphasis on religion, community, kinship, language, and other similar loyalties' - and especially the bonds of family - transformed these relations into sites of exploitation and expropriation for western capitalists engaged in colonial industry.

Initially, the kangany system was relatively informal, and "most *kanganies* were simply the appointed spokesmen of gangs of laborers who traveled together from South India."³⁸ They were common among the Tamil workers on coffee and tea plantations in Ceylon, and when planters traveled from Ceylon to Malaya in the late 19th century, they carried with them a preference for Tamil labor, organized under kanganies. As the distance from Madras to Malaya was far greater than the distance to Ceylon, the kanganies began to absorb more of the role of recruiter in addition to overseer. They were particularly valuable in this role precisely because of the connections they maintained with their villages and communities back home in southern India. Upon being sent back home, they offered advances to family and community members who were able then to "free themselves from servitude, or to pay off debts incurred in times of dearth."³⁹ These advances created new debts however, and although "relations of caste and kinship could moderate the kanganies' power ... such intimacy could also make the bonds of debt

³⁸Amrith, *Crossing the Bay of Bengal*, 83.

³⁹Amrith, *Crossing the Bay of Bengal*, 83.

more pervasive."⁴⁰ Social and traditional bonds were therefore not maintained, exactly, but ruptured and reconfigured to serve the needs of plantation labor.

The exploitation of family connections between India and Malaya led to significant abuses by kanganyes. Although the companies (and later the Tamil Immigration Fund) would pay the kangany's travel costs and expenses as well as a commission for each recruited worker, many kanganyes resorted to exploitative practices to further enrich themselves. Once workers reached India, they were formally free, but this freedom only refers to the lack of a legal compulsion to work - the economic compulsion of debt was not only present, but was reinforced by the personal compulsion of hierarchical family and community ties. This is visible in the complaints of the behavior of kanganyes during their recruiting trips to India. They exploited family and community connections in order to make commissions, drawing the ire of Indian officials:

A Madras government officer in 1925 listed the irregular recruiting practices as: obtaining recruits from professional recruiters or hotel-keepers; forging the signature of village *munsifs* who from 1923 were required to see each intending emigrant; exploiting family quarrels to get some members of the family to emigrate; promising young men that they could find wives in Malaya; catching recruits at weekly "shandies;" matching strangers in order to circumvent the rule that minors and women be accompanied by a parent or relative; misrepresentation of work and wages; substitution of an unfit person for one already passed.⁴¹

Despite these abuses, the kangany system was the preferred method of labor recruitment for both planters and the Malayan government. In fact, these abuses, insofar as they made the recruit dependent on the kangany, seem to have been implicitly condoned. After recruiting a batch of workers, the kangany would travel back to Malaya with them and serve as their overseer or foreman on the estate. Since the workers were formally free apart from their debt, planters were

⁴⁰Amrith, *Crossing the Bay of Bengal*, 83.

⁴¹Parmer, *Colonial Labor Policy*, 58.

forced to "rel[y] on a combination of reasonable terms and the kangany's influence over the laborer to keep the laborer on the estate."⁴²

Amrith argues that this blurring of the lines between free and bonded labor was a central element in the system of labor migration across the Bay of Bengal. The manipulation of family and community ties allowed the governments to resolve the legal category of free labor with the planters need for compelled labor. The use of the kangany system and the creation of the legal fiction of the free Tamil laborer did not, however, stifle the outcry against the working conditions in Malaya. Both Indian Nationalists and British trade unionists decried the unfreedom of the labor in Malaya. In 1926 the Labour Research Department, a research organization affiliated with UK trade unions, published a report called "British Imperialism in Malaya" which demanded solidarity between domestic industrial workers and migrant labor on the colonies: "British workers have to realize that the exploitation of cheap imported labor in Malaya is not a separate episode, but is part of the effort of capitalism to maintain itself by depressing the standards of all workers."⁴³ This depressed standard included dangerous working conditions, low wages, and compulsion to work. It is clear that British unionists considered the freedom of the laborer under the kangany system to be a mirage, noting that "[e]ven after the 'free' immigration scheme took the place of indentured labour, the deathrate remained appallingly high."⁴⁴ The assisted immigration scheme indeed served to undermine labor, and it contained the contradiction of unfree free labor in the colonies: "The Admiralty must co-operate with the planters in bringing over more wage workers; and some method other than an increase of wages

⁴²Parmer, *Colonial Labor Policy*, 22.

⁴³"British Imperialism in Malaya," Colonial Series (London: Labour Research Department, 1926), 5.

⁴⁴"British Imperialism in Malaya," 37.

must be found to induce them to come, because 'the forcing up of rates of pay will not help planters,' and a rising standard either of wages or leisure may stiffen the workers' demands." The end result: "the use of some form of compulsion, direct or indirect, such as has been applied in other parts of the British Empire."⁴⁵

Though reviled by the both Indian Nationalists and the British Left, the kangany system was strongly supported for decades by both the planters and the government of Malaya. However, this support began to erode towards the middle of the 20th century. This erosion was effected by several forces: changing conditions in the global economy (especially the effect of the Great Depression on rubber demand), growing opposition in India to assisted emigration, and a desire to develop a domestic population of Tamils that could be exploited without the cost of importation. The issue of establishing a permanent Tamil laboring population was first raised in the 1920s, but was met with ambivalence from the planters and the government who were at that moment satisfied with the kangany system. In the thirties, however, the price of rubber dropped, and with it the wages paid to plantation workers. This resulted in a mass exodus of Tamil laborers who desired to return home, as well as Indian opposition to emigration due to the low wages. Although assisted emigration was restored in the early 1930s, quotas were asserted by the Indian government and from 1935 they demanded a partial restoration of wages. This was effected in 1937, and then rolled back the next year in 1938 due to production restrictions from the International Rubber Regulation Agreement.⁴⁶

Throughout the fracas between the Malayan and Indian governments, planters revisited the possibility of establishing a domestic population of Tamil workers. From 1922, Indian had

⁴⁵"British Imperialism in Malaya," 38.

⁴⁶Parmer, *Colonial Labor Policy*, 76.

established a sex ratio rule, which declared that "the number of male emigrants, unmarried or unaccompanied by wives, being assisted to emigrate must not exceed one in every five persons to any one country in any one year."⁴⁷ Although Malaya was at times partially exempt from this rule, "[t]he Controller of Labor was of the opinion in 1935 that Malaya had already got a 'large permanent labour population' and that with an increasing number of Indian and Chinese female immigrants the tendency for laborers to remain in Malaya was increasing."⁴⁸ Indeed, during the depression when the Indian Immigration Committee was offering to cover some of the costs of repatriation, many Tamils chose instead to stay in Malaya. Although planters were anxious about India cutting off the migrant labor supply, they evidently did not feel compelled to accept the Indian government's wage demands. In June of 1938, India halted all assisted emigration to Malaya. Negotiations for its resumption lasted until 1941 to no avail, at which point the Indian Immigration Committee and the Tamil Immigration Fund became effectively defunct.

A Permanent Population

With the end of assisted migration from India, the rubber industry in Malaya was forced to adjust to a different system of labor reproduction. The two systems of labor 'recruitment' that dominated the industry in the Amazon and Africa, migration and the enslavement of the local population, were now unworkable, and so planters resorted to a system that more closely resembled the family-based generational reproduction that was promoted among auto workers in Michigan. During the twentieth century, rubber plantations established a stable, domestic community of wage-laborers that would self-reproduce, obviating the need for continuous labor migration. This community shared several characteristics with the well paid auto workers of

⁴⁷Parmer, *Colonial Labor Policy*, 71.

⁴⁸Parmer, *Colonial Labor Policy*, 75.

Michigan: it was generally organized into nuclear family units, the planters made concessions for education and childcare, and workers were paid in cash wages, rather than in merchandise. There were also, however, several significant differences: nuclear families were organized into larger hierarchies, often under the paternalistic management of a kangany, households required all members to work (including children) rather than relying on the wage of a single earner, and rather than promoting homeownership, planters housed their workers in uniform, barracks-style residences. This section will examine the conditions of reproduction within the established, permanent laboring communities.

The transition from a migration-based system of labor reproduction to one based on local generational reproduction did not happen overnight. The planters did not build a community of working class families from scratch, but drew on existing population dynamics. As highlighted in previous sections, the family and gender dynamics of migrating workers had long been debated. Although Malaya was partly exempt from India's rules mandating that a certain proportion of assisted migrants be women, some plantations had actively been recruiting whole families from South India as early as 1904.⁴⁹ In some cases, kanganyes were "given infrastructural aid to ensure that they recruited more single female coolies by being paid extra for every woman they recruited."⁵⁰ As a result, "[b]y the 1930s, immigrant Indian women formed 30 to 40 percent of the total coolie force on European rubber estates in Malaya," but despite the essential role they played in creating a stable labor force for plantations, "their active engagement in the socioeconomic and political life of the estates has been occluded and sometimes misrepresented

⁴⁹Ravindra K. Jain, *South Indians on the Plantation Frontier in Malaya* (Yale University Press, 1970), 238-239.

⁵⁰Arunima Datta, *Fleeting Agencies: A Social History of Indian Coolie Women in British Malaya* (Cambridge University Press, 2020), 39.

in labor studies."⁵¹ Arunima Datta argues in *Fleeting Agencies* that both colonial and nationalist perspectives viewed coolie women as "useful pawns to be categorized, stereotyped, and deployed in the realm of colonial power politics,"⁵² and their migration treated as a footnote to the more important migration of men.

In reality, coolie women were central to the plantation industry in Malaya. Not only did they work alongside men in the fields - Datta remarks that "female coolies were counted and recorded as individuals whom the planters employed as laborers in their own right and not as parts of family units"⁵³ and that "[t]here is no evidence of any significant gendered division of labor"⁵⁴ - but they were also valued for their domestic labor and for their ability to bear and rear a new generation of workers. Coolie women thus held a position in Malayan society that mirrored that of Black women in Detroit during the early years of Fordism: they were required to contribute to household income both as wage earners and as unwaged care workers.⁵⁵ Similarly to these racialized semi-proletarian families in Michigan, Malayan families also relied on child laborers to contribute to family income, and "once female coolies began settling in Malaya, child labor on estates became an increasingly visible sight on colonial plantations."⁵⁶

In the regimented communities of Malayan rubber plantations the distinction between domestic labor and plantation labor was blurred, particularly as regards Coolie women, whose reproductive labor "was not solely a part of the coolie domestic economy," but "formed a crucial

⁵¹Datta, *Fleeting Agencies*, 8-9.

⁵²Datta, *Fleeting Agencies*, 4.

⁵³Datta, *Fleeting Agencies*, 48.

⁵⁴Datta, *Fleeting Agencies*, 51.

⁵⁵Kathleen Stanley and Joan Smith, "The Detroit Story: The Crucible of Fordism," in *Creating and Transforming Households: The Constraints of the World-Economy*, ed. Joan Smith and Immanuel Wallerstein, Studies in Modern Capitalism (Cambridge University Press, 1992).

⁵⁶Datta, *Fleeting Agencies*, 33.

part of the colonial plantation system and its future."⁵⁷ Although domestic labor was generally unpaid, planters established nurseries where "older coolie women who were no longer physically fit for strenuous work on the estates"⁵⁸ were paid a wage to care for young children while their mothers returned to work. These nurseries were free and also supplied free food to the children. In addition to nurseries, workers were also supplied with free health care, schooling for children, sick leave and maternity leave for women.⁵⁹

Plantations provided housing to their laborers as well. Ravindra K. Jain describes one plantation, Pal Melayu, as a "miniature township with its own roads and houses, its own school, temples, dispensary, and club."⁶⁰ Although the houses were rudimentary - in the 1950s Pal Melayu's homes did not have individual bathrooms, running water, or electricity - the workers did not pay rent. Further, they had garden plots and were encouraged to plant subsistence crops.⁶¹ This is like Ford's workers in Michigan, who were encouraged to garden at home, but unlike rubber collectors in the Amazon who were disallowed or otherwise unable to tend to gardens, or who had their crops stolen by rubber company management. The workers in Pal Melayu were forced to conform to strict living standards and had very limited freedom in how they kept up their homes: "[f]or instance, no laborer occupying a line may plant flowers in front of his residence. Similarly, a laborer risks his job on the estate if he puts up a tin shed to serve as a bathroom."⁶²

⁵⁷Datta, *Fleeting Agencies*, 60.

⁵⁸Datta, *Fleeting Agencies*, 62.

⁵⁹ Jain, *South Indians on the Plantation Frontier in Malaya*, 99.

⁶⁰Jain, *South Indians on the Plantation Frontier in Malaya*, 13.

⁶¹Jain, *South Indians on the Plantation Frontier in Malaya*, 18-19.

⁶² Jain, *South Indians on the Plantation Frontier in Malaya*, 20-21.

In this strict requirement can be seen something of the paternalism of Ford. Planters were clearly interested in controlling more than just the working hours of laborers, and the suppression of personality via a ban on flowers brings to mind Ford's rationalization of workers and the evacuation from working tasks any element of creativity. But whereas in Michigan, the paternalism was impersonal, mediated by a sociological department armed with lists of qualities they wanted to reward, the paternalism of Malaya's rubber plantations was decidedly personal: both familial and economic compulsion were combined in the figure of the kangany who was often related by blood to his underlings.

Even when the kangany's role as recruiter was made obsolete by the outlawing of assisted migration from India, he retained his influence among workers on the plantations. As Jain notes, "[n]o change in the industrial subsystem could immediately affect the existing networks of kinship and economic ties into which a kangany found it advantageous to enmesh a new recruit."⁶³ Sometimes their interactions with laborers were positive, as they retained ceremonial roles in community events, such as in weddings. In other cases, however, they had a harsher role:

In times of such acute need for a stable labor force as the mid- and late 1920s, the manager on Pal Melayu manipulated roles in such a way that for the coolie hand his European assistants became the benevolent father figure while the head kangany and his kanganies represented their punitive counterparts. The capture and chastisement of bolters as well as the field supervision of laborers was entrusted increasingly to the kangany, head kangany, probationer conductor (*kanakka-p-pillai*), and Asian field conductors.⁶⁴

In other words, the personal relationship between the kangany and the workers in his gang served the purpose of social cohesion, but was also utilized by plantation managers to maintain labor discipline.

⁶³ Jain, *South Indians on the Plantation Frontier in Malaya*, 231-232.

⁶⁴ Jain, *South Indians on the Plantation Frontier in Malaya*, 218.

Kanganies also served as money lenders in the community, providing loans to their workers as well as to local craftsmen. The most common form of lending was the *cittu*, or savings group, in which members contributed some small amount each month to a general fund which would be distributed to members on a rotating basis.⁶⁵ In this manner, each member of the group would become debtors and creditors to each other in turn. The leader of the *cittu*, who was often but not always a kangany, would be responsible for collecting the contributions and distributing the payouts. The organizer would often take the first payout and use it to fund some entrepreneurial venture: Jain mentions *cittus* organized to fund a goldsmith, a furniture maker and a bicycle shop.⁶⁶ The *cittu* allowed the kangany to profit off of his unique position embodying both personal and economic compulsion. The *cittu* benefited the plantation as well, as both debtors and creditors were less likely or able to leave for work elsewhere.

These ventures - goldsmithing and furniture and bicycle shops - hint at an important difference in the operation of debt between Malaysia and the Amazon, as well as a difference in patterns of consumption. In both places debt was used as a tool to discipline and capture labor, but in the Amazon the debt primarily came from merchandise that was advanced to tappers by their managers - including food and tools that were manufactured thousands of miles away. In Malaya the debt was interpersonal. Although initially workers were indebted to their kanganies for recruiting costs, as time went on a robust debt economy developed between workers. This was part of a growing domestic market that allowed workers to buy locally produced luxuries and daily goods.

⁶⁵ Jain, *South Indians on the Plantation Frontier in Malaya*.

⁶⁶ Jain, *South Indians on the Plantation Frontier in Malaya*, 169-171.

This raises one final question: if domestic commodity consumption is essential to the realization of value in the global economy, why did the European-owned plantations allow and even encourage a robust economy of locally-made goods? Why did they not substitute foreign goods as in the Amazon? A possible answer can be found in the annual reports compiled by the colonial government, which tracks import and export data alongside domestic production: unlike in the extractive economy based on depletion of an already extant resources, rubber plantations had to be actively developed and maintained. The report on the Federated Malay States for 1936 mentions several small crops that supplemented rubber and contributed to social stability. Pineapples were initially grown alongside rubber to defray the costs of the long waiting maturation period. The crop was so successful that eventually pineapple was processed at local canning factories for export.⁶⁷ It was also promoted during downturns in the rubber market. However, most other crops were not consumed locally: the report notes that "[p]ractically none of the coffee produced in Malaya is exported"⁶⁸ while in 1936 almost 800,000 pounds were imported, and with regard to fruit other than pineapples, "the demand is by no means satisfied by local production."⁶⁹ However rice, the largest food crop under cultivation dwarfs these others. The total acreage of coffee, pineapples, and other fruit in the Federated Malay States amounted to a little over 60,000 acres, whereas rice was cultivated on over 175,000 acres, yielding approximately 75,000 tons. Despite this significant cultivation, rice remained a major import

⁶⁷ Colonial Office Great Britain, *Annual Report on the Social and Economic Progress of the People of the Federated Malay States* (H.M. Stationary Office, 1936), 35.

⁶⁸ Great Britain, *Federated Malay States*, 1936, 36

⁶⁹ Great Britain, *Federated Malay States*, 1936, 44

commodity for Malaya: in 1933, Malaya imported over half a million tons of rice, largely from what are now Thailand and Myanmar.⁷⁰

The import and export data of Malaya paint a very different picture than we have seen in the Amazon. On the one hand it is true that the economy of Malaya was based on the production of rubber and tin for export - the annual report for the Straits Settlements in 1933 states that "the trade of Malaya is dominated by imports of produce from Netherlands India, exports of rubber and tin to America, and imports of manufactured goods from the United Kingdom and Japan. On the other hand, the imported goods were either supplements to local production, or were heavy manufactures intended for industry. The 1935 annual report for the Straits Settlement lists "rice, copra, kerosene, motor cars, dredges and dredging materials" as the fastest growing imports.⁷¹ It is clear that despite being dominated by two large export industries, the domestic economy of Malaya was much more diversified than in the Amazon. Many of the commodities consumed by workers in Malaya were produced locally - and most of those which came from abroad such as fruit and rice were imported not because local production was undermined, but in spite of its promotion. The imported population simply required more food than could be produced locally.

The Aftermath: Coolie-hood and Repatriation

It is easy to read into the history of Tamil labor in Malaya a certain familiar teleology. Indentured labor gave way to the nominally free (though indebted) labor of the kangany system, which in the mid-century resolved into a fully capitalist system of overseeing mediated not by traditional forms of authority, but by the wage. I want to resist this reading by looking at the

⁷⁰ Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1933), 34.

⁷¹ Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1935), 41.

underside of this 'progression.' The primary effect of the slow development of formally free wage-labor among plantation workers occurred away from the plantations, in the creation of what to capital's standpoint appears to be a great mass of human detritus. For most workers, the category of 'coolie' persisted long after the brief period of plantation labor.

Coolie, like 'kangany', comes into English from Tamil: the word *kūli* being a payment for menial labor. According to Bremen and Daniel, "[b]eing a menial alone did not qualify a person to be called a 'coolie'; s/he had to be a menial who did not have any customary rights." The recipient of the *kūli* was totally severed from a traditional system of social connections: "[t]here was no one who accepted any responsibility for the survival of a menial who received *kūli*, and *kūli* never served as a token of such responsibility. *Kūli*, by its sheer paltriness, came to index the fact that the recipient belonged to 'the lowest level of the industrial labour market'."⁷²

This seems at first to simply be the process of proletarianization, but Bremen and Daniel note that the word 'coolie' carries also the connotation of the Gujarati word *Kuli*, a tribe whose "members were described as 'thieves, robbers, and plunderers, as degenerate and inferior,' in short, 'as a "villainous race"'."⁷³ In the coolie is embodied both the *kūli*, the wage which separates the laborer from communal patterns of subsistence, and the *Kuli*, a member of a community marked as racially inferior. Planters in search of cheap and malleable labor found the Tamil coolie to be "not so much a proletarian who stood ready to be transported to the rural and urban enclaves created by the new economy, but one who was available and vulnerable for recruitment by virtue of being cast adrift."⁷⁴ What was to become of the coolie when the work was up?

⁷²Jan Bremen and E. Valentine Daniel, "Conclusion: The Making of a Coolie," *The Journal of Peasant Studies* 19, no. 3–4 (1992), 269.

⁷³Bremen and Daniel, "Conclusion: The Making of a Coolie," 269.

⁷⁴Bremen and Daniel, "Conclusion: The Making of a Coolie," 270.

Most of the Tamil workers who migrated to Malaya did so only temporarily. Even in the years of indenture, they served fixed terms, returning home after their contract was up. After the abolition of indentured servitude, the majority of workers still returned home after working for a few years. The reasons for return were various, but perhaps the most significant driver was the global rubber market. With the onset of the great depression and the consequent collapse of rubber production in the early 1930s, previous immigration patterns were flipped on their head. The number of laborers whose passages to Malaya were paid for by the Tamil Immigration Fund from southern India fell from 42,771 in 1930 to only 111 the next year. Meanwhile in 1931 94,918 adults and children returned to India from Malaya, over half of whose passage was paid for by the Labour Department in Malaya.⁷⁵

Repatriation was a controversial topic in colonial India, and prominent Indian nationalists such as Gandhi, though first supporting the repatriation of emigrant workers, began to advocate for the rights of Indians to remain abroad in the communities that they had established. In a survey of the 'Repatriation Debate', Heena Mistry notes that opposition to repatriation often came from the repatriates themselves. Upon returning to their home villages, former immigrant workers sometimes found their position within the caste system questioned. In one such case, Bhawani Dayal Sannyasi, who was born to an indentured father in South Africa and "saw himself as having an 'appearance, education, culture, and bearing' above the villagers, thus not meriting any treatment as an outcaste," found himself ostracized in his father's home village in India. As Sannyasi had been born abroad, and as his mother had died before their return, his caste status could not be verified upon his return to Bihar in East India. Rather than advocate on

⁷⁵Colonial Office Great Britain, "Annual Report on the Social and Economic Progress of the People of the Straits Settlements" (London: H.M. Stationary Office, 1931), 7.

his behalf, his father told the "village *panchayat* that he would label his South-African born children as illegitimate in exchange for acceptance into the community."⁷⁶ Similarly Mistry notes that returnees from indentured labor in Fiji "were targeted by scammers and thieves upon their arrival or were ostracized in their ancestral villages."⁷⁷ Through the process of migrating abroad to work as a coolie, workers both lost their status in their old communities and, as discussed before, were considered by Indian planters to be unfit for work.

The situation for Tamil repatriates in South India was much the same as in the rest of the country. Amrith quotes the Indian Agent of Malaya's complaints of the effect of repatriation on migrants, saying that in the process "'no relief is thereby afforded' to Tamil workers, 'but that their suffering is merely transferred from Malaya to South India'."⁷⁸ Jan Bremen goes further and states that returned migrant workers "discovered that they were not 'peasants' but only coolies who had given up their chance of becoming peasants again."⁷⁹ Their ties to the land had been cut, and they were not able to reenter life in their villages on the old terms. He notes that returnees from the Malayan plantations were called by the epithet 'sucked oranges' - their vital forces used up. They were not at an intermediate step between peasant and free wage worker, but had been shunted into a permanent position of precarity.

The persistence of coolie-hood beyond the contract term represents a break with the logic of capitalism, as traditionally understood. The arrival of Fordism in the world-economy did not bring proletarianization to South Asia, but destitution. Family and community structures were

⁷⁶Heena Mistry, "The Repatriation Debate After the Abolition of Indenture," *Journal of Indian Ocean World Studies* 5, no. 1 (2021), 127.

⁷⁷Mistry, "The Repatriation Debate After the Abolition of Indenture," 132-133.

⁷⁸Amrith, "Indians Overseas? Governing Tamil Migration to Malaya 1870-1941," 256.

⁷⁹Bremen and Daniel, "Conclusion: The Making of a Coolie," 278.

exploited and used up in the production process, but no stable wage form was established in their place. If we accept that the production of rubber was a capitalist phenomenon, and that it was a constituent part of the rise of Fordist industry then it becomes clear that the archetype of the Fordist worker is not the 5 dollar a day earning family man, but the footloose lumpen.

Conclusion

Today, the majority of rubber produced for the market is synthetic, totaling 15 million metric tons to natural rubbers 13.5 million in 2017.¹ Experimentation in synthetic rubber is almost as old as the period of industrial rubber itself, but it was not produced in commercial quantities until after World War I when the USSR and Germany took an interest in it.² However, due to the low price of natural rubber during the 1930s, synthetic rubber was not produced in truly great quantities until the Second World War. In 1941, Japan occupied Malaya and effectively cut off the West from its main source of rubber.³ By the next year, "Standard and Firestone reached full-scale production of synthetic rubber based on petroleum and coal, making 1 million tons of rubber per year."⁴ After the war, there was a renewal of rubber plantations in Asia, but synthetic rubber production grew 3 times as fast as natural rubber production in the 25 years following the war.⁵ The era of natural rubber was over.

This dissertation has shown what happened to the workers who powered the natural rubber industry: most of them died. In the Amazon workers died by the thousands, and in the Congo by the millions - they were murdered, they were worked to death, or they succumbed to

¹Natalie Aster, "Natural vs. Synthetic Rubber: Key Market Trends & Statistics," Market Publishers, July 31, 2018, <https://marketpublishers.com/lists/23821/news.html>.

²Maurice Morton, *History of Synthetic Rubber*, 15, no. 7 (1981), 1293.

³Richard Tucker, "Rubber," in *The Cambridge World History*, The Cambridge World History, ed. J. R. McNeill and Kenneth Pomeranz (Cambridge University Press, 2015), 430.

⁴Tucker, "Rubber," 435.

⁵Tucker, "Rubber," 436.

tropical diseases. A small portion of Indian migrants to Malaya remained and established a permanent population of Tamils there, but the vast majority either died from overwork or returned to India as "sucked oranges," alive but drained of their vital energies and unable to work. I have referred to these people in the introduction as "proletarian detritus," the dross cast off by the process of proletarianization, and I believe that they are the real legacy of Fordism.

But these workers in the Global South are not alone. Returning to the initial question which started this investigation in 2019 - "Why are all of the houses in Michigan so terrible?" - it is clear that the people in the former center of Fordism have been abandoned by capital as well. The automotive companies were, for a time, invested in improving the lives of their employees. Whereas in the Global South it was more profitable to squeeze workers for every drop of labor they had in them, in the North companies were briefly incentivized to give them a comfortable life. High pay and social programs solved several problems for capital: it reduced employee turnover, it reduced the training costs of skilled positions, and it allowed the working class to become consumers, expanding the market. Over time, this math changed and in the 1970s capital began to leave the Midwest, looking for places with cheaper labor, and more relaxed labor laws. Their former workers, tied down by mortgages and families, were not so flexible, and they remained behind in dying cities. This was especially true for Black workers who had been excluded from many of the benefits of the Fordist era.

The Flint Water Crisis in 2014 demonstrates this dynamic quite well. In the 1960s General Motors employed over 70,000 in the Flint area.⁶ In 2023, that number had fallen to 6,500.⁷ With the collapse of the manufacturing industry in Flint, the city's tax base was eroded

⁶Highsmith, *Demolition Means Progress*, 246.

⁷Ryan Jeltema, "GM Planning 'positive Plant Manufacturing Announcement' in Flint," abc12 News, January 19, 2023, <https://www.abc12.com/news/business/gm-planning-positive-plant-manufacturing-announcement-in->

and it entered bankruptcy. To save money, the city switched its water source from Detroit to the Flint River. Because the river water was much more corrosive than the water from Detroit, it leached lead from the old pipes and poisoned the homes of Flint's residents. Despite immediate outcry from the residents, it took 18 months for the state to act and provide funds to reconnect the city to Detroit water.⁸ In the meantime, "nearly 9,000 children were supplied lead-contaminated water for 18 months."⁹ Parallel to this, the river water was linked to a Legionnaires' disease outbreak that killed 12 people in the city.¹⁰

While the state and city officials were sitting on their hands, General Motors sprang into action - to protect itself. The corporation noticed that the new water was corroding the engine blocks at its engine factory. While the state was lying to the public, stating that the water in Flint was safe to drink, General Motors "began working through the bureaucratic red tape of extracting itself from its Flint water contract and hooking into [Flint] township, which use[d] water treated by Detroit."¹¹ The residents of the city of Flint did not have this option. In a cruel twist of fate, General Motors had abandoned its previous mission of investing in the quality of life of its employees, only donating \$50,000 to purchase water filters for Flint residents - about 63 cents per person.¹² The company stepped back and watched as people got sick and died in their homes from the water system in the city that it had built.

[flint/article_da598552-981c-11ed-a28a-2f3f6df08e46.html](https://www.flint/article_da598552-981c-11ed-a28a-2f3f6df08e46.html).

⁸Kenyon P. Cavender, "Predatory Inclusion and the Flint Water Crisis," *Science & Society* 87, no. 1 (2023): 8–20.

⁹Melissa Denchak, "Flint Water Crisis: Everything You Need to Know," Natural Resources Defense Council, July 14, 2026, <https://www.nrdc.org/stories/flint-water-crisis-everything-you-need-know>.

¹⁰Denchak, "Flint Water Crisis: Everything You Need to Know."

¹¹Mike Colias, "How GM Saved Itself from Flint Water Crisis," Auto News, January 31, 2016, <https://www.autonews.com/article/20160131/OEM01/302019964/how-gm-saved-itself-from-flint-water-crisis/>.

¹²Colias, "How GM Saved Itself from Flint Water Crisis."

In this dissertation, I have tried to challenge some myths surrounding the history of Fordism, especially its supposed benevolence towards workers. The benefits granted to some portion of the working classes in the early automotive industry were made possible by the immiseration of thousands, both at home and abroad. I have insisted that the phenomenon of Fordism must be analyzed within the context of its global commodity chain, which includes the rubber workers who were forced to labor under unimaginable conditions for the benefit of capitalists thousands of miles away. Finally, I have demonstrated that the preference that capital displayed for nuclear families in early 20th century Michigan is not a universal feature of capitalist society, but a specific solution to a historical set of problems. When the conditions of production changed, so did the preferred household configuration.

Unfortunately the entirety of the global rubber industry is beyond the scope of a single dissertation. I have had to be selective in choosing my cases. I have said nothing of the rubber factories of the US and Europe, and I have said too little of rubber collection in Africa, Central America, Sumatra, and other places. The rubber industry in these places, whether based on wild collection or plantation production, will provide fertile ground for further study. In future work, I aim to examine these places as well to widen our understanding of the different patterns of social reproduction around the globe, and their intimate relation to industrial production in the core countries.

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